

Split Zero and the Arithmetic Trace

A reconstruction from the complete support geometry

Split-Zero research programme

23 September 2026; edition SZ-20260922-031

This manuscript constructs the programme from its coefficient operations, all relevant prime spectra, and the maps connecting them. Every assertion below has its stated mathematical scope. The manuscript is being written cumulatively; it does not assert completion of the RH argument or of the entire requested reconstruction.

Contents

1	The coefficient operations and their prime spectra	3
1.1	Every multiplicative prime, before imposing addition	4
1.2	The complete lattice of zero-amplitude support	5
2	Supported zero as the central arithmetic prime	6
3	The absolute-base maps and the role of every integer	7
4	Exact localizations and contraction maps on the full support spectrum	9
4.1	1. Full coefficient object and its primes	9
4.2	2. The exact lattice localization at a prime	10
4.3	3. Pure-support prime stalks, including all surviving levels	11
4.4	4. Amplitude prime stalks retain the entire lattice	11
4.5	5. Every generalization and localization contraction	12
4.6	6. Scalar inclusion, branch maps, and their local forms	13
4.7	7. Finite Boolean support as an exact fibre product	13
4.8	8. Complete contractions from the independent finite product	14
5	The full ideal arithmetic, divisor coordinates, and norms	15
5.1	Cancellation and the localization retaining both zeros	15
5.2	The number-field ideal calculation used in the comparison	15
5.3	Every fractional ideal, with both zero ideals retained	17
5.4	The original class relation and its exact comparison	18
5.5	Codimension and the divisor map	19
5.6	Finite norms and the ideal zeta function	21
6	The full adelic valuation space and its derived geometry	22
6.1	Objects and source scope	22
6.2	The entire space of split pseudo-absolute values	23
6.3	Finiteness, residue, and the complete prime space	24
6.4	Integral structures and the support point	25
6.5	The exact additive bridge and its limits	27
6.6	Valuation stalks and the sheaf that retains support	27

6.6.1	The full spectral extension, including the extra generic point	29
6.7	The exact scope of the Tor assertion and its refinement	30
6.7.1	The whole-scheme submodel limit does not have the claimed bound . . .	30
6.7.2	The valuation refinement is a new space with an exact map	30
6.7.3	Where the derived contribution goes under a blowup	32
6.8	Adelic data transported without changing the product formula	33
7	Prime faces and the complete Weil functional	34
7.1	The exact semilocalization associated with a monoid prime	34
7.2	Every prime-power, Gamma and pole term	34
8	The support defect of the Fourier map	35
8.1	All mixed support faces	36
9	All adelic boundary faces and the two pole terms	37
9.1	The exact finite boundary representation	37
9.2	Original generators, projectors and higher cohomology	39
9.3	Pairing, Fourier duality and the actual pole receiver	40
9.4	Passing to every prime with the original reference functions	40
10	The mixed-support extension and its arithmetic determinant	42
10.1	A nonzero class carried by all the mixed faces	42
10.2	The determinant of the actual local differential	43
10.3	An exact receiver in the original Weil prime term	44
11	Relative cohomology at the full supported prime	46
11.1	The full space and the receiving categories	46
11.2	The six elementary functors and their derived terms	47
11.3	Cross extensions, relative traces, and exact pairing maps	49
11.4	What the actual split structure sheaf retains	50
11.5	Principal parts at the supported primes	51
11.6	The exact filtration, finite pairings, and length weights	53
11.7	A trace operator receiving the original prime-power Weil term	55
11.8	Provenance and exact dependence	56
12	The prime-degree operator and its exact Weil pairing	56
12.1	The actual filtration, degree and finite character pairing	56
12.2	The square-root factor is fixed by the prime degree	57
12.3	The original prime term is the energy of this operator	58
12.4	The same determinant and the exact finite support guard	59
12.5	The precise local weight identity	59
13	All integers, including supported zero, on the relative object	60
13.1	Simultaneous arithmetic action and the original integral vector	61
13.2	Supported zero acts in an explicitly computed distribution space	62
14	Complete support cohomology of the original maps	62
14.1	The two zeros and cancellation paths	63
14.2	Every compatible support trajectory	64
14.3	Reconstruction and the full $G_{\mathcal{L}_D}(R)$ base	65
14.4	Gluing, matrix changes, and the space defined by curvature	67
14.5	The original source, conductor and invariant observation	68
14.6	Exact return to the original metric and arithmetic action	70

14.7 One support base for the four original cutoffs	71
14.8 Sources and verification scope	72
15 The attained metric on every support cohomology fibre	72
15.1 Return through all four original cutoffs	74

1 The coefficient operations and their prime spectra

Definition 1.1. For a nonzero commutative unital ring R , let $G(R) = R \sqcup \{\tau\}$. Supported elements have their original ring sum and product; $x + \tau = x$ and $x\tau = \tau$. Write $e = 0_R$ for the supported zero. Thus $e \neq \tau$, $e^2 = e$, and $er = e$ for every supported r .

The injection $\tau \mapsto (0, 0)$, $r \mapsto (r, 1)$ into $R \times \mathbb{B}$ proves all semiring laws, including distributivity. Here addition in $\mathbb{B}$ is join and multiplication is meet. The two projections are

$$p : G(R) \rightarrow R, \quad p(\tau) = 0, \quad p(r) = r; \quad \chi : G(R) \rightarrow \mathbb{B}, \quad \chi(\tau) = 0, \quad \chi(r) = 1. \quad (1)$$

They jointly retain the entire original element. Every map to a ring kills e since $e + e = e$, and then factors uniquely through p . The supported inclusion of R does not preserve the specified additive zero and is not a semiring section of p .

Theorem 1.2 (The scalar spectrum and its stalks). *The ideals of $G(R)$ are $\{\tau\}$ and $J \cup \{\tau\}$ for ring ideals J . Its primes are*

$$P_\tau = \{\tau\}, \quad P_{\mathfrak{p}} = \mathfrak{p} \cup \{\tau\} \quad (\mathfrak{p} \in \text{Spec } R).$$

The map induced by p identifies $\text{Spec } R$ with $V(e)$, while $D(e) = \{P_\tau\}$ is open and dense. The localization stalks are

$$G(R)_{P_\tau} = \mathbb{B}, \quad G(R)_{P_{\mathfrak{p}}} = G(R_{\mathfrak{p}}).$$

For $R = \mathbb{Z}$, all strict maximal prime chains are

$$\{\tau\} \subsetneq \{\tau, e\} \subsetneq p\mathbb{Z} \cup \{\tau\}. \quad (2)$$

In particular its dimension is two.

Proof. Every ideal contains τ . If an ideal contains a supported element, multiplication by e places e in the ideal. Its supported intersection is a ring ideal because multiplication by -1 supplies additive inverses. This proves the ideal classification and its converse. A product equals τ only if a factor does, proving primality of P_τ . Testing supported products proves primality of $J \cup \{\tau\}$ exactly when J is prime.

For supported a , $D(a) = \{P_\tau\} \cup D_R(a)$, whereas $D(\tau)$ is empty. These formulas prove the topology assertions, and P_τ is contained in every prime. At P_τ , every supported element is inverted, including e . Its idempotence forces $e = 1$, and $er = e$ then forces every supported r to equal 1. The character χ factors through the localization and distinguishes 1 from τ , proving that the result is exactly $\mathbb{B}$. At $P_{\mathfrak{p}}$ the denominators are $R \setminus \mathfrak{p}$. The fraction map $r/s \mapsto r/s$, $\tau/s \mapsto \tau$ is an isomorphism: equality of supported fractions is precisely the usual localization relation; an absent fraction cannot equal a supported fraction since denominators and the extra multiplier remain supported. The list of ring primes of $\mathbb{Z}$ now proves (2) and excludes longer chains. $\square$

Proposition 1.3 (The direction of the absolute-point maps). *The maps in (1) induce*

$$\text{Spec } R \xrightarrow{\text{closed}} \text{Spec } G(R) \xleftarrow{\text{open}} \text{Spec } \mathbb{B}.$$

There is no unital semiring map $\mathbb{B} \rightarrow G(R)$. The pointed multiplicative monoid $\{0, 1\}$ does map to the multiplicative monoid of $G(R)$ by $0 \mapsto \tau$, $1 \mapsto 1$.

Proof. The first assertion is the localization and prime contraction already proved. A map from $\mathbb{B}$ would send $1 + 1 = 1$ to $1_R + 1_R = 1_R$, which contradicts $1_R \neq 0_R$. The two-element pointed monoid has no such additive relation, and its product table verifies the last map. Thus the two arrows and the pointed-monoid map retain their exact categories. $\square$

1.1 Every multiplicative prime, before imposing addition

Let M be the pointed multiplicative monoid underlying $G(\mathbb{Z})$ and let $\mathcal{P}$ be the rational primes. A monoid ideal is absorbing under multiplication; no additive closure is required.

Theorem 1.4 (The complete monoid spectrum). *For $\Sigma \subseteq \mathcal{P}$, put*

$$J_\Sigma = \{\tau, e\} \cup \{n \in \mathbb{Z} \setminus \{0\} : p \mid n \text{ for some } p \in \Sigma\}.$$

Then

$$\text{Spec}_{\text{mon}} M = \{\{\tau\}\} \sqcup \{J_\Sigma : \Sigma \subseteq \mathcal{P}\}, \quad J_\Sigma \subseteq J_{\Sigma'} \iff \Sigma \subseteq \Sigma'.$$

This spectrum has infinite dimension. The map

$$j : \text{Spec } G(\mathbb{Z}) \longrightarrow \text{Spec}_{\text{mon}} M \tag{3}$$

that keeps each prime ideal as a subset is a continuous topological embedding. Its image is exactly $\{\tau\}$, $J_\emptyset$, and the $J_{\{p\}}$.

Proof. The displayed sets are proper absorbing ideals. Products in J_Σ either involve τ , vanish in the integral domain $\mathbb{Z}$, or have a prime factor in Σ , so they satisfy primality. If a prime monoid ideal contains a supported element it contains e . It contains neither unit $1, -1$. Every nonzero element in it has an integer prime factor in it, by repeated primality and integer factorization. Conversely it contains every multiple of every such prime factor. This proves exhaustion, including $J_\emptyset = \{\tau, e\}$. Membership of a rational prime p tests whether $p \in \Sigma$, proving the inclusion statement. Arbitrarily long strictly nested finite subsets give arbitrarily long prime chains.

The scalar theorem identifies the image of j . For every $a \in M$, the inverse image under j of the monoid basic open $D_M(a)$ is the semiring basic open $D_{G(\mathbb{Z})}(a)$. These basics also generate the topology on the source, proving that the injection is an embedding. Finally if Σ contains distinct primes p, q , Bézout integers u, v give $up + vq = 1$. Both summands belong to J_Σ , so additive closure would force 1 into it. Such a monoid prime therefore cannot be a semiring prime. This also proves exactly which additive relation removes it. $\square$

Proposition 1.5 (Reconstruction retains the missing relations). *Let $\mathbb{N}_0[M]$ be the monoid semiring with the basis symbol $[\tau]$ identified with its additive zero. Then the quotient by*

$$[a] + [b] \sim [a + b] \quad (a, b \in \mathbb{Z})$$

is canonically $G(\mathbb{Z})$.

Proof. Evaluation $[a] \mapsto a$ respects all displayed relations. Every nonempty supported sum reduces to the single basis symbol of its integer sum, while the empty sum remains the symbol τ . Conversely $a \mapsto [a]$, $\tau \mapsto 0$ preserves multiplication, every supported addition, and addition with the specified zero. These maps are inverse. The contraction of $[\tau]$ is essential: in the uncontracted free semiring, the relations $[x] + [\tau] = [x]$ alone do not identify $[\tau]$ with the empty sum. Indeed they hold in the semiring obtained by adjoining another external zero below the already specified zero of $G(\mathbb{Z})$, where the two remain distinct. This gives an explicit separating model for the missing relation. $\square$

1.2 The complete lattice of zero-amplitude support

Let L be a nontrivial bounded distributive lattice and put

$$S_L = G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L.$$

Operations are $(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu)$ and $(r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu)$. Set $z_\lambda = (0, \lambda)$, $\tau_L = z_0$, $e_L = z_1$, and $Z_L = \{z_\lambda\}$.

Theorem 1.6 (All support and arithmetic primes). *The primes of S_L are exactly*

$$P_{\mathfrak{a}} = \{z_\lambda : \lambda \in \mathfrak{a}\}, \quad \mathfrak{a} \in \text{Spec}_{\text{lat}} L; \quad Q_{\mathfrak{p}} = Z_L \cup \{(r, 1) : r \in \mathfrak{p}\}, \quad \mathfrak{p} \in \text{Spec } R.$$

Within each family, inclusions retain the original prime order; every $P_{\mathfrak{a}}$ is strictly contained in every $Q_{\mathfrak{p}}$. Topologically

$$\text{Spec } S_L = \text{Spec}_{\text{lat}} L \blacktriangleright \text{Spec } R, \quad D(e_L) = \text{Spec}_{\text{lat}} L, \quad V(e_L) = \text{Spec } R, \quad (4)$$

where opens are opens of the first stratum and sets consisting of that entire stratum together with an open of the second. If both dimensions are finite, their dimensions add with one further step between the strata.

Proof. An ideal containing a supported element contains e_L by multiplication by it and therefore all of Z_L , since $e_L z_\lambda = z_\lambda$. Its supported elements form a ring ideal. Every remaining ideal is $\{z_\lambda : \lambda \in \mathfrak{a}\}$ for a proper lattice ideal: closure under join follows from addition, and downward closure follows from multiplication by z_μ for $\mu \leq \lambda$. Conversely these are all ideals. Primality in the first family is the meet-prime condition on $\mathfrak{a}$; in the second it is the ring-prime condition. Products involving a zero-fibre element obey exactly these tests. The inclusions follow from the displayed sets. Finally

$$D(z_\lambda) = \{P_{\mathfrak{a}} : \lambda \notin \mathfrak{a}\}, \quad D((r, 1)) = \text{Spec}_{\text{lat}} L \cup D_R(r)$$

prove the topology and the two strata. Every chain is a support chain followed by an arithmetic chain, and maximal finite chains concatenate, proving the dimension formula. $\square$

Proposition 1.7 (Two support channels and their exact comparison). *For $L = \mathcal{P}(\{1, 2\})$, $\{\tau_L\}$ is not prime, since $z_{\{1\}}z_{\{2\}} = \tau_L$ although neither factor is absent. There are two support primes*

$$P_1 = \{\tau_L, z_{\{2\}}\}, \quad P_2 = \{\tau_L, z_{\{1\}}\}.$$

For $R = \mathbb{Z}$, each specializes to the common $Q_{(0)}$ and then to every $Q_{(p)}$. Moreover

$$G_{\mathbb{B}^2}(R) \simeq G(R) \times_R G(R), \quad (5)$$

where the two maps to R are ring reflections. The two factor-generic primes contract to P_1, P_2 , while their arithmetic primes above the same $\mathfrak{p}$ both contract to $Q_{\mathfrak{p}}$.

Proof. The two coordinate maps $L \rightarrow \mathbb{B}$ have precisely the displayed prime kernels; a proper prime ideal must contain exactly one of the two atoms, proving exhaustion. For (5), send z_A to the pair with entry e on A and τ off A , and send $(r, \{1, 2\})$ to (r, r) . Equal nonzero amplitudes force both coordinates to be that supported value, whereas common amplitude zero allows exactly the four masks. Hence the map is onto the fibre product and injective; the operations verify it is an isomorphism. Taking inverse images of the factor primes proves their contractions. $\square$

Proposition 1.8 (What a Boolean branch retains). *A bounded lattice homomorphism $\epsilon : L \rightarrow \mathbb{B}$ induces $\beta_\epsilon : G_L(R) \rightarrow G_{\mathbb{B}}(R)$ by $z_\lambda \mapsto z_{\epsilon(\lambda)}$ and $(r, 1) \mapsto (r, 1)$. Its spectral image is*

$$\{P_{\epsilon^{-1}(0)}\} \blacktriangleright \text{Spec } R.$$

Proof. The map preserves joins, meets and supported arithmetic, hence is a semiring homomorphism. The inverse image of the scalar absent prime is $\{z_\lambda : \epsilon(\lambda) = 0\}$, and the inverse image of each arithmetic prime contains the full zero fibre and exactly its original amplitudes. This proves the asserted image. In particular a branch selects one support prime; the formula does not recover the other primes. $\square$

These proofs reconstruct the scalar and multiplicative results of [1, 2] and the lattice-support results of [3]. They give explicit maps whenever their spectra differ. The following section uses the multiplicative prime faces to recover the finite sets of places occurring in the arithmetic trace, retaining the support geometry of (4) for the cohomological coefficient construction.

2 Supported zero as the central arithmetic prime

The nonzero prime element e is part of the starting geometry, not a notation for the semiring zero. The assertion already appears explicitly in [1, `thm:zero-prime-not-irr`]. Here its ideal, height, localization, quotient, and power filtration are reconstructed on the full lattice-support base.

Theorem 2.1 (The supported-zero prime and the arithmetic closed stratum). *Let R be a nonzero integral domain and L a nontrivial bounded distributive lattice. In $S = G_L(R)$ the supported zero $e = z_1$ is a nonzero nonunit prime element. Precisely,*

$$eS = Z_L = \ker p_L =: E, \quad e \mid x \iff p_L(x) = 0, \quad e \mid xy \implies e \mid x \text{ or } e \mid y. \quad (6)$$

Its principal ideal E is prime, $E^2 = E$, and it is not irreducible because $e = e \cdot e$ with neither factor a unit. Every support prime lies strictly below E . The primes above E are exactly the arithmetic primes Q_p , and

$$V(E) \simeq \operatorname{Spec} R.$$

For finite-dimensional $\operatorname{Spec}_{\text{lat}} L$,

$$\operatorname{ht}(E) = \dim \operatorname{Spec}_{\text{lat}} L + 1.$$

For $R = \mathbb{Z}$ and finite Boolean $L = \mathbb{B}^I$ with $I \neq \emptyset$, E has height one and each $Q_{(p)}$ has height two.

Proof. For $z_\lambda \in Z_L$, $ez_\lambda = z_\lambda$, so every zero-amplitude element belongs to eS . Conversely a product with e has zero amplitude, proving the first equality and the divisibility equivalence. Since R is a domain, $p_L(x)p_L(y) = 0$ implies that one factor has zero amplitude; this proves the prime-element condition and primality of its ideal. The element e differs from $\tau = z_0$, and it is not a unit since every product with it has amplitude zero, whereas 1 has nonzero amplitude. Its idempotence proves both $E^2 = E$ and the failure of irreducibility.

The complete prime classification places every support prime below $Q_{(0)} = E$, and a domain has no arithmetic prime strictly below (0). Thus every chain ending at E consists of a support chain followed by this final step. Taking suprema of lengths proves the height formula. A finite Boolean lattice has only its coordinate prime ideals, which are pairwise incomparable: primality and disjointness of atoms force exactly one atom outside a prime ideal. Its lattice spectrum has dimension zero, proving the last assertions. $\square$

Theorem 2.2 (The stalk and residue at the supported-zero prime). *If $K = \operatorname{Frac} R$, then*

$$S_E \simeq G_L(K), \quad S_E/ES_E \simeq K. \quad (7)$$

The first map retains every support element z_λ ; the second is the ring reflection and uses the ideal's Bourne congruence. Inverting the element e , in contrast, gives

$$S[e^{-1}] \simeq L. \quad (8)$$

Proof. The complement of E consists exactly of supported nonzero amplitudes. Inverting them creates their original field fractions. For a zero-fibre element, multiplication by any such denominator fixes z_λ , so its fraction still equals the same z_λ . Different zero-fibre labels remain distinct: the support map $S \rightarrow L$ sends every denominator to 1 and therefore distinguishes them in the localization. A nonzero field fraction cannot become a zero-fibre element because the ring-reflection map to K distinguishes their amplitudes. These observations prove the first isomorphism, including all fraction identifications.

The Bourne relation for E identifies x, y when $x + i = y + j$ for some $i, j \in E$. It preserves amplitudes. Conversely if amplitudes agree, adding e to either element gives the same supported amplitude; hence the relation identifies exactly equal amplitudes. This proves the quotient assertion. For (8), invertibility of the idempotent e forces $e = 1$. Each supported r then satisfies $r = er = e = 1$. Every z_λ survives, and its inherited join and meet give L with unit e . The support map realizes this identification, proving its universal property. $\square$

Proposition 2.3 (The complete power filtration of the prime E). *For every $n \geq 1$, $E^n = E$. The ideal-congruence quotients and their inverse limit are*

$$S/E^n \simeq R, \quad \varprojlim_n S/E^n \simeq R. \quad (9)$$

Every transition map is the identity on R , and the completion map is p_L . The support-only ideal E is isomorphic to L with its own multiplicative identity e ; its inclusion in S is not unit preserving.

Proof. Idempotence $E^2 = E$ proves the power identity by induction. The Bourne-quotient argument in the preceding proof applies before localization and identifies S/E exactly with R . Thus all levels and transition maps are as stated. The last assertion follows from $z_\lambda z_\mu = z_{\lambda \wedge \mu}$, $z_\lambda + z_\mu = z_{\lambda \vee \mu}$, and $ez_\lambda = z_\lambda$. Since $e \neq 1_S$, its internal unit is not the unit of S . $\square$

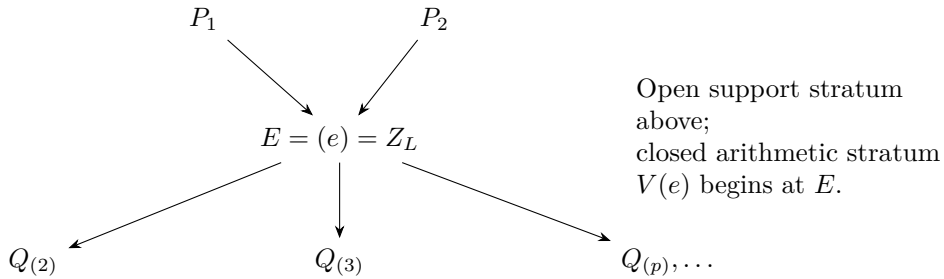


Figure 1: Exact specialization diagram for $G_{\mathbb{B}^2}(\mathbb{Z})$, with arrows from a smaller prime to a larger one. The two minimal support primes specialize to the supported-zero prime E ; every finite arithmetic prime lies one step below it in this drawing. Theorem 2.1 proves that their heights are 0, 1, 2. The two branches share the arithmetic stratum.

The coordinate E is simultaneously a nonzero prime ideal of S and the generic point of its closed arithmetic subspace. These facts describe its actual position; calling it the old zero prime alone loses its nonzero principal generator and its height in the new space. The arithmetic constructions transported through p_L live over $V(E)$. Extending their conclusions to the whole support spectrum requires the support diagram and its connecting maps, which the rebuilt cohomology retains.

3 The absolute-base maps and the role of every integer

This section specifies the actual arrows used by the proposed geometry over τ . It retains supported integer zero throughout. Monoid schemes and blueprints in the original source are attributed to

Deitmar and Lorscheid [11, 12]; the elementary maps needed here are proved below.

Proposition 3.1 (Integer coefficients and their vanishing loci). *Let $S = G(\mathbb{Z})$ and $X = \text{Spec } S$. The supported inclusion $j : \mathbb{Z} \hookrightarrow S$, $n \mapsto n$, preserves addition, multiplication and one. Its zero is $j(0) = e$. The ring reflection satisfies $pj = I_{\mathbb{Z}}$. The initial unital semiring map $\iota : \mathbb{N} \rightarrow S$ is instead*

$$\iota(0) = \tau, \quad \iota(n) = j(n) \quad (n > 0). \quad (\text{AZ1})$$

Writing $T = (\tau)$, $E = (e)$, and $P_p = \{\tau\} \cup p\mathbb{Z}$, one has

$$\iota^{-1}(T) = \iota^{-1}(E) = \{0\}, \quad \iota^{-1}(P_p) = p\mathbb{N}. \quad (\text{AZ2})$$

Every supported integer defines a function on the semiring spectrum and its exact vanishing set is

$$V_X(j(0)) = V_X(E) \simeq \text{Spec } \mathbb{Z}, \quad V_X(j(n)) = \{P_p : p \mid n\} \quad (n \neq 0). \quad (\text{AZ3})$$

For $n = \pm 1$ the latter set is empty. The zero integer has its own nonzero prime principal ideal E , at height one.

Proof. On supported elements the two operations are exactly integer operations, proving the claims about j and p . A unital map from $\mathbb{N}$ sends its zero to the target additive identity and sends a positive n to a sum of n copies of one. This proves (AZ1), including preservation of the two operations when an input is zero. Testing its values against the explicit prime sets proves (AZ2). The ideals T contain no supported integers, E contains exactly the supported integer zero, and P_p contains exactly the integers divisible by p together with τ . This proves every case of (AZ3). The height of E is the already proved foundational theorem. $\square$

Thus the two zero conventions have an exact relation: j keeps the integer zero as an element with support, and ι is the unital semiring structure map. Neither substitutes one zero for the other in a calculation.

Proposition 3.2 (Generic localization and the monoidal base). *The support morphism*

$$\chi : S \longrightarrow \mathbb{B}, \quad \chi(\tau) = 0, \quad \chi(j(n)) = 1 \quad (n \in \mathbb{Z}) \quad (\text{AZ4})$$

is localization at e . Its spectral arrow is the open immersion

$$\text{Spec } \mathbb{B} \longrightarrow X, \quad \text{image} = D_X(e) = \{T\}. \quad (\text{AZ5})$$

There is no unital semiring homomorphism $\mathbb{B} \rightarrow S$. There is a unital pointed-monoid map

$$\mathbb{F}_1^{\text{mon}} = \{0, 1\} \longrightarrow (S, \cdot), \quad 0 \longmapsto \tau, \quad 1 \longmapsto 1, \quad (\text{AZ6})$$

and hence a structure morphism from the entire multiplicative spectrum to the one-point monoidal base. Give $(S, \cdot)$ its actual additive relations: a formal sum is declared equivalent to another exactly when their sums are equal in S , and the empty sum equals τ . Then (AZ6) preserves the relations of the initial blueprint and gives a blueprint structure map. The subset of multiplicative primes closed under these additive relations is exactly $\text{Spec } S$; its induced map to the one-point base still includes every support and arithmetic prime.

Proof. Support of a sum is the join of its supports, even when supported amplitudes cancel to e . Support of a product is their meet. Hence χ is a unital semiring morphism. Inverting the idempotent e makes $e = 1$; then $ej(n) = e$ makes every supported integer one. The resulting two elements are still distinguished by χ . This proves localization and (AZ5). For a unital semiring map from $\mathbb{B}$, its relation $1 + 1 = 1$ would become $j(2) = j(1)$ in S , contradicting the injectivity of the supported inclusion. This proves the specific obstruction to a reverse semiring arrow.

For (AZ6), the multiplication tables agree, including absorption by zero, so the indicated map exists and is unique. The only proper prime of $\{0, 1\}$ is $\{0\}$, proving the base assertion. The initial blueprint has no additional nontrivial additive identifications; the empty-sum relation is respected because it is sent to τ . A multiplicative ideal closed under the declared additive relations is closed under addition, and is thus a semiring ideal. Conversely every semiring ideal has precisely that closure. The multiplicative prime condition is the same product test in both settings, proving the asserted spectral identification and the restriction of the structure map. $\square$

Equations (AZ4)–(AZ6) retain two different useful arrows with their directions proved: generic localization inside the full semiring spectrum, and the absolute monoidal/blueprint structure map of the whole object. The full-support cohomology diagram in this manuscript is attached to that whole object. It is not defined by replacing it with the stalk at T .

4 Exact localizations and contraction maps on the full support spectrum

22 September 2026. This derivation starts with the full bounded distributive support lattice. It does not select a Boolean branch before localizing. Its original provider is **The Clankers**, *Split Support Geometry*, version 11, `split_support_geometry_arithmetic_curve_v11.tex`: `def:lattice-split` and `thm:lattice-normal-form` at lines 1210–1321. The localization theorem `thm:lattice-localization` is at lines 1457–1487; the complete ideal/prime/ordinal-sum proofs at lines 1607–1771; and the branch construction and inverse-prime map at lines 1875–1987. Those sections were read, and the stalks, finite Boolean fibre product, and all contractions below are derived explicitly.

4.1 1. Full coefficient object and its primes

Let $R \neq 0$ be a commutative unital ring, and let L be a nontrivial bounded distributive lattice. Set

$$S = G_L(R) = \{z_\lambda = (0, \lambda) : \lambda \in L\} \cup \{\widehat{r} = (r, 1_L) : r \in R\} \subset R \times L. \quad (\text{TL1})$$

The two parts meet exactly at $e_L = z_{1_L} = \widehat{0}$. The semiring zero is $\tau_L = z_{0_L}$, and its one is $\widehat{1}$. The operations are

$$\begin{aligned} z_\lambda + z_\mu &= z_{\lambda \vee \mu}, & z_\lambda z_\mu &= z_{\lambda \wedge \mu}, & \widehat{r} + z_\lambda &= \widehat{r}, & \widehat{r} z_\lambda &= z_\lambda, \\ \widehat{r} + \widehat{s} &= \widehat{r + s}, & \widehat{r} \widehat{s} &= \widehat{rs}. \end{aligned} \quad (\text{TL2})$$

The formulas agree at the overlap. Distributivity follows from that of the ring and lattice. Define $Z_L = \{z_\lambda : \lambda \in L\}$.

A lattice ideal is lower, contains 0_L , and is closed under finite joins. It is prime if it is proper and $\lambda \wedge \mu$ in the ideal forces one factor to be in the ideal. For a prime lattice ideal $\mathfrak{a}$ and a ring prime $\mathfrak{p}$, put

$$P_{\mathfrak{a}} = \{z_\lambda : \lambda \in \mathfrak{a}\}, \quad Q_{\mathfrak{p}} = Z_L \cup \{\widehat{r} : r \in \mathfrak{p}\}. \quad (\text{TL3})$$

These are every prime of S . Here is a complete verification of the needed classification. An ideal containing a supported $\widehat{r}$ contains e_L : multiply by -1 and add when $r \neq 0$, while $r = 0$ already gives it. Multiplication of e_L by z_λ then places every z_λ in the ideal. Its supported amplitudes form a ring ideal, by TL2 and multiplication by $\widehat{-1}$. An ideal containing no supported element is contained in Z_L , and its support labels form a proper lattice ideal by TL2 and the identity $z_\mu z_\lambda = z_\mu$ for $\mu \leq \lambda$. Conversely these two constructions give ideals. For the second family, primality is precisely lattice primality; no product of supported elements belongs to a proper pure-support ideal. For the first family, primality reduces on supported factors to ring primality; every product with a zero-fibre factor is already in the amplitude ideal. This proves the assertion.

The complete inclusions are

$$P_{\mathfrak{a}} \subseteq P_{\mathfrak{b}} \iff \mathfrak{a} \subseteq \mathfrak{b}, \quad Q_{\mathfrak{p}} \subseteq Q_{\mathfrak{q}} \iff \mathfrak{p} \subseteq \mathfrak{q}, \quad P_{\mathfrak{a}} \subsetneq Q_{\mathfrak{p}} \text{ for every pair.} \quad (\text{TL4})$$

Indeed the pure-support ideals omit e_L , while every amplitude ideal contains it. The basic opens are

$$D(z_\lambda) = \{P_{\mathfrak{a}} : \lambda \notin \mathfrak{a}\}, \quad D(\widehat{r}) = \text{SpecLat } L \sqcup D_R(r). \quad (\text{TL5})$$

These formulas retain the whole support-prime space as an open stratum and the whole arithmetic spectrum as a closed stratum. They also prove the stated spectral ordinal-sum topology directly.

4.2 2. The exact lattice localization at a prime

Fix a prime lattice ideal $\mathfrak{a}$, and let $F = L \setminus \mathfrak{a}$. This is an upward-closed meet filter containing 1_L and excluding 0_L . Define

$$\lambda \sim_F \mu \iff \exists f \in F : f \wedge \lambda = f \wedge \mu. \quad (\text{TL6})$$

This is a bounded-lattice congruence. Reflexivity uses $f = 1_L$, and symmetry is immediate. If witnesses f, g give two consecutive equalities, their meet $f \wedge g \in F$ witnesses transitivity. For congruence under join and meet, replace two witnesses by their meet. Distributivity then gives equality after meeting either join with that common witness; associativity gives the corresponding equality for meets. Therefore the quotient

$$L_{\mathfrak{a}} = L / \sim_F, \quad \ell_{\mathfrak{a}}(\lambda) = [\lambda] \quad (\text{TL7})$$

is a bounded distributive lattice, regarded as a semiring with join and meet. Its bottom and top are distinct: equality $[0_L] = [1_L]$ would force some $f \in F$ to equal zero.

The map $\ell_{\mathfrak{a}}$ is the semiring localization inverting F . Every $f \in F$ becomes one because $f \wedge f = f \wedge 1_L$. Conversely, let $u : L \rightarrow T$ be a semiring map to any semiring, inverting every $f \in F$. Each $u(f)$ is multiplicatively idempotent and invertible, hence equals one. Equality in TL6 therefore implies $u(\lambda) = u(\mu)$, so u factors uniquely through TL7. This proves the full universal property, not only a description of representatives.

The top class has the exact preimage

$$[\lambda] = 1_{L_{\mathfrak{a}}} \iff \lambda \in F. \quad (\text{TL8})$$

For the nontrivial direction, $f \wedge \lambda = f$ implies $f \leq \lambda$, and upward closure gives $\lambda \in F$. For the reverse direction use witness $f = \lambda$. In any nontrivial lattice semiring the only unit is the top: $x \wedge y = 1$ forces $x = y = 1$. Hence $L_{\mathfrak{a}}$ is local, with unique maximal ideal

$$\mathfrak{m}_{\mathfrak{a}} = \{[\lambda] : \lambda \in \mathfrak{a}\} = L_{\mathfrak{a}} \setminus \{1\}. \quad (\text{TL9})$$

It is an ideal because $\mathfrak{a}$ is closed under joins and downward containment, and all proper ideals must omit the unit. The branch character

$$\epsilon_{\mathfrak{a}}(\lambda) = \begin{cases} 0, & \lambda \in \mathfrak{a}, \\ 1, & \lambda \notin \mathfrak{a} \end{cases} \quad (\text{TL10})$$

is a bounded lattice homomorphism: join lies in $\mathfrak{a}$ exactly when both factors do, while meet lies there exactly when at least one factor does. It factors through $L_{\mathfrak{a}}$, and its zero fibre is TL9. Quotienting the local lattice by that maximal ideal gives $\mathbb{B}$. Explicitly every nonunit is identified with zero by the ideal quotient, while one cannot be identified with zero because an equality $1 \vee i = j$, with $i, j \in \mathfrak{m}_{\mathfrak{a}}$, would force $j = 1$. Thus the Boolean object is the local residue of this support stalk; the entire stalk need not be Boolean.

4.3 3. Pure-support prime stalks, including all surviving levels

The complement $S \setminus P_a$ contains every supported $\widehat{r}$, including e_L , and the elements z_f for $f \in F$. Inverting e_L first gives L by the exact support map

$$\chi_L : S \longrightarrow L, \quad \widehat{r} \mapsto 1_L, \quad z_\lambda \mapsto \lambda. \quad (\text{TL11})$$

Indeed an invertible idempotent image of e_L equals one; then $e_L \widehat{r} = e_L$ forces every supported amplitude to map to one. The images of the z_λ preserve join, meet, and both endpoints. This proves the universal property of TL11 as localization at e_L .

After TL11, the remaining denominators are precisely F . Combining its universal property with TL7 proves

$$\boxed{S_{P_a} \cong L_a, \quad \widehat{r} \mapsto 1, \quad z_\lambda \mapsto [\lambda].} \quad (\text{TL12})$$

Every denominator maps to one. The numerator fraction has the concrete value

$$\widehat{r}/t \mapsto 1, \quad z_\lambda/t \mapsto [\lambda] \quad (t \notin P_a),$$

because every inverted denominator is idempotent after TL11 and therefore equal to one. Conversely any map inverting the original complement must factor first through TL11 and then through TL7, which proves that no additional support identifications have been omitted.

When L is finite, define

$$f_a = \bigwedge_{f \in F} f \in F. \quad (\text{TL13})$$

Then TL6 is equivalent to $f_a \wedge \lambda = f_a \wedge \mu$. One implication follows by meeting a witnessing equality with $f_a \leq f$, and the reverse uses f_a itself as witness. Therefore

$$\boxed{L_a \xrightarrow{\sim} \downarrow f_a, \quad [\lambda] \mapsto f_a \wedge \lambda.} \quad (\text{TL14})$$

This map is onto because any $\nu \leq f_a$ is its own image. Its bottom is 0_L , its unit is f_a , and its operations are the original join and meet restricted to that principal interval.

For the complete chain $C_n = \{0 < 1 < \dots < n\}$, every prime ideal has the form $\mathfrak{a}_j = \{0, \dots, j-1\}$, $1 \leq j \leq n$. Lowerness forces every proper ideal to be such an initial segment, and primality follows because a minimum lies below j exactly when one input does. Its complement has least element j , so

$$\boxed{G_{C_n}(R)_{P_{\mathfrak{a}_j}} \cong C_j, \quad z_\lambda \mapsto \min(\lambda, j), \quad \widehat{r} \mapsto j.} \quad (\text{TL15})$$

For $j > 1$, this stalk retains the $j-1$ distinct support elements strictly between absence and its unit. The residue map to $\mathbb{B}$ sends all of them to zero and sends j to one. This is an explicit counterexample to replacing every pure-support prime stalk by $\mathbb{B}$.

Every ring-valued realization of L_a is the zero ring, because $1 \vee 1 = 1$ forces $1 + 1 = 1$ in the target. This arithmetic reflection statement does not identify the distinct support elements of the stalk itself.

4.4 4. Amplitude prime stalks retain the entire lattice

Fix $\mathfrak{p} \in \text{Spec } R$. The complement of $Q_{\mathfrak{p}}$ consists exactly of supported $\widehat{s}$ with $s \notin \mathfrak{p}$. In particular it contains no element of Z_L . The exact localization is

$$\boxed{S_{Q_{\mathfrak{p}}} \cong G_L(R_{\mathfrak{p}}), \quad \widehat{r}/\widehat{s} \mapsto \widehat{r/s}, \quad z_\lambda/\widehat{s} \mapsto z_\lambda.} \quad (\text{TL16})$$

The second formula follows from $z_\lambda \widehat{s} = z_\lambda$, after $\widehat{s}$ becomes invertible. For the first, if $r/s = r'/s'$, there is $u \notin \mathfrak{p}$ with $us'r = usr'$. Any semiring map inverting the denominators gives equal images

of these fractions after cancellation of the invertible factors. Thus the formulas are well-defined. Their compatibility with TL2 proves the mixed addition and multiplication cases. Conversely a map from S that inverts $R \setminus \mathfrak{p}$ extends by those same formulas, uniquely, so TL16 has the required universal property.

No two distinct support labels become equal: in the fractional equivalence relation, multiplication of z_λ and z_μ by any supported denominator leaves both unchanged, so equality forces $\lambda = \mu$. A supported fraction can coincide with a zero-fibre element only at the top label 1_L , when its ring fraction equals zero. This retains precisely the original overlap in TL1.

The unit group of $G_L(A)$, for any nonzero ring A , is $A^\times$: a product equal to $\widehat{1}$ must have nonzero amplitude and hence both factors are supported, where invertibility is ordinary ring invertibility. Thus TL16 is local with maximal ideal

$$Z_L \cup \{\widehat{a} : a \in \mathfrak{p}R_{\mathfrak{p}}\}. \quad (\text{TL17})$$

The support-preserving residue and its ordinary ring shadow are

$$G_L(R_{\mathfrak{p}}) \longrightarrow G_L(\kappa(\mathfrak{p})) \xrightarrow{p_L} \kappa(\mathfrak{p}), \quad \kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}. \quad (\text{TL18})$$

The first map keeps every z_λ and reduces the amplitude; the second kills every z_λ . The composite is the ideal quotient by TL17. Indeed killing the ideal kills e_L , and $e_L + z_\lambda = e_L$ then kills every support element; the surviving supported ring quotient is exactly $\kappa(\mathfrak{p})$. Conversely TL18 has that exact zero fibre. Thus ordinary residue-field norms remain available without discarding the full support residue $G_L(\kappa(\mathfrak{p}))$.

4.5 5. Every generalization and localization contraction

For lattice primes $\mathfrak{a} \subseteq \mathfrak{b}$, the filters satisfy $F_{\mathfrak{b}} \subseteq F_{\mathfrak{a}}$, so the exact generalization map is

$$S_{P_{\mathfrak{b}}} = L_{\mathfrak{b}} \longrightarrow L_{\mathfrak{a}} = S_{P_{\mathfrak{a}}}, \quad [\lambda]_{\mathfrak{b}} \mapsto [\lambda]_{\mathfrak{a}}. \quad (\text{TL19})$$

It is well-defined because every equality witness from the smaller filter remains a witness in the larger filter. If L is finite, $f_{\mathfrak{a}} \leq f_{\mathfrak{b}}$, and in TL14 coordinates this is the explicit map

$$\downarrow f_{\mathfrak{b}} \longrightarrow \downarrow f_{\mathfrak{a}}, \quad \lambda \mapsto \lambda \wedge f_{\mathfrak{a}}. \quad (\text{TL20})$$

For ring primes $\mathfrak{q} \subseteq \mathfrak{p}$, the generalization map is $G_L(R_{\mathfrak{p}}) \rightarrow G_L(R_{\mathfrak{q}})$, obtained by the ring fraction map and the identity on all support labels. For every pure-support/amplitude pair, TL4 gives the generalization

$$S_{Q_{\mathfrak{p}}} = G_L(R_{\mathfrak{p}}) \xrightarrow{\chi_L} L \xrightarrow{\ell_{\mathfrak{a}}} L_{\mathfrak{a}} = S_{P_{\mathfrak{a}}}. \quad (\text{TL21})$$

It sends $\widehat{r/s}$ to one and z_λ to $[\lambda]$. It inverts every element in the larger complement, so the universal property proves it is exactly the stalk map. These maps compose according to the inclusions in TL4, as is also immediate from their formulas.

The prime contractions of TL12 are exactly

$$\text{Spec } L_{\mathfrak{a}} \longleftrightarrow \{\mathfrak{b} \in \text{SpecLat } L : \mathfrak{b} \subseteq \mathfrak{a}\}, \quad \mathfrak{b} \longmapsto \ell_{\mathfrak{a}}(\mathfrak{b}), \quad (\text{TL22})$$

and the corresponding contraction to S is $P_{\mathfrak{b}}$. For a direct proof, if $\mathfrak{b} \subseteq \mathfrak{a}$, it is disjoint from $F_{\mathfrak{a}}$. If $\lambda \sim_F \mu$ and $\lambda \in \mathfrak{b}$, then a witnessing $f \notin \mathfrak{b}$ satisfies $f \wedge \mu = f \wedge \lambda \in \mathfrak{b}$. Primality gives $\mu \in \mathfrak{b}$. Thus membership in $\mathfrak{b}$ is well-defined on equivalence classes, and its image is prime. Conversely the inverse image of any proper prime in the quotient is prime in L , omits every $f \in F$ because they map to one, and is therefore contained in $\mathfrak{a}$. The constructions are inverse by surjectivity of $\ell_{\mathfrak{a}}$.

The prime contractions of TL16 are exactly all the pure-support primes and the amplitude primes below $Q_{\mathfrak{p}}$:

$$\begin{aligned} P_{\mathfrak{b}} \subset G_L(R_{\mathfrak{p}}) &\longmapsto P_{\mathfrak{b}} \subset S, \\ Q_{\mathfrak{q}R_{\mathfrak{p}}} &\longmapsto Q_{\mathfrak{q}}, \quad \mathfrak{q} \in \operatorname{Spec} R, \quad \mathfrak{q} \subseteq \mathfrak{p}. \end{aligned} \quad (\text{TL23})$$

The support contraction follows because the labels are unchanged and no supported element belongs to a pure-support ideal. For the ring part, a prime ideal $\mathfrak{q} \subseteq \mathfrak{p}$ has extension consisting of fractions with numerator in $\mathfrak{q}$. Membership is well-defined because a denominator outside $\mathfrak{p}$ lies outside $\mathfrak{q}$, and primality cancels it from the fractional equality. That extension is prime and contracts to $\mathfrak{q}$. Conversely a prime of $R_{\mathfrak{p}}$ contracts to a ring prime disjoint from $R \setminus \mathfrak{p}$, and every fraction is a numerator times an invertible denominator, proving extension and contraction inverse. Combined with TL3, this proves that TL23 lists every contraction.

4.6 6. Scalar inclusion, branch maps, and their local forms

The scalar inclusion is

$$\iota : G(R) = G_{\mathbb{B}}(R) \hookrightarrow G_L(R), \quad \tau \mapsto z_{0_L}, \quad r \mapsto \widehat{r}. \quad (\text{TL24})$$

For a lattice prime $\mathfrak{a}$, use its branch TL10 to define $\beta_{\mathfrak{a}} : G_L(R) \rightarrow G(R)$: supported amplitudes remain unchanged, while z_{λ} maps to τ for $\lambda \in \mathfrak{a}$ and to e otherwise. TL2 and the lattice homomorphism property prove it is a zero- and one-preserving homomorphism, and $\beta_{\mathfrak{a}}\iota = \text{id}$. Its complete prime inverse images and the inclusion contractions are

$$\begin{aligned} \beta_{\mathfrak{a}}^{-1}(P_{\tau}) &= P_{\mathfrak{a}}, \quad \beta_{\mathfrak{a}}^{-1}(P_{\mathfrak{p}}) = Q_{\mathfrak{p}}, \\ \iota^{-1}(P_{\mathfrak{b}}) &= P_{\tau} \text{ for every } \mathfrak{b}, \quad \iota^{-1}(Q_{\mathfrak{p}}) = P_{\mathfrak{p}}. \end{aligned} \quad (\text{TL25})$$

These equalities follow by evaluating the maps on supported elements, bottom, and top. Every pure-support prime omits the top and contains the bottom. At the chosen pure-support point the induced local map is

$$L_{\mathfrak{a}} \longrightarrow \mathbb{B}, \quad [\lambda] \mapsto \epsilon_{\mathfrak{a}}(\lambda). \quad (\text{TL26})$$

It is exactly the residue map of TL9–10, and can have a nontrivial kernel containing several distinct support levels. At an amplitude point it is $G_L(R_{\mathfrak{p}}) \rightarrow G(R_{\mathfrak{p}})$ with the same chosen branch. Thus the scalar stalk computation is recovered through a displayed local morphism; it is not substituted for the entire full-support stalk.

4.7 7. Finite Boolean support as an exact fibre product

Let I be a finite nonempty set and $L = 2^I$, with union and intersection. For every $i \in I$ let $\pi_R : G(R) \rightarrow R$ be the ring reflection. Form the fibre product in the category of unital zero-preserving semirings,

$$\mathcal{F}_I(R) = \prod_{i \in I, R} G(R) = \{(x_i) \in G(R)^I : \pi_R(x_i) = \pi_R(x_j) \text{ for all } i, j\}. \quad (\text{TL27})$$

It is a subsemiring because equal amplitudes remain equal under coordinatewise operations. There is an exact isomorphism

$$\Phi : G_{2^I}(R) \xrightarrow{\sim} \mathcal{F}_I(R), \quad z_A \mapsto (x_i), \quad x_i = \begin{cases} e, & i \in A, \\ \tau, & i \notin A, \end{cases} \quad \widehat{r} \mapsto (r)_{i \in I}. \quad (\text{TL28})$$

To prove surjectivity, let r be the common amplitude of a fibre-product tuple. If $r \neq 0$, each coordinate must be the supported r , since the reflection fibre of a nonzero amplitude is a singleton.

If $r = 0$, every coordinate is either τ or e , so it determines a unique subset A . These cases also prove injectivity, including the overlap $r = 0, A = I$. Coordinatewise union/intersection on zero amplitudes and the supported rules TL2 prove preservation of every operation and both identities.

The coordinate maps of TL28 are exactly the branch maps β_i for $\epsilon_i(A) = \mathbf{1}_{i \in A}$. Its scalar inclusion TL24 becomes the diagonal $G(R) \rightarrow G(R)^I$. Thus one has the actual diagram of semiring maps

$$G(R) \xrightarrow{\iota} G_{2I}(R) \xrightarrow{\Phi} \mathcal{F}_I(R) \hookrightarrow G(R)^I \xrightarrow{\text{pr}_i} G(R), \quad \text{pr}_i \Phi \iota = \text{id}. \quad (\text{TL29})$$

This is the amplitude fibre product. The different synchronized-support identity $G(R^I) = \prod_{i \in I, \mathbb{B}} G(R)$ is verified by the same coordinate reasoning: equal support means either all coordinates absent, or all present with independent amplitudes. The independent product $G(R)^I$ contains both subobjects. In TL28 amplitudes are equal and the whole zero-support cube is retained; in $G(R^I)$ support is equal and amplitudes are independent. These exact fibre-product maps state their relation without identifying them.

Every lattice prime of 2^I is

$$\mathfrak{a}_i = \{A \subseteq I : i \notin A\}, \quad i \in I. \quad (\text{TL30})$$

Indeed the top is the finite join of singleton atoms, so a prime ideal must omit at least one atom. It cannot omit two distinct atoms, because their meet is bottom and primality would fail. If it omits $\{i\}$, every subset not containing i is a join of included atoms and lies in the ideal, while no subset containing i can lie in it by lowerness. This proves TL30. Its complement has least element $\{i\}$, so every pure-support stalk in this finite Boolean case is $\downarrow \{i\} \cong \mathbb{B}$. This conclusion depends on the Boolean atoms and does not apply to TL15's longer chains.

4.8 8. Complete contractions from the independent finite product

Every prime of the independent product $G(R)^I$ is $\text{pr}_i^{-1}(\mathfrak{P})$ for exactly one $i \in I$ and $\mathfrak{P} \in \text{Spec } G(R)$. To prove this, let d_i have supported one at i and absent zero elsewhere. Then $\sum_i d_i = 1$ and $d_i d_j = 0$ for $i \neq j$. A proper prime must omit some d_i and can omit at most one. Every x decomposes as $\sum_j d_j x$; all terms except the omitted coordinate lie in the prime. Hence x lies in the prime exactly when $d_i x$ does: one direction uses multiplication by d_i , and the other uses the displayed finite sum. The values at that coordinate form a proper prime of $G(R)$, proving existence and uniqueness of the claimed description.

Contract these primes along the embedding in TL29. For all $i \in I$ and ring primes $\mathfrak{p}$, the complete formulas are

$$\boxed{\Phi^{-1}(\text{pr}_i^{-1}(P_\tau)) = P_{\mathfrak{a}_i}, \quad \Phi^{-1}(\text{pr}_i^{-1}(P_{\mathfrak{p}})) = Q_{\mathfrak{p}}.} \quad (\text{TL31})$$

For the first, a zero-mask coordinate equals τ precisely when $i \notin A$, and no supported diagonal amplitude equals τ . For the second, both τ and supported e belong to $P_{\mathfrak{p}}$, so every zero mask belongs to the contraction, while $\widehat{r}$ belongs precisely when $r \in \mathfrak{p}$. TL3 and TL30 show that these formulas exhaust every prime of $G_{2I}(R)$.

Thus the spectral map from the independent product onto the amplitude fibre product has singleton fibres over each pure-support prime $P_{\mathfrak{a}_i}$, and exactly $|I|$ points over each amplitude prime $Q_{\mathfrak{p}}$, one from each coordinate. The map to the scalar subobject then contracts all $P_{\mathfrak{a}_i}$ to P_τ while keeping the ring-prime labels, as in TL25. These are contractions of prime ideals under the actual displayed homomorphisms, with every support branch present.

At a product prime $\text{pr}_i^{-1}(\mathfrak{P})$, localization inverts d_i , whose principal factor kills every other coordinate. Its stalk is therefore $G(R)_{\mathfrak{P}}$. The local maps induced by TL31 are the identity $\mathbb{B} \rightarrow \mathbb{B}$ at $P_{\mathfrak{a}_i}$, and

$$G_{2I}(R_{\mathfrak{p}}) \xrightarrow{\beta_i} G(R_{\mathfrak{p}}) \quad (\text{TL32})$$

at an amplitude prime. In the latter map the full zero-support cube is mapped through its specified i -th branch; the full source stalk remains TL16. This completes the local and spectral contraction diagram without prematurely replacing the full coefficient object by a branch.

The classical ideal arithmetic is reproved in the following chapter; its human-source context is Neukirch [10], retained from the foundational bibliography.

5 The full ideal arithmetic, divisor coordinates, and norms

Let K be a number field, let $A = \mathcal{O}_K$, and retain

$$S = G(A) = A \sqcup \{\tau\}, \quad e = 0_A, \quad T = \{\tau\}, \quad E = \{\tau, e\}.$$

Thus the zero ideal of the semiring is T , whereas $E = eS$ is a nonzero prime ideal. The supported-zero prime theorem is prior work: [1, `thm:zero-prime-not-irr`] states it explicitly. The programme's originator attributes its original derivation to Gemini 2.5 Pro. The present section calculates the complete fractional ideal structure and the exact class quotients using that theorem. Ideal products mean finite sums of elementwise products; an empty sum is τ . The integral ideal classification and Dedekind zeta identity are also retained prior results of [1, `thm:OK-ideals-spectrum`, `thm:Dedekind-zeta`]. The arguments below give their full fractional-ideal setting and the comparison for the exact displayed class relation.

5.1 Cancellation and the localization retaining both zeros

Proposition 5.1 (The exact regular elements). *Multiplication $x \mapsto ax$ on S is injective exactly for $a \in A \setminus \{0_A\}$. The units of S are $A^\times$. The only element whose addition map is injective is τ . Nevertheless every $a \neq \tau$, including e , satisfies*

$$ax = \tau \implies x = \tau.$$

Thus absence of a nontrivial product equal to τ is a weaker condition here than multiplicative cancellation.

Proof. For supported nonzero a , a supported product cannot equal an absent product, and supported equality cancels in the domain A . Multiplication by e identifies 1 and e ; multiplication by τ is constant. A product is 1 exactly when its factors are inverse elements of A , proving the unit assertion. For supported a , $a + \tau = a = a + e$, whereas addition by τ is the identity. Finally a product equals τ exactly when one factor equals τ . $\square$

Call injective multiplication scalars *regular*. Their localization is

$$Q_{\text{reg}}(S) = (A \setminus \{0_A\})^{-1}S = G(K). \quad (10)$$

Indeed τ/d maps to τ and supported a/d to the corresponding field fraction. The localization relation identifies supported fractions precisely by equality in K , and cannot identify one with τ . The units of $G(K)$ are $K^\times$. Localization at all of $S \setminus \{\tau\} = A$ instead inverts e and yields $\mathbb{B}$, as in [1, `thm:frac-collapse`]. Every fractional ideal below uses exactly (10).

5.2 The number-field ideal calculation used in the comparison

Lemma 5.2 (Integral lattice and local valuation rings). *The ring A is free abelian of rank $n = [K : \mathbb{Q}]$, is Noetherian and integrally closed, and every nonzero prime $\mathfrak{p}$ is maximal with finite residue field. For each such $\mathfrak{p}$, $A_{\mathfrak{p}}$ is a discrete valuation ring. Its valuation $v_{\mathfrak{p}} : K^\times \rightarrow \mathbb{Z}$ has $v_{\mathfrak{p}}(\pi_{\mathfrak{p}}) = 1$ for a generator of its maximal ideal.*

Proof. Choose a $\mathbb{Q}$ -basis $\alpha_1, \dots, \alpha_n$ of K of algebraic integers by scaling any basis by suitable nonzero integers. The trace matrix $B = (\text{Tr}_{K/\mathbb{Q}}(\alpha_i \alpha_j))$ has integer entries and nonzero determinant. To check nondegeneracy, choose a primitive element θ of the separable extension. The matrix of its n distinct embeddings on $1, \theta, \dots, \theta^{n-1}$ is an invertible Vandermonde matrix; the trace matrix is its transpose times itself. Change of basis preserves nondegeneracy. For $x \in A$, each $\text{Tr}(x \alpha_j)$ is integral, being rational and a sum of algebraic integers. Thus if $x = \sum_i c_i \alpha_i$, then $c_i \in (\det B)^{-1} \mathbb{Z}$. Consequently

$$\sum_i \mathbb{Z} \alpha_i \subseteq A \subseteq (\det B)^{-1} \sum_i \mathbb{Z} \alpha_i.$$

It follows that A is free of rank n . Each ideal is a subgroup of this lattice, hence has finitely many abelian generators, which also generate it as an A -module. This proves Noetherianity.

If $x \in K$ is integral over A , adjoin to $\mathbb{Z}$ the finitely many coefficients of its monic equation and then x . The resulting ring is a finitely generated $\mathbb{Z}$ -module: powers of each successive integral generator reduce using its monic equation. Multiplication by x , applied to a finite set of module generators, gives a matrix; the determinant of $xI - C$ annihilates that module and hence its element 1. This is a monic integer equation for x , proving $x \in A$. For $0 \neq a \in A$, the nonzero constant term of its monic minimal polynomial belongs to aA . Every nonzero ideal therefore contains a nonzero integer d , and its quotient is a quotient of the finite group A/dA . A finite domain is a field. Hence every nonzero prime is maximal with finite residue field.

Fix $\mathfrak{p} \neq 0$, and write $B_{\mathfrak{p}} = A_{\mathfrak{p}}$ and $\mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}$. This is a Noetherian local domain with only primes $0, \mathfrak{m}$. It is integrally closed: given an integral equation for x over $A_{\mathfrak{p}}$, choose $s \notin \mathfrak{p}$ clearing every coefficient. Substituting $y = sx$ gives a monic equation over A , since its j -th coefficient is s^j times the old one. Thus $sx \in A$ and $x \in A_{\mathfrak{p}}$.

For $0 \neq t \in \mathfrak{m}$, $\sqrt{tB_{\mathfrak{p}}} = \mathfrak{m}$. Finite generation implies $\mathfrak{m}^r \subseteq tB_{\mathfrak{p}}$ for some r : a power of each generator lies in $tB_{\mathfrak{p}}$, and a sufficiently long product includes one of those powers. Choose the least r . If $r = 1$, then $\mathfrak{m} = tB_{\mathfrak{p}}$. Otherwise choose $y \in \mathfrak{m}^{r-1} \setminus tB_{\mathfrak{p}}$ and set $z = y/t$. Then $z \notin B_{\mathfrak{p}}$ and $z\mathfrak{m} \subseteq B_{\mathfrak{p}}$. If $z\mathfrak{m} \subseteq \mathfrak{m}$, express multiplication by z on finitely many generators of the nonzero ideal $\mathfrak{m}$. The determinant of $zI - C$ annihilates them. Cancellation of a nonzero generator in the field makes this determinant zero, forcing z integral over $B_{\mathfrak{p}}$, a contradiction. Thus the ideal $z\mathfrak{m}$ contains a unit and equals $B_{\mathfrak{p}}$. Some $\pi \in \mathfrak{m}$ satisfies $z\pi = 1$, and $u = \pi(zu)$ for every $u \in \mathfrak{m}$; therefore $\mathfrak{m} = \pi B_{\mathfrak{p}}$.

A nonzero $b \in B_{\mathfrak{p}}$ cannot lie in every $\pi^j B_{\mathfrak{p}}$. Otherwise the ascending sequence (b/π^j) of ideals stabilizes, forcing π to be a unit after cancellation of b . Thus $b = \pi^j u$ for a unique $j \geq 0$ and a unit u . Taking the least exponent occurring in a nonzero ideal proves that the ideal is a power of $\mathfrak{m}$. Extending exponents to fractions gives the claimed discrete valuation. $\square$

Lemma 5.3 (Fractional factorization and norm). *Let $\mathcal{I}(A)$ be the nonzero fractional ideals in K . Every such ideal is invertible and has a unique expression*

$$J = \prod_{\mathfrak{p} \neq 0} \mathfrak{p}^{v_{\mathfrak{p}}(J)}, \quad v_{\mathfrak{p}}(J) \in \mathbb{Z}, \quad (11)$$

with finite support. Its ideal operations satisfy

$$\begin{aligned} J \subseteq H &\iff v_{\mathfrak{p}}(J) \geq v_{\mathfrak{p}}(H) \quad \text{for every } \mathfrak{p}, \\ v_{\mathfrak{p}}(J + H) &= \min(v_{\mathfrak{p}}(J), v_{\mathfrak{p}}(H)), \\ v_{\mathfrak{p}}(J \cap H) &= \max(v_{\mathfrak{p}}(J), v_{\mathfrak{p}}(H)), \\ v_{\mathfrak{p}}(JH) &= v_{\mathfrak{p}}(J) + v_{\mathfrak{p}}(H). \end{aligned}$$

For integral $J \neq 0$,

$$|A/J| = \prod_{\mathfrak{p}} |A/\mathfrak{p}|^{v_{\mathfrak{p}}(J)}. \quad (12)$$

Proof. Every bounded A -submodule of K is finitely generated, since a common denominator embeds it in the Noetherian module A . Localizing J gives a power of the maximal ideal in each discrete valuation ring. For integral J , its exponents are nonnegative and vanish outside the finitely many primes containing J ; there are finitely many because A/J is finite. Clearing a denominator proves finite support for fractional J .

Fractional ideals are determined by these localizations. If $x \notin J$, the proper ideal $(J : x) = \{a \in A : ax \in J\}$ lies in a maximal $\mathfrak{p}$. Membership $x \in J_{\mathfrak{p}}$ would give an $s \notin \mathfrak{p}$ with $sx \in J$, a contradiction. Define $\mathfrak{p}^{-1} = \{x \in K : x\mathfrak{p} \subseteq A\}$. For $0 \neq a \in \mathfrak{p}$, this module lies in $a^{-1}A$, so it is fractional. At any maximal $\mathfrak{q}$, its localization equals the inverse of $\mathfrak{p}A_{\mathfrak{q}}$: to prove the reverse inclusion, clear denominators outside $\mathfrak{q}$ for finitely many generators of $\mathfrak{p}$. At $\mathfrak{q} \neq \mathfrak{p}$, $\mathfrak{p}A_{\mathfrak{q}} = A_{\mathfrak{q}}$; at $\mathfrak{p}$ it is the principal maximal ideal. Thus $(\mathfrak{p}\mathfrak{p}^{-1})_{\mathfrak{q}} = A_{\mathfrak{q}}$ for every $\mathfrak{q}$. The equality test proves $\mathfrak{p}\mathfrak{p}^{-1} = A$. The product in (11) now has the same localizations as J , which proves existence and uniqueness. The same local test with powers in each valuation ring proves the formulas for inclusion, sum, intersection, and product.

Put $q_{\mathfrak{p}} = |A/\mathfrak{p}|$. Each module $\mathfrak{p}^j/\mathfrak{p}^{j+1}$ is killed by $\mathfrak{p}$, so localizing it at $\mathfrak{p}$ changes nothing: every scalar outside $\mathfrak{p}$ already acts invertibly. Its localization is $\pi^j A_{\mathfrak{p}}/\pi^{j+1} A_{\mathfrak{p}}$, one-dimensional over $A/\mathfrak{p}$. Hence $|A/\mathfrak{p}^r| = q_{\mathfrak{p}}^r$. Powers of distinct maximal ideals are comaximal, because a maximal ideal containing their sum would contain both radicals. For comaximal ideals, writing $1 = u + v$ with u, v in the two ideals proves the surjectivity of the Chinese remainder map and that intersection equals product. Induction gives

$$A/\prod_{\mathfrak{p}} \mathfrak{p}^{r_{\mathfrak{p}}} \simeq \prod_{\mathfrak{p}} A/\mathfrak{p}^{r_{\mathfrak{p}}}.$$

Cardinalities prove (12). □

5.3 Every fractional ideal, with both zero ideals retained

Definition 5.4. A fractional S -ideal in $G(K)$ is an S -subsemimodule $F \subseteq G(K)$ with $dF \subseteq S$ for some $d \in A \setminus \{0_A\}$. It contains τ and is closed under addition and multiplication by S . No condition removes E or T . Let $\mathcal{F}(S)$ be all such ideals and put $I_J = J \cup \{\tau\}$ for $J \in \mathcal{I}(A)$. A fractional ideal is regular when it contains a regular element.

Theorem 5.5 (The entire fractional ideal structure). *Every fractional S -ideal is finitely generated, and*

$$\mathcal{F}(S) = \{T, E\} \sqcup \{I_J : J \in \mathcal{I}(A)\}. \quad (13)$$

Conversely, every finitely generated S -subsemimodule of $G(K)$ is a fractional ideal in this list. Its regular ideals are exactly the I_J , and its operations are

$$I_J I_H = I_{JH}, \quad I_J + I_H = I_{J+H}, \quad I_J \cap I_H = I_{J \cap H}, \quad (14)$$

$$TF = T, \quad T + F = F, \quad T \cap F = T, \quad (15)$$

$$EF = E \ (F \neq T), \quad E + F = F \ (F \neq T), \quad E \cap F = E \ (F \neq T). \quad (16)$$

Thus $E^r = E$ for every $r \geq 1$. The unit group of the multiplicative monoid $\mathcal{F}(S)$ consists exactly of the regular ideals, with $I_J^{-1} = I_{J^{-1}}$. Multiplication by F on $\mathcal{F}(S)$ is injective exactly when F is regular.

Proof. If $F \cap K = \emptyset$, then $F = T$. Otherwise multiplication by e puts e in F , and multiplication by -1 supplies additive inverses on $F \cap K$. Thus $J = F \cap K$ is an A -submodule. A common denominator makes dJ an ideal of A . If $J = 0$, then $F = E$; otherwise $J \in \mathcal{I}(A)$, so $F = I_J$. Conversely every listed set has the required closure and denominator. The ideals T, E are generated by τ, e . If $a_1, \dots, a_r$ generate J over A , their S -span is I_J : supported sums give the A -linear combinations and the absent sum gives τ . Hence all fractional ideals are finitely generated.

Conversely finitely many elements of $G(K)$ have finitely many supported field denominators. Their product is a nonzero common denominator for their entire S -span.

Every supported product from I_J, I_H lies in JH , and every element of JH is a finite sum of such products. This proves the product formula, including supported zero. Sum and intersection follow from their subsets. The relations involving T follow from its being the specified zero subsemimodule. Every $F \neq T$ contains E , and multiplication by e sends its supported elements to e ; hence (16). By Lemma 5.3, $I_J I_{J^{-1}} = S$. Neither E nor T is invertible, because its products stay in E . Invertible ideals act injectively; E identifies E and S under multiplication, and T acts constantly. $\square$

The entire ideal monoid therefore has an exact divisor presentation. Set

$$D_K = \bigoplus_{\mathfrak{p} \neq 0} \mathbb{Z}[\mathfrak{p}], \quad \widehat{D}_K = D_K \sqcup \{\epsilon, \perp\},$$

with extended addition

$$\epsilon + D = \epsilon, \quad \epsilon + \epsilon = \epsilon, \quad \perp + x = \perp \quad (x \in \widehat{D}_K).$$

Unique ideal factorization and the theorem give a monoid isomorphism

$$\mathcal{F}(S) \xrightarrow{\sim} \widehat{D}_K, \quad I_J \mapsto \sum_{\mathfrak{p}} v_{\mathfrak{p}}(J)[\mathfrak{p}], \quad E \mapsto \epsilon, \quad T \mapsto \perp. \quad (17)$$

Integral ideals correspond to effective divisors together with $\epsilon, \perp$. The inclusion order keeps the strict levels

$$T \subsetneq E \subsetneq I_J, \quad I_J \subseteq I_H \iff v_{\mathfrak{p}}(J) \geq v_{\mathfrak{p}}(H) \text{ for all } \mathfrak{p}.$$

Every homomorphism from the full monoid to an abelian group is trivial: $E I_J = E$ forces each I_J to the identity, and idempotence forces E, T there too. The object $\widehat{D}_K$ retains all three kinds of ideals.

Proposition 5.6 (Extended coordinates and support). *Define*

$$\tilde{v}_{\mathfrak{p}}(I_J) = v_{\mathfrak{p}}(J), \quad \tilde{v}_{\mathfrak{p}}(E) = \tilde{v}_{\mathfrak{p}}(T) = +\infty.$$

With $m + \infty = \infty$, these coordinates send product to addition, sum to minimum, and intersection to maximum. Together with $\sigma(T) = 0$, $\sigma(F) = 1$ for $F \neq T$, they determine every fractional ideal. Also $\sigma(FH) = \sigma(F)\sigma(H)$.

Proof. For regular ideals this is Lemma 5.3; the two other cases follow from (15)–(16). Regular ideals have integer exponent tuples with finite support. Only E, T have the all-infinite tuple, and σ distinguishes them. A product of supported ideals contains e , whereas a product with T is T . $\square$

5.4 The original class relation and its exact comparison

Write $\text{Cl}(A) = \mathcal{I}(A)/\{cA : c \in K^\times\}$ for the quotient of the fractional ideal group by its principal subgroup. The source [1, definition before `thm:class-semigroup`] defines, on nonzero semiring ideals,

$$F \sim_{\text{src}} H \iff \exists a, b \in S \setminus \{\tau\} : aF = bH. \quad (18)$$

Its displayed scalar set is A , so it contains e . We retain this relation exactly as written.

Theorem 5.7 (Both class quotients and their map). *On nonzero integral ideals, or on $\mathcal{F}(S) \setminus \{T\}$, the source quotient is the one-element monoid. Define a distinct relation with regular scalars:*

$$F \sim_{\text{reg}} H \iff \exists a, b \in A \setminus \{0_A\} : aF = bH. \quad (19)$$

On either of those two domains its quotient is

$$\text{Cl}_{\text{reg}}^+(S) \simeq \text{Cl}(A) \sqcup \{\epsilon\}, \quad \epsilon c = \epsilon, \quad \epsilon^2 = \epsilon. \quad (20)$$

The isomorphism is $[I_J]_{\text{reg}} \mapsto [J]$, $[E]_{\text{reg}} \mapsto \epsilon$. On regular ideals alone it is a group isomorphism $\text{Cl}_{\text{reg}}(S) \simeq \text{Cl}(A)$. The identity on ideals induces the exact comparison

$$q : \text{Cl}(A) \sqcup \{\epsilon\} \longrightarrow \text{Cl}_{\text{src}}(S) = \{*\}, \quad q([J]) = q(\epsilon) = *. \quad (21)$$

If T is included, the regular quotient adds a second absorbing level $\perp$, while the source quotient has exactly two classes, represented by T, E , with the Boolean multiplicative table.

Proof. For every $F \neq T$, $eF = E$. Choosing $a = b = e$ in (18) relates every two such ideals, including E, S , and proves the first assertion. For regular scalars, $aI_J = bI_H$ is equivalent to $aJ = bH$, or $J = (b/a)H$ in K . Conversely every multiplier in $K^\times$ has this form with $a, b \in A \setminus \{0\}$. Thus on regular ideals the relation is exactly ordinary ideal-class equivalence. Every fractional ideal scales to an integral one, so integral ideals have the same class set. Regular scalars fix E and cannot send a regular ideal to E . Its class is therefore a distinct absorber, and all other classes form the ordinary ideal class group. The implication $\sim_{\text{reg}} \Rightarrow \sim_{\text{src}}$ gives the map q with the displayed values. Finally every supported scalar sends T to T and a supported ideal to a supported ideal, so T is a separate class in both relations. The product table follows from $TF = T$. The explicit descriptions also verify transitivity and compatibility with multiplication on each stated domain. $\square$

Thus the regular-scalar class group is an exact comparison object within the complete ideal calculation, and q records the precise quotient imposed by the original scalars. The comparison construction in [3, `cor:dedekind-preservation`] must likewise specify its regular ideals and regular principal scalars; it is not used to rename (18).

5.5 Codimension and the divisor map

Theorem 5.8 (The complete codimension filtration). *Put $X = \text{Spec } S$, $Y = V(E)$. Every point, height, closure, and residue object is in this table; residue objects use the Bourne congruence of the stalk's maximal ideal.*

point	height in X	closure	$\mathcal{O}_{X,x}$	$\kappa(x)$
$\eta = T$	0	X	$\mathbb{B}$	$\mathbb{B}$
$\xi = E$	1	Y	$G(K)$	K
$x_{\mathfrak{p}} = I_{\mathfrak{p}}$	2	$\{x_{\mathfrak{p}}\}$	$G(A_{\mathfrak{p}})$	$A/\mathfrak{p}$

The reflection induces a closed immersion $i : \text{Spec } A \rightarrow X$ with image Y , sending the ordinary generic point to E and increasing every codimension by one. Precisely,

$$X^{(0)} = \{\eta\}, \quad X^{(1)} = \{E\}, \quad X^{(2)} = \{x_{\mathfrak{p}} : \mathfrak{p} \neq 0\}, \quad X^{(r)} = \emptyset \quad (r > 2). \quad (22)$$

Hence $X^{\geq 0} = X$, $X^{\geq 1} = Y$, $X^{\geq 2} = \{x_{\mathfrak{p}}\}$, and $X^{\geq 3} = \emptyset$. The subset $X^{\geq 2}$ is dense in Y and is not closed in X .

Proof. Every ideal containing a supported element contains e , and its supported intersection is a ring ideal. Thus the integral ideals are T and $J \cup \{\tau\}$ for integral ring ideals J , including 0. Product testing shows that T is prime and $J \cup \{\tau\}$ is prime exactly when J is prime. By Lemma 5.2, all maximal strict chains are $T \subsetneq E \subsetneq I_{\mathfrak{p}}$. Closures consist of primes containing the given prime. This proves the heights, closures, and filtration.

At T , localization inverts e ; its idempotence forces $e = 1$, and then $a = ea = e = 1$ for every supported a . The support map to $\mathbb{B}$ distinguishes these from τ , proving the first stalk. At E , localization is (10). At $I_{\mathfrak{p}}$, the same fraction calculation with denominators $A \setminus \mathfrak{p}$ gives $G(A_{\mathfrak{p}})$.

For any ring R and ideal J , the Bourne relation of $J \cup \{\tau\} \subseteq G(R)$ is $x \sim y$ when $x + u = y + v$ for some $u, v \in J \cup \{\tau\}$. It implies equality of amplitudes in R/J . Conversely adding e makes any absent element supported; if the two amplitudes differ by an element of J , add that difference to one side. This proves that the quotient is exactly R/J . For T , the relation is equality. Applying these facts at the three stalks proves the residues in the table. Contraction along reflection sends J to $J \cup \{\tau\}$; the formula $D_X(a) \cap Y = D_A(a)$ proves its asserted topology.

For every rational prime ℓ , the nonzero ring $A/\ell A$ has ℓ^n elements and a maximal ideal. Its inverse image is a prime above ℓ ; distinct rational primes give distinct maximal ideals of A . A fixed nonzero a belongs to only finitely many such ideals, since A/aA is finite. Thus every nonempty $D_A(a)$ contains a closed point. The closed points are dense in Y but omit E , so their set is not closed. $\square$

Proposition 5.9 (Arithmetic divisor sequence). *The free cycle groups on points of codimension r are*

$$Z^0(X) = \mathbb{Z}[\eta], \quad Z^1(X) = \mathbb{Z}[E], \quad Z^2(X) = \bigoplus_{\mathfrak{p} \neq 0} \mathbb{Z}[x_{\mathfrak{p}}].$$

The inclusion i identifies $Z^0(Y)$ with $Z^1(X)$, and $Z^1(Y)$ with $Z^2(X)$, point by point. There is an exact sequence

$$1 \longrightarrow A^\times \longrightarrow K^\times \xrightarrow{\text{div}_{\text{ar}}} Z^2(X) \longrightarrow \text{Cl}_{\text{reg}}(S) \longrightarrow 0, \quad \text{div}_{\text{ar}}(a) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(a)[x_{\mathfrak{p}}]. \quad (23)$$

A regular fractional ideal I_J has the unique divisor with coefficients $v_{\mathfrak{p}}(J)$. Its extension at an arithmetic point is the fractional ideal

$$I_J G(A_{\mathfrak{p}}) = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}(J)} A_{\mathfrak{p}} \cup \{\tau\} \subseteq G(K).$$

At E , it localizes to $G(K)$, while E itself localizes to the proper ideal $\{\tau, e\}$. At T , every nonzero ideal localizes to the whole stalk $\mathbb{B}$.

Proof. The cycle groups and i 's maps follow from the point classification. Unique factorization identifies regular fractional ideals with $Z^2(X)$, carrying $I_{aA} = aS$ to the displayed principal divisor. The quotient by these principal ideals is exactly $\text{Cl}_{\text{reg}}(S)$. If every $v_{\mathfrak{p}}(a)$ is zero, then $aA = A$, equivalent to $a \in A^\times$, proving exactness. At $I_{\mathfrak{p}}$, adjoining τ to the ordinary equality $JA_{\mathfrak{p}} = \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}(J)} A_{\mathfrak{p}}$ gives the displayed local ideal. Every nonzero fractional ring ideal localizes to K at its generic point, whereas E keeps its two elements at the stalk $G(K)$. At T , a supported ideal contains e , which becomes 1, so its extension is the whole stalk. $\square$

The point E is therefore a codimension-one prime with an idempotent principal ideal. Any group-valued homomorphism on ideals sends E to the group identity because $E^2 = E$; the full divisor object $\widehat{D}_K$ retains it as ϵ . A regular a is a unit of $G(K)$, and direct prime membership gives

$$V_X(a) = \{x_{\mathfrak{p}} : v_{\mathfrak{p}}(a) > 0\} \quad (0 \neq a \in A), \quad V_X(e) = Y.$$

Thus the vanishing set of a regular principal nonunit ideal lies in codimension two, whereas the codimension-one ideal E has the nonregular generator e .

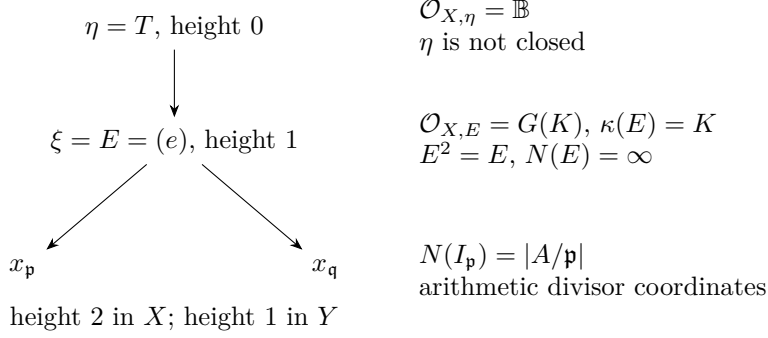


Figure 2: Complete types of specialization in $X = \text{Spec } G(\mathcal{O}_K)$. Arrows mean strict inclusion of primes. The bottom points represent arbitrary distinct finite primes, not a two-point spectrum. Theorems 5.8 and 5.10 prove heights and norms; Proposition 5.9 proves the divisor map. The supported-zero prime is the prior result [1, thm:zero-prime-not-irr].

5.6 Finite norms and the ideal zeta function

Theorem 5.10 (All integral ideal norms and closed points). *For an integral $F \subseteq S$, let $N_B(F)$ be the cardinality of its Bourne-congruence quotient. Then*

$$N_B(T) = \infty, \quad N_B(E) = \infty, \quad N_B(I_J) = |A/J| \quad (0 \neq J \subseteq A). \quad (24)$$

The finite-norm ideals are exactly the regular integral ideals. The closed points are exactly $I_{\mathfrak{p}}$ for nonzero primes $\mathfrak{p}$, all with finite norm $|A/\mathfrak{p}|$. The point E is not closed in X , has infinite quotient A , and infinite residue field K . It is closed in its local spectrum $\text{Spec } G(K)$, still with infinite residue field. The generic point T has residue object $\mathbb{B}$ of size two, but is not closed and has infinite global ideal quotient.

Proof. The Bourne relation of T is equality, giving quotient S . The quotient calculation in Theorem 5.8 gives $S/E = A$ and $S/I_J = A/J$. The first two quotients are infinite, and nonzero ring ideals have finite quotient by Lemma 5.2. Closed points are maximal primes, as already classified. The spectrum of $G(K)$ consists of $T \subsetneq E$, so E is closed there, with residue K . The remaining assertions follow from the stalk and closure table. $\square$

Proposition 5.11 (Norm homomorphism and Euler product). *On regular fractional ideals the norm is*

$$N(I_J) = \prod_{\mathfrak{p}} |A/\mathfrak{p}|^{v_{\mathfrak{p}}(J)} \in \mathbb{Q}_{>0}.$$

It is multiplicative, with $N(cS) = |\text{Norm}_{K/\mathbb{Q}}(c)|$ for $c \in K^\times$. The sum over all finite Bourne-norm ideals is

$$\zeta_{S,B}(s) = \sum_{N_B(F) < \infty} N_B(F)^{-s} = \sum_{0 \neq J \subseteq A} |A/J|^{-s} = \prod_{\mathfrak{p} \neq 0} (1 - |A/\mathfrak{p}|^{-s})^{-1} = \zeta_K(s) \quad (\text{Re } s > 1). \quad (25)$$

Equation (24) accounts explicitly for both zero ideals in this summation rule.

Proof. The ideal norm formula is Lemma 5.3. For $0 \neq c \in A$, multiplication by c on a $\mathbb{Z}$ -basis of A is a full-rank integer matrix with image cA . Its determinant's absolute value is the lattice index $[A : cA]$: integer elementary operations put the matrix in Smith diagonal form, where each quantity is the product of the positive diagonal entries. The rational determinant is the field norm by definition. For $c = a/b \in K^\times$, multiplicativity gives the quotient of these two identities.

Theorem 5.10 proves the termwise equality of the sums. Unique ideal factorization gives the Euler product as a sum of nonnegative terms for real $s > 1$. To prove convergence, let

$n = [K : \mathbb{Q}]$. Primes above a rational prime ℓ have residue sizes ℓ^{f_p} with $f_p \geq 1$. There are at most n such primes: the Chinese remainder surjection

$$A/\ell A \longrightarrow \prod_{p|\ell} A/p$$

gives $\sum_{p|\ell} f_p \leq n$. Every nonzero prime lies over a rational prime, since its residue field is finite. Hence for real $\sigma > 1$,

$$\prod_p (1 - |A/p|^{-\sigma})^{-1} \leq \prod_{\ell \text{ prime}} (1 - \ell^{-\sigma})^{-n} = \left(\sum_{m \geq 1} m^{-\sigma} \right)^n < \infty.$$

The last sum converges by the integral test; its product expression follows from integer unique factorization and increasing finite products. Thus all rearrangements are valid for complex s with real part greater than one. The final ideal sum is the defining Dedekind zeta series, proving (25). $\square$

The quotient retaining a separate absent element has its own exact comparison map. For $0 \neq J \subseteq A$, set

$$x \equiv_J^{\text{sp}} y \iff (x = y = \tau) \text{ or } (x, y \in A, x - y \in J).$$

Its quotient is $G(A/J)$, of size $|A/J| + 1$, with zero class T , rather than I_J . The natural morphism

$$G(A/J) \longrightarrow A/J, \quad \tau \mapsto 0, \quad [a] \mapsto [a],$$

identifies exactly its two zero-amplitude elements and has kernel $\{\tau, 0_{A/J}\}$. This proves the exact relation between the two quotient constructions. Their weights obey

$$(N(J) + 1)(N(H) + 1) - (N(JH) + 1) = N(J) + N(H).$$

The support-preserving cardinality therefore has its stated counting rule, while the Bourne norm retains the multiplicative formula (25).

6 The full adelic valuation space and its derived geometry

6.1 Objects and source scope

Throughout, a ring is commutative and has an identity. For a nonzero ring A , define the semiring

$$G(A) = \{\tau\} \sqcup A.$$

The symbol τ is the additive identity and multiplicative absorber. Addition and multiplication of two supported elements of A are their original ring operations, with their results remaining in the supported copy of A . We write e_A for its element 0_A . Thus

$$\tau \neq e_A, \quad e_A + e_A = e_A, \quad a + (-a) = e_A, \quad e_A a = e_A \quad (a \in A). \quad (\text{AV1.1})$$

The last formula does not apply to $a = \tau$: $e_A \tau = \tau$. All semiring homomorphisms below preserve the additive identity and the multiplicative identity. Associativity and commutativity follow on supported inputs from those of A and otherwise from the rules for τ . Distributivity follows from that of A when all three inputs are supported; if the multiplier is τ , both sides are τ ; if either summand is τ , the identity reduces to $ab = ab + \tau$. These checks prove that $G(A)$ is a semiring. A ring homomorphism $f : A \rightarrow B$ gives a semiring homomorphism $G(f)$ by $\tau \mapsto \tau$ and $a \mapsto f(a)$, including $e_A \mapsto e_B$.

The human source used here is Antoine Sédillot, *Topological adelic curves: Zariski-Riemann spaces, algebraic coverings, Harder-Narasimhan filtrations and heights*, arXiv:2503.20156v2. The exact author file is `topological_adelic_curves_V2.tex`. Its definition of pseudo-absolute values is at source lines 623–629; integral structures at lines 674–689, label `subsub:prelim_integral_structures`; the algebraic Zariski–Riemann construction at lines 819–848, labels `subsub:Zariski-Riemann_space_algebraic` and `example:ZR_algebraic`; the general coherence and Tor assertion at lines 942–985, label `prop:coherence_algebraic_ZR_sheaf`. The spelling of the latter label is retained so the locator matches the source. The statements about split objects below are proved here. The general Tor assertion of the source is examined in Section 6.7; it is not used as a substitute for the proofs in this note.

6.2 The entire space of split pseudo-absolute values

Equip $[0, \infty]$ with its compact order topology. Extended addition has $r + \infty = \infty$. Extended multiplication is used only away from the two ordered pairs $(0, \infty)$ and $(\infty, 0)$.

Definition 6.1. For a field K , let $\mathcal{P}(\mathbf{G}(K))$ consist of maps $w : \mathbf{G}(K) \rightarrow [0, \infty]$ such that

$$w(\tau) = 0, \quad w(1) = 1, \quad w(a + b) \leq w(a) + w(b), \quad (\text{AV2.1})$$

and

$$w(ab) = w(a)w(b) \quad \text{if } \{w(a), w(b)\} \neq \{0, \infty\}. \quad (\text{AV2.2})$$

Give this set its topology of pointwise convergence. Let M_K be the space of pseudo-absolute values in Sédillot’s definition: the same conditions on K , with $v(0_K) = 0$ and $v(1) = 1$.

Theorem 6.2 (Exact component decomposition). *Define*

$$\chi_K(\tau) = 0, \quad \chi_K(a) = 1 \ (a \in K), \quad j_K(v)(\tau) = 0, \quad j_K(v)(a) = v(a) \ (a \in K).$$

Then

$$\mathcal{P}(\mathbf{G}(K)) = \{\chi_K\} \sqcup j_K(M_K) \quad (\text{AV2.3})$$

is a homeomorphism of a disjoint union of two compact Hausdorff spaces. The two components are respectively $w(e_K) = 1$ and $w(e_K) = 0$ and are both open and closed. In particular, the full space is not identified with M_K by restriction to the supported coefficients.

Proof. Since $(-1)^2 = 1$, multiplicativity applies to the repeated value $w(-1)$ and gives $w(-1)^2 = 1$. Neither zero nor infinity has square one, so $w(-1) = 1$. The equality $1 + (-1) = e_K$ implies $w(e_K) \leq 2$. The equality $e_K^2 = e_K$ then gives $w(e_K)^2 = w(e_K)$, whence $w(e_K) \in \{0, 1\}$.

If $w(e_K) = 1$, then for every supported a the pair $(w(e_K), w(a))$ is allowed in (AV2.2); hence $1 = w(e_K a) = w(a)$. Thus $w = \chi_K$. This map satisfies the triangle inequality: the sum of two supported elements is supported, so its value one is at most two; a sum involving τ is immediate. It is multiplicative because a product is supported exactly when both factors are supported.

If $w(e_K) = 0$, its restriction to K satisfies precisely Sédillot’s three axioms. Conversely, any such v extends by $v(\tau) = 0$: the new triangle inequalities are equalities and each newly required product identity has value zero on both sides. This proves the bijection. Both restriction and extension are continuous in the product topologies. The continuous evaluation at e_K takes only the values zero and one, which have disjoint open neighbourhoods. It therefore makes both components open and closed.

For completeness, M_K is a closed subset of $[0, \infty]^K$. The two value constraints and every triangle inequality are closed conditions. For a convergent net satisfying multiplicativity, if its limiting pair is allowed, it has a neighbourhood avoiding the two forbidden pairs; extended multiplication is continuous at that pair, so the limiting identity holds. Thus its multiplicativity condition is closed as well. The product is compact Hausdorff and the trivial absolute value supplies a point. This proves the compactness and Hausdorff assertions, and then those of (AV2.3). $\square$

Proposition 6.3 (Field maps preserve both components). *A field embedding $i : K \hookrightarrow L$ induces a continuous restriction $\mathcal{P}(\mathbf{G}(L)) \rightarrow \mathcal{P}(\mathbf{G}(K))$. It sends χ_L to χ_K and $j_L(v)$ to $j_K(v|_K)$. Every extension of χ_K is χ_L ; extensions of $j_K(v)$ correspond exactly to extensions of v to L .*

Proof. Precomposition preserves every displayed axiom and is continuous in the pointwise topology. It preserves e , so preserves its value zero or one. Theorem 6.2 then gives all assertions. $\square$

6.3 Finiteness, residue, and the complete prime space

Proposition 6.4 (Finiteness and residue proved from the axioms). *For $v \in M_K$ put*

$$A_v = \{a \in K : v(a) < \infty\}, \quad \mathfrak{m}_v = \{a \in K : v(a) = 0\}, \quad \kappa_v = A_v / \mathfrak{m}_v.$$

Then A_v is a valuation ring of K , its unique maximal ideal is $\mathfrak{m}_v$, and v induces an absolute value $\bar{v}$ on κ_v . The finite locus of $j_K(v)$ is exactly $\mathbf{G}(A_v)$.

Proof. As above, $v(-1) = 1$, so $v(-a) = v(a)$. The triangle and product axioms make A_v a subring and $\mathfrak{m}_v$ an ideal in A_v . For $a \neq 0$, the identity $aa^{-1} = 1$ gives the following exhaustive possibilities. If $0 < v(a) < \infty$, then $v(a^{-1}) = v(a)^{-1}$: all pairs with the positive finite value $v(a)$ are allowed. If $v(a) = 0$, then $v(a^{-1})$ must be infinity, since every finite alternative would give product value zero. If $v(a) = \infty$, then $v(a^{-1})$ must be zero, since every nonzero alternative would give product value infinity. Consequently a or a^{-1} belongs to A_v , proving the valuation property, and its units are precisely the elements of positive finite value. This proves the assertion about the maximal ideal.

If $a - b \in \mathfrak{m}_v$, the triangle inequality gives $v(a) \leq v(b)$ and $v(b) \leq v(a)$; hence the value depends only on the residue class. It vanishes precisely on the zero class, is finite there, and satisfies the ordinary triangle and multiplication laws. Thus it is an absolute value of the residue field. The final assertion follows from the value zero of τ . $\square$

Proposition 6.5 (All ideals and all primes). *For every nonzero ring A , the ideals of $\mathbf{G}(A)$ are $\{\tau\}$ and*

$$I^+ = \{\tau\} \sqcup I \quad (I \text{ an ideal of } A). \quad (\text{AV3.1})$$

Its proper prime ideals are $\{\tau\}$ and $\mathfrak{p}^+$ for $\mathfrak{p} \in \text{Spec } A$. The map $\mathfrak{p} \mapsto \mathfrak{p}^+$ identifies $\text{Spec } A$ with the closed subset $V(e_A)$ of $\text{Spec } \mathbf{G}(A)$. The remaining point $\{\tau\}$ is open and is a generalization of every point. For supported a ,

$$D_{\mathbf{G}(A)}(a) = \{\{\tau\}\} \sqcup D_A(a), \quad D_{\mathbf{G}(A)}(e_A) = \{\{\tau\}\}. \quad (\text{AV3.2})$$

If A is a domain and $\mathfrak{p}$ a nonzero prime, its original chain is retained as

$$(\tau) = \{\tau\} \subsetneq (e_A) = \{\tau, e_A\} \subsetneq \mathfrak{p}^+. \quad (\text{AV3.3})$$

Proof. An ideal containing a supported element contains e_A by multiplication by e_A . Its supported elements are closed under addition, multiplication by every element of A , and negatives by multiplication by -1 ; hence they form a ring ideal. The converse follows directly from the operations. A product of two supported elements is always supported, even when its coefficient is zero, proving primality of $\{\tau\}$. Primality of $\mathfrak{p}^+$ is exactly primality of $\mathfrak{p}$ in A . The formulas for the basic opens follow by membership in these ideals; they prove the topological assertions and the strict chain. $\square$

Theorem 6.6 (Residue map with supported zero). *For $v \in M_K$ there is a commuting diagram of semirings*

$$\begin{array}{ccc} \mathbf{G}(A_v) & \xrightarrow{\mathbf{G}(\pi_v)} & \mathbf{G}(\kappa_v) \\ q_{A_v} \downarrow & & \downarrow q_{\kappa_v} \\ A_v & \xrightarrow{\pi_v} & \kappa_v. \end{array} \quad (\text{AV3.4})$$

Here $q_A(\tau) = q_A(e_A) = 0_A$ and $q_A(a) = a$ on supported coefficients. The top map is surjective and local: its inverse image of the unique maximal ideal $\{\tau, e_{\kappa_v}\}$ is $\mathfrak{m}_v^+$. Its fibre over the additive identity is only $\{\tau\}$. Its full congruence is

$$\tau \sim \tau, \quad a \sim b \iff a - b \in \mathfrak{m}_v \quad (a, b \in A_v), \quad (\text{AV3.5})$$

with no equivalence of τ and a supported element. On $\mathbf{G}(A_v)$ one has

$$j_K(v) = j_{\kappa_v}(\bar{v}) \circ \mathbf{G}(\pi_v). \quad (\text{AV3.6})$$

Proof. Every map in (AV3.4) respects its stated operations; on a supported input both composites are its residue coefficient and on τ both are zero. The assertions concerning fibres, congruence, and localness follow respectively from preservation of support, equality of residue coefficients, and Proposition 6.5. The last formula is Proposition 6.4 on supported elements and the value zero on τ . $\square$

The restriction to the finite locus is essential. If $0 \neq a \in \mathfrak{m}_v$, then $v(a) = 0$, $v(a^{-1}) = \infty$, and $v(aa^{-1}) = 1$. Thus $\{w = 0\}$ is generally not an ideal of $\mathbf{G}(K)$; it becomes the prime ideal $\mathfrak{m}_v^+$ of $\mathbf{G}(A_v)$. The precise defect is the set of forbidden products

$$\mathcal{D}_v = \{(a, b) \in K^2 : v(a) = 0, v(b) = \infty\} \cup \{(a, b) : v(a) = \infty, v(b) = 0\}.$$

The restriction map to $\mathbf{G}(A_v)$ removes every such pair and gives the exact residue factorization (AV3.6); it does not identify the two zeros.

6.4 Integral structures and the support point

Let A be a Prüfer domain with fraction field K . Here this means that every $A_{\mathfrak{p}}$ is a valuation ring. Let $\|\cdot\|$ be a Banach ring norm as in Sédillot's integral-structure definition. A bounded multiplicative seminorm α on A means a finite map to $[0, \infty)$ with $\alpha(0) = 0$, $\alpha(1) = 1$, triangle inequality, multiplicativity, and $\alpha(a) \leq \|a\|$.

Proposition 6.7 (Exact extension from the integral spectrum). *Restriction is a bijection between such α and the subset*

$$M_{K,A} = \{v \in M_K : v(a) \leq \|a\| \text{ for all } a \in A\}. \quad (\text{AV4.1})$$

For $\mathfrak{p} = \ker \alpha$, the inverse map has finiteness ring $A_{\mathfrak{p}}$ and is given by

$$v_{\alpha}(x) = \begin{cases} \bar{\alpha}(x \bmod \mathfrak{p}A_{\mathfrak{p}}), & x \in A_{\mathfrak{p}}, \\ \infty, & x \notin A_{\mathfrak{p}}, \end{cases} \quad (\text{AV4.2})$$

where $\bar{\alpha}$ is the absolute value on $\kappa(\mathfrak{p}) = \text{Frac}(A/\mathfrak{p})$ defined by $\bar{\alpha}(\bar{a}/\bar{b}) = \alpha(a)/\alpha(b)$. The subset (AV4.1) is closed in M_K , and this bijection is a homeomorphism for the pointwise topologies.

Proof. Multiplicativity makes $\mathfrak{p}$ prime. The triangle inequality proves that α depends only on a residue class, as in Proposition 6.4, and its quotient norm extends uniquely by the indicated fraction formula. The residue field of $A_{\mathfrak{p}}$ is exactly $\text{Frac}(A/\mathfrak{p})$ because its nonzero residue denominators are the elements outside $\mathfrak{p}$.

For any valuation ring $V \subset K$ with maximal ideal $\mathfrak{m}$ and residue absolute value β , the map equal to β on residues of V and infinity outside V satisfies the pseudo-absolute-value axioms. To verify this, sums of two finite inputs use the residue triangle inequality and other sums have infinite right side. Products of finite inputs use the residue multiplication law. An element outside V has inverse in $\mathfrak{m}$. Multiplication by a unit of V preserves its being outside V , and the product of two outside elements is outside because its inverse is a product in $\mathfrak{m}$. The sole remaining case is one residue-zero factor and one infinite factor, for which no identity is imposed. This proves existence of (AV4.2).

For uniqueness let v restrict to α . Since every element of A is finite, $A \subset A_v$. All elements of $A \setminus \mathfrak{p}$ are units of A_v , giving $A_{\mathfrak{p}} \subset A_v$ with the displayed residue values. If $x \in A_v \setminus A_{\mathfrak{p}}$, the valuation property implies $x^{-1} \in \mathfrak{p}A_{\mathfrak{p}}$, whose value is zero. But then multiplicativity for the finite pair (x, x^{-1}) would give $v(1) = 0$, a contradiction. Thus the finiteness ring and every value are forced.

Each inequality in (AV4.1) is closed in the pointwise topology. This subset is compact by Theorem 6.2. Restriction is a continuous bijection into a Hausdorff product space, so its inverse is continuous: a closed subset of a compact domain has compact, hence closed, image in a Hausdorff space. $\square$

Corollary 6.8 (What a bound does to the split point). Extend the norm bound by $\|\tau\| = 0$ and $\|e_A\| = \|0_A\| = 0$. The maps in $\mathcal{P}(\mathbf{G}(K))$ obeying this bound on $\mathbf{G}(A)$ are precisely $j_K(M_{K,A})$. In particular χ_K does not obey it. The full split extension of the integral space is instead the explicit disjoint union $\{\chi_K\} \sqcup j_K(M_{K,A})$.

Proof. The character has $\chi_K(e_A) = 1 > 0$, whereas every lifted valuation has value zero there. On every other supported input the bound is exactly (AV4.1). Apply Theorem 6.2. $\square$

Proposition 6.9 (The center map includes the additional prime). *For the explicit space $\widetilde{M}_{K,A} = \{\chi_K\} \sqcup j_K(M_{K,A})$, there is a continuous map*

$$c_A : \widetilde{M}_{K,A} \longrightarrow \text{Spec } \mathbf{G}(A), \quad c_A(\chi_K) = \{\tau\}, \quad c_A(j_K(v)) = (\mathfrak{m}_v \cap A)^+. \quad (\text{AV4.3})$$

At $j_K(v)$, writing $\mathfrak{p} = \mathfrak{m}_v \cap A$, the associated local semiring is $\mathbf{G}(A)_{\mathfrak{p}^+} = \mathbf{G}(A_{\mathfrak{p}}) = \mathbf{G}(A_v)$. At the additional prime one instead has $\mathbf{G}(A)_{\{\tau\}} \cong \mathbb{B}$, where $\mathbb{B} = \{0, 1\}$ has $1 + 1 = 1$.

Proof. The values of v are finite on A , so its zero locus there is a prime ideal, by multiplication of finite nonnegative values. For a supported $a \in A$, the inverse image of the basic open $D(a)$ is $\{\chi_K\} \sqcup \{j_K(v) : v(a) > 0\}$, which is open in the stated pointwise topology. The inverse image of $D(\tau)$ is empty. These opens prove continuity.

For the local semiring at $\mathfrak{p}^+$, denominators are exactly supported $s \in A \setminus \mathfrak{p}$. Send a supported fraction a/s to its supported coefficient in $A_{\mathfrak{p}}$ and the fraction τ/s to τ . The usual cross-multiplication criterion proves equality of two supported fractions exactly when their ring fractions agree; no supported fraction equals a τ fraction since multiplication of supported elements stays supported. This proves the first localization isomorphism. The equality with A_v was proved in Proposition 6.7.

At $\{\tau\}$ every supported element, including e_A , is inverted. An invertible multiplicative idempotent equals one, so $e_A = 1$ in that localization. Then $e_A a = e_A$ forces every supported a to equal one. The two classes remain distinct because the support map to $\mathbb{B}$ sends all supported denominators to a unit, and therefore factors through the localization while distinguishing their class from τ . Its addition is $1 + 1 = 1$ by the image of $e_A + e_A = e_A$. $\square$

6.5 The exact additive bridge and its limits

Theorem 6.10 (Universal ring recovery). *The map $q_A : \mathbf{G}(A) \rightarrow A$ is universal among semiring homomorphisms from $\mathbf{G}(A)$ to rings. It identifies exactly τ and e_A and keeps all other coefficients distinct. Consequently the category of $\mathbf{G}(A)$ -semimodules whose additive monoid is an abelian group is canonically equivalent to the category of A -modules, by restriction along q_A . Under this equivalence tensor products, resolutions, and Tor are exactly the ordinary A -module constructions.*

Proof. If $f : \mathbf{G}(A) \rightarrow R$ is a semiring homomorphism with R a ring, then $f(e_A) + f(e_A) = f(e_A)$ implies $f(e_A) = 0$. It already sends τ to zero. Its restriction to supported A preserves addition, multiplication, and identities, hence is a ring homomorphism. It is uniquely determined and factors f through q_A .

On an abelian-group semimodule M , the relation $e_{Am} + e_{Am} = e_{Am}$ similarly gives $e_{Am} = 0_M$ for all m . The action therefore factors through A , and all semimodule homomorphisms are precisely A -linear maps. Conversely every A -module gives such an action. These are inverse functors and preserve each addition and scalar action. A balanced biadditive map for one description is exactly one for the other, so the tensor universal properties agree. Free and projective objects, exact sequences, and tensor resolutions in this abelian category therefore agree, proving the Tor assertion. $\square$

There is also a way to retain the two zeros on modules. For an A -module M , put $\mathbf{G}(M) = \{\tau_M\} \sqcup M$, with supported addition inherited from M , and the scalar action induced from A , with either τ input absorbing. The semimodule laws follow by the same cases as in (AV1.1).

Proposition 6.11 (Support-preserving module category). *A homomorphism $h : \mathbf{G}(M) \rightarrow \mathbf{G}(N)$ is either the constant map to τ_N , or is $\mathbf{G}(f)$ for a unique A -linear map $f : M \rightarrow N$. The morphisms preserving supported elements form a category equivalent to A -modules. Its additive zero morphism is the supported map $m \mapsto 0_N$, extended by $\tau_M \mapsto \tau_N$; it differs from the constant- τ_N homomorphism in the ambient semimodule category. For every linear f ,*

$$(\mathbf{G}(f))^{-1}(\{\tau_N\}) = \{\tau_M\}, \quad \mathbf{G}(\ker f) = \{\tau_M\} \sqcup \{m : f(m) = 0_N\}. \quad (\text{AV5.1})$$

The second expression, with its inclusion, is the kernel in the support-preserving category; the first is the categorical zero fibre in the ambient category.

Proof. The only additive idempotents in $\mathbf{G}(N)$ are τ_N and 0_N : a supported idempotent satisfies $n + n = n$ in a group, so $n = 0_N$. Hence $h(0_M)$ is one of these. If $h(0_M) = \tau_N$, then $h(m) + h(-m) = \tau_N$ forces both summands to be τ_N , since any supported summand makes a sum supported. This is the constant map. If $h(0_M) = 0_N$, the identity $0_M + m = m$ rules out $h(m) = \tau_N$, because $0_N + \tau_N = 0_N \neq \tau_N$. Thus all supported elements have supported images. The homomorphism laws become exactly linearity of $f = h|_M$. Conversely every linear map extends, proving the categorical claim. The two kernel statements now follow by direct inverse-image calculation, including the distinct images of 0_M and τ_M . $\square$

In particular, no assertion about derived functors in the entire nonadditive semimodule category follows merely by renaming ordinary module Tor . Theorem 6.10 and Proposition 6.11 specify two exact functors and the exact categories in which the asserted transfer holds.

6.6 Valuation stalks and the sheaf that retains support

Proposition 6.12 (Tor bound over a valuation ring). *Let V be a valuation ring with fraction field F . Every torsion-free V -module is flat. Every V -module has a flat resolution of length at most one. Every finitely presented V -module has a finite free resolution of length at most one. Consequently*

$$\text{Tor}_i^V(M, N) = 0 \quad (i \geq 2) \quad (\text{AV6.1})$$

for all V -modules M, N .

Proof. First a finitely generated torsion-free module is free. Given generators $m_1, \dots, m_n$, if they are dependent choose a nonzero relation $\sum a_i m_i = 0$. Among its nonzero coefficients choose a_j dividing all the others; this is possible because principal ideals in a valuation ring are totally ordered. Write $a_i = a_j b_i$, with $b_j = 1$. Torsion-freeness gives $\sum b_i m_i = 0$, expressing m_j in the other generators. Repeating decreases the number of generators until an independent generating set remains. This is a free basis.

An arbitrary torsion-free module is the filtered union of its finitely generated submodules, all free by the preceding argument. It is flat: tensor commutes with filtered colimits, and filtered colimits preserve injections and exactness of modules. Explicitly, a finite tensor relation already lies in some finitely generated submodule, and if its image vanishes in the colimit it vanishes at a later stage; exactness there gives exactness in the colimit. Thus no relation obstructing flatness appears in the union.

Choose a free surjection $P \rightarrow M$. Its kernel is a submodule of a free module over a domain and so is torsion-free, hence flat. This is the claimed length-one flat resolution. If M is finitely presented, choose P finite free and its kernel finitely generated; that kernel is then finite free by the first paragraph. Tensoring either resolution proves (AV6.1). $\square$

If $(Z, \mathcal{O}_Z)$ is a locally ringed space with valuation-ring stalks, then for all sheaves of modules $\mathcal{F}, \mathcal{H}$,

$$\mathcal{T}or_i^{\mathcal{O}_Z}(\mathcal{F}, \mathcal{H}) = 0 \quad (i \geq 2). \quad (\text{AV6.2})$$

Indeed stalks commute with tensor and homology; applying a flat resolution and taking stalks identifies the stalk of this sheaf with $\text{Tor}_i^{\mathcal{O}_{Z,z}}(\mathcal{F}_z, \mathcal{H}_z)$, which vanishes by Proposition 6.12. A sheaf with all stalks zero is zero, since every section whose germs vanish vanishes on a neighbourhood of each point and hence globally by the sheaf axiom.

There is a small sheaf issue that matters for retaining support. The assignment $U \mapsto \mathbf{G}(\mathcal{O}_Z(U))$ need not itself be a sheaf: on two disjoint nonempty opens, a supported section on one and τ on the other must be able to glue. Define instead

$$\mathcal{S}_Z(U) = \{(C, s) : C \subset U \text{ is open and closed, } s \in \mathcal{O}_Z(C)\}. \quad (\text{AV6.3})$$

For $C = \emptyset$ the unique pair is τ . Addition has support $C \cup D$ and is the original sum on $C \cap D$, the first section on $C \setminus D$, and the second on $D \setminus C$. Multiplication has support $C \cap D$ and is the original product there. These pieces are open and closed in the union, so the sheaf axiom for $\mathcal{O}_Z$ defines each operation. In particular cancellation on $C \cap D$ leaves a supported zero.

Proposition 6.13 (Sheaf-level split and recovery). *The construction (AV6.3) is a sheaf of semirings with stalk $\mathcal{S}_{Z,z} = \mathbf{G}(\mathcal{O}_{Z,z})$. Extension by zero on $U \setminus C$ gives a morphism $q : \mathcal{S}_Z \rightarrow \mathcal{O}_Z$ which has the universal ring-recovery property stalkwise and as a sheaf. Sheaves of $\mathcal{S}_Z$ -semimodules with abelian-group addition are exactly sheaves of $\mathcal{O}_Z$ -modules. If the stalks of $\mathcal{O}_Z$ are valuation rings, their Tor sheaves in this abelian category obey (AV6.2).*

Proof. Compatible supports on an open cover glue to a subset whose membership is locally constant, hence to an open-and-closed subset. The supported sections then glue uniquely by the original sheaf axiom. All semiring identities hold locally according to whether each support is present, and there reduce to the verified identities of $\mathbf{G}$. A germ either has absent support on a neighbourhood or has full support on a neighbourhood. These alternatives give respectively τ and a unique supported coefficient germ, proving the stalk identification. The extension-by-zero map is compatible with restrictions and operations by the same local cases. Any map to a sheaf of rings factors uniquely on stalks by Theorem 6.10; equivalently, a local section splits on its support and its complement, where the factorizations are unique and so glue. The action on every additive-group semimodule likewise factors locally and uniquely, giving the stated equivalence. Its tensor products agree on stalks by Theorem 6.10, and hence as sheaves; the Tor conclusion follows from (AV6.2). $\square$

6.6.1 The full spectral extension, including the extra generic point

Let $(Z, \mathcal{O}_Z)$ be a nonempty irreducible locally ringed space whose local rings are nonzero. Form the space

$$Z^+ = Z \sqcup \{\eta_\tau\} \quad (\text{AV6.4})$$

whose opens are the empty set, $\{\eta_\tau\}$, and $U_W = \{\eta_\tau\} \cup W$ for nonempty open $W \subset Z$. These sets form a topology: arbitrary unions are of the same form, and a finite intersection is obtained by intersecting the corresponding opens in Z , with empty intersection there giving $\{\eta_\tau\}$. Define

$$\mathcal{O}_Z^+(U_W) = \mathbf{G}(\mathcal{O}_Z(W)), \quad \mathcal{O}_Z^+(\{\eta_\tau\}) = \mathbb{B}. \quad (\text{AV6.5})$$

Restriction between two opens with nonempty arithmetic parts is the split extension of the original restriction map. Restriction to $\{\eta_\tau\}$ sends τ to 0 and every supported section to 1. This is a semiring homomorphism to $\mathbb{B}$: supported sums and products remain supported, including supported zero coefficients. On the empty open take the one-element semiring. The restriction maps compose because ordinary restriction never changes the support tag.

Theorem 6.14 (Full spectral gluing and stalks). *The construction (AV6.5) is a sheaf of local semirings. Its stalks are*

$$\mathcal{O}_{Z, \eta_\tau}^+ = \mathbb{B}, \quad \mathcal{O}_{Z, z}^+ = \mathbf{G}(\mathcal{O}_{Z, z}) \quad (z \in Z). \quad (\text{AV6.6})$$

The inclusion $i : Z \hookrightarrow Z^+$ is a closed embedding of spaces. The inverse-image sheaf $i^{-1}\mathcal{O}_Z^+$ is the split sheaf $\mathcal{S}_Z$ of Proposition 6.13; its universal ring quotient is $\mathcal{O}_Z$. If $Z = \text{Spec } A$ for a domain A , this construction is canonically the entire semiring spectrum $\text{Spec } \mathbf{G}(A)$ with its localization sheaf, not only its closed arithmetic subset.

Proof. Every nonempty open of Z^+ contains η_τ . On any cover of such an open, compatible sections must therefore have the same image in the stalk $\mathbb{B}$ at η_τ . If that image is zero, each section on an open with arithmetic part is τ , and every section on the singleton is zero; they glue uniquely to the absent section. If that image is one, the sections on the arithmetic parts are supported. Their coefficients agree wherever the arithmetic parts overlap, because supported equality in $\mathbf{G}$ is coefficient equality. The original sheaf axiom glues these coefficients to one section on W , giving the unique supported glue on U_W . Sections on a singleton member of the cover must be one and impose exactly that support condition. This also proves the sheaf axiom for a cover of the singleton. In the irreducible case, nonempty arithmetic opens intersect, so the coefficient compatibility can equivalently be checked directly on their nonempty intersections.

The singleton is a smallest neighbourhood of η_τ , yielding its stalk. Neighbourhoods of an arithmetic point z are exactly U_W for neighbourhoods W of z . In their colimit no restriction identifies τ with a supported section, and supported germs are precisely $\mathcal{O}_{Z, z}$. This proves (AV6.6). If $(R, \mathfrak{m})$ is a local ring, Proposition 6.5 shows that $\mathbf{G}(R)$ has the unique maximal ideal $\mathfrak{m}^+$. The Boolean stalk has the unique maximal ideal $\{0\}$. Thus all stalks are local semirings.

The complement of $i(Z)$ is the open singleton. Its relative topology is the original topology of Z . The presheaf computing the inverse image of $\mathcal{O}_Z^+$ on an open $W \subset Z$ has the restriction of $\mathbf{G}(\mathcal{O}_Z(W))$ as a cofinal representative, because U_W is the smallest open of Z^+ containing $i(W)$. Sheafifying this presheaf allows the support tag to vary locally, giving exactly the open-and-closed supported subsets and coefficient sections of (AV6.3). The construction in Proposition 6.13 proves the ring quotient assertion.

For a domain A , Proposition 6.5 identifies the points and opens with Z^+ , taking η_τ to the prime $\{\tau\}$. For a nonzero supported a , the basic open $D_{\mathbf{G}(A)}(a)$ has arithmetic part $D_A(a)$, and its local-fraction semiring is $\mathbf{G}(A)_a \cong \mathbf{G}(A_a)$, by the same cross-multiplication calculation as Proposition 6.9. This agrees with $\mathbf{G}(\mathcal{O}_Z(D_A(a)))$. The remaining nonempty basic open $D_{\mathbf{G}(A)}(e_A) = \{\eta_\tau\}$ has localization $\mathbb{B}$ by that proposition. All restriction maps agree: between arithmetic charts they are localization maps, and to the singleton they retain exactly the support. Since these form a basis, gluing identifies their localization sheaf with (AV6.5). $\square$

Proposition 6.15 (Every local morphism extends with its support point). *Let $f : (Z, \mathcal{O}_Z) \rightarrow (X, \mathcal{O}_X)$ be a morphism of nonempty irreducible locally ringed spaces with nonzero local rings. Then*

$$f^+ : Z^+ \longrightarrow X^+, \quad f^+(z) = f(z), \quad f^+(\eta_{\tau, Z}) = \eta_{\tau, X} \quad (\text{AV6.7})$$

is canonically a morphism of locally semiringed spaces. At an arithmetic point its stalk map is $G(f_z^\#)$, and at the extra point it is the identity of $\mathbb{B}$. The construction respects identities and composition, and its restriction followed by coefficient recovery is the original morphism f .

Proof. The inverse image of U_W is $\{\eta_{\tau, Z}\} \cup f^{-1}(W)$, an open of Z^+ even when the arithmetic inverse image is empty. If it is nonempty, use $G(f^\#) : G(\mathcal{O}_X(W)) \rightarrow G(\mathcal{O}_Z(f^{-1}W))$ on sections. If it is empty, use the support homomorphism to $\mathbb{B}$. On the singleton use its identity. These homomorphisms respect restrictions, including restrictions that make the arithmetic part empty, because coefficient maps preserve the support tag. They therefore define a sheaf morphism and give the stated stalk maps. A local ring map has inverse image of the maximal ideal equal to the maximal ideal. Formula (AV3.1) shows the same for its split extension. The Boolean identity is local as well. Composition and recovery hold on every supported coefficient and on τ , proving every remaining assertion. $\square$

6.7 The exact scope of the Tor assertion and its refinement

6.7.1 The whole-scheme submodel limit does not have the claimed bound

In the supplied source a projective submodel of a projective K -scheme X over $k \subset K$ is a factorization $X \rightarrow \mathcal{X} \rightarrow \text{Spec } k$ with schematically dominant first map and projective second map. Its Zariski–Riemann space is the inverse limit of these submodels.

Proposition 6.16 (An explicit counterexample to the unrestricted claim). *If $k = K$ and X is a projective K -scheme, this submodel limit is X . For $X = \mathbf{P}_K^2$, a coherent point sheaf has nonzero Tor_2 with itself. Thus the source’s assertion of Tor-dimension at most one for all coherent sheaves on its general $\text{ZR}(X/k)$, with only its stated hypotheses, fails in this case.*

Proof. The identity factorization makes X itself a submodel, and the defining map $X \rightarrow \mathcal{X}$ is the unique arrow from this object to each submodel compatible with the factorization. For any compatible cone, its map to the X component determines every other component. Therefore the inverse limit is X , also as a locally ringed space.

Take a K -rational point with affine coordinates $x = y = 0$. Its local ring is $R = K[x, y]_{(x, y)}$, with residue field K . The sequence

$$0 \longrightarrow R \xrightarrow{c \mapsto (-yc, xc)} R^2 \xrightarrow{(a, b) \mapsto xa + yb} R \longrightarrow K \longrightarrow 0 \quad (\text{AV7.1})$$

is exact. The left map is injective because R is a domain. If $xa + yb = 0$, reduction modulo x in the domain $K[y]_{(y)}$ gives $b \equiv 0 \pmod{x}$, so $b = xc$, and then $a = -yc$. The image at R is the ideal (x, y) , giving the last exactness. Tensoring with K makes both differentials zero, so $\text{Tor}_2^R(K, K) = K \neq 0$. The coherent skyscraper sheaf at the point therefore has a nonzero second Tor stalk with itself. All the stated hypotheses hold: K is stably coherent, and $\mathbf{P}_K^2$ is projective and integral. $\square$

6.7.2 The valuation refinement is a new space with an exact map

Let X be an integral separated scheme of finite type over a field and $F = K(X)$ its actual function field. Consider integral schemes Y with a projective birational morphism $Y \rightarrow X$, and compatible morphisms over X . Their category is cofiltered: the closure of the common generic point in $Y_1 \times_X Y_2$ is another such integral model dominating both. Define

$$\mathfrak{Z}_X = \varprojlim_{Y \rightarrow X} Y, \quad \mathcal{O}_{\mathfrak{Z}_X} = \varprojlim_{Y \rightarrow X} p_Y^{-1} \mathcal{O}_Y. \quad (\text{AV7.2})$$

The limit topology and the indicated sheaf are used. Stalks are the filtered colimits of the local rings at compatible centers; these colimits are local because all transition maps between these local rings are local. The projection to the identity model is a morphism of locally ringed spaces $\rho : \mathfrak{Z}_X \rightarrow X$.

Theorem 6.17 (Valuation refinement and the proved amplitude bound). *Points of $\mathfrak{Z}_X$ are in bijection with pairs (V, c) where V is a valuation ring of F and $c : \text{Spec } V \rightarrow X$ extends the generic point $\text{Spec } F \rightarrow X$. At the corresponding point z , the stalk is exactly V . If $x = c(\mathfrak{m}_V)$, the stalk map of ρ is the inclusion $\mathcal{O}_{X,x} \hookrightarrow V$. All sheaves of modules on $\mathfrak{Z}_X$ satisfy*

$$\mathcal{T}or_i^{\mathcal{O}_{\mathfrak{Z}_X}}(\mathcal{F}, \mathcal{H}) = 0 \quad (i \geq 2). \quad (\text{AV7.3})$$

The split sheaf from (AV6.3) has stalks $\mathbf{G}(V)$ and maps to this ringed space by the exact coefficient quotient; its abelian-group module category has the same bound.

Proof. Given (V, c) and a model $Y \rightarrow X$, its generic map lifts uniquely to $\text{Spec } V \rightarrow Y$. This is the valuative property of a projective map; it can be checked here using homogeneous coordinates: write the generic lift as $[a_0 : \cdots : a_n]$ in a projective embedding of Y , divide by an a_j whose value is least among these finitely many nonzero coordinates, and obtain coordinates in V with one equal to one. The homogeneous equations hold over V because they hold over F . Uniqueness follows from separatedness, since two maps agreeing at the generic point agree on the closed equations of the diagonal. These unique lifts give compatible centers z_Y and therefore z . Each local ring $\mathcal{O}_{Y,z_Y}$ embeds locally into V .

Conversely, a compatible family gives a union $R_z = \varinjlim_Y \mathcal{O}_{Y,z_Y} \subset F$. For each $f \in F^*$, take the closure of the graph of its rational map to $\mathbf{P}^1$ inside $X \times \mathbf{P}^1$. It is an integral projective birational model. Its center lies in at least one of the two standard affine charts, so either f or f^{-1} belongs to its local ring and hence to R_z . Therefore R_z is a valuation ring. Each center map is local, and the inclusion of the X local ring gives the required center on X .

For the family constructed from V , the union of center rings equals V , rather than just being contained in it. If $f \in V$, the graph model just used has center in its finite-coordinate chart: its unique V -lift has coordinate f , which is defined at the closed point of $\text{Spec } V$. Thus f lies in that center's local ring. Every element of V is obtained. The construction is inverse in the other direction because the inclusions of center rings into the union give the unique lifts to each separated model. This proves both the bijection and the stalk and morphism formulas. The Tor assertions follow from (AV6.2) and Proposition 6.13. $\square$

The distinction between (AV7.2) and the source's whole-scheme submodels is an explicit morphism ρ and a change of modification problem: the former models the generic map $\text{Spec } F \rightarrow X$, whereas the latter requires the whole map $X \rightarrow \mathcal{X}$ to survive. For $X = \text{Spec } K$ over a subring, valuation centers recover the valuation stalk setting recalled in Sédillot's `example:ZR_algebraic`, source lines 831–835. The bound is therefore proved on the valuation objects needed by that field-level route without asserting it on every whole-scheme submodel limit.

Corollary 6.18 (The entire spectrum survives valuation refinement). The space $\mathfrak{Z}_X$ in Theorem 6.17 is irreducible with generic point represented by the valuation ring F . The refinement induces the exact morphism

$$\rho^+ : \mathfrak{Z}_X^+ \longrightarrow X^+. \quad (\text{AV7.3a})$$

Its stalk map at a valuation center is $\mathbf{G}(\mathcal{O}_{X,x}) \rightarrow \mathbf{G}(V)$, induced by the original inclusion; its stalk map at the new generic point is the Boolean identity. The ordinary generic point ξ remains in the closed arithmetic stratum and remains distinct from η_τ . At ξ , the stalk is $\mathbf{G}(F)$; the two primes that record the two generic points are $\{\tau\}$ and $\{\tau, e_F\}$. The latter is the supported-zero prime of the ordinary generic point.

Proof. The compatible generic points of all integral birational models form a point of $\mathfrak{Z}_X$, with stalk F by Theorem 6.17. Every nonempty basic open in the inverse-limit topology is a finite intersection of inverse images of opens on models. A common dominating model replaces these finitely many conditions by one open condition. If the intersection is nonempty, that open is nonempty in an integral scheme, so contains its generic point. Thus every nonempty open contains the compatible generic point, proving irreducibility. Proposition 6.15 applies to ρ and gives the stated map and stalk formulas. The arithmetic stratum is closed and omits the new point by Theorem 6.14, so the two are distinct. At ξ , Proposition 6.5 for the field F gives exactly the two listed primes; their localizations are respectively $\mathbb{B}$ and $\mathbb{G}(F)$, as proved in Proposition 6.9. This identifies their roles without identifying e_F and τ . $\square$

6.7.3 Where the derived contribution goes under a blowup

Theorem 6.19 (Complete blowup-chart calculation). *Let $A = K[x, y]$, $M = A/(x, y)$, and let*

$$B = K[u, t], \quad A \longrightarrow B, \quad x \longmapsto u, \quad y \longmapsto ut \quad (\text{AV7.4})$$

be the first chart of the blowup of (x, y) . Ordinary pullback is $B \otimes_A M = B/(u)$, whose projective dimension over B is one. Derived pullback is represented, in homological degrees 2, 1, 0, by

$$0 \longrightarrow B \xrightarrow{c \mapsto (-ut, uc)} B^2 \xrightarrow{(a, b) \mapsto u(a+tb)} B \longrightarrow 0. \quad (\text{AV7.5})$$

Its homology is

$$H_0 = B/(u), \quad H_1 = (B/(u)) [(-t, 1)], \quad H_2 = 0. \quad (\text{AV7.6})$$

Consequently the kernel and cokernel relevant to the middle map are

$$\ker d_1 = B(-t, 1), \quad \text{im } d_2 = uB(-t, 1), \quad \text{im } d_1 = uB, \quad \text{coker } d_1 = B/(u). \quad (\text{AV7.7})$$

On the whole blowup with exceptional curve $E \cong \mathbf{P}_K^1$, the two homology sheaves are $\mathcal{O}_E$ in degree zero and $\mathcal{O}_E(-1)$ in degree one. These are the complete contributions; no degree-two term remains on either chart.

Proof. The polynomial-ring version of the exact Koszul sequence (AV7.1) has the same proof: reducing modulo x in $K[y]$ shows that a relation $xa + yb = 0$ has $(a, b) = (-yc, xc)$. It is a free A -resolution of M . Tensoring it with the specified A -algebra B gives (AV7.5), which therefore computes derived pullback.

Because u is not a zero divisor, d_2 is injective. The equation $u(a + tb) = 0$ gives $a = -tb$, so $\ker d_1 = B(-t, 1)$. The image of d_2 is precisely u times this free rank-one kernel. The image of d_1 is uB because a can be arbitrary. This proves all four formulas in (AV7.7) and their homology quotients. The sequence $0 \rightarrow B \xrightarrow{u} B \rightarrow B/(u) \rightarrow 0$ is a length-one free resolution. Its cokernel is not projective: after tensoring this resolution with $B/(u)$ the degree-one homology is $B/(u) \neq 0$, whereas a projective module has vanishing positive Tor.

On the second chart $B' = K[v, s]$ with $y = v, x = vs$, the same calculation gives a degree-one generator $(1, -s)$ and quotient $B'/(v)$. On the overlap $s = t^{-1}$ and $v = ut$, and the first generator satisfies

$$(-t, 1) = -t(1, -s). \quad (\text{AV7.8})$$

The tautological line on $E = \mathbf{P}^1$ has local frames $(1, t)$ and $(s, 1)$ with first frame t times the second. Multiplying one frame by -1 changes no line bundle. Thus (AV7.8) is exactly the transition of $\mathcal{O}_E(-1)$. Degree-zero homology is the structure sheaf of the exceptional divisor on both charts, and degree-two homology is zero on this open cover, proving the global assertions. $\square$

$$\begin{array}{ccc}
3\mathbf{A}_K^2 & \longrightarrow & \mathrm{Bl}_{(x,y)} \mathbf{A}_K^2 \\
& \searrow \rho & \downarrow (u,t) \mapsto (u,ut) \\
& & \mathbf{A}_K^2
\end{array}$$

Figure 3: Exact maps for the valuation refinement and one blowup chart. The exceptional equation is $u = 0$. The supported point module pulls back to $B/(u)$, while its derived pullback also retains $(B/(u))[-t, 1]$ in degree one. Theorems 6.17 and 6.19 prove every map and homology label. The source comparison is Sédillot, arXiv:2503.20156v2, author-TeX labels `lemma:Tor-dimension-pullback` and `lemma:Tor-dimension-flattening`, lines 950–982.

For clarity, even a valuation target keeps the derived defect rather than erasing it. Let $A \rightarrow V$ be a map to a valuation ring contained in $K(x, y)$, with $u = x \neq 0$, $t = y/x \in V$, and u noninvertible. The same calculation over the domain V gives

$$V \otimes_A M = V/(u), \quad \mathrm{Tor}_1^A(V, M) = V/(u), \quad \mathrm{Tor}_i^A(V, M) = 0 \ (i \geq 2). \quad (\text{AV7.9})$$

The nonzero first group is precisely the defect of flatness on this module. It is compatible with (AV6.1), whose base ring is V : that latter formula bounds derived tensor over the valuation ring, while (AV7.9) computes base change from the surface ring A . The connecting operation is the explicit base-change complex (AV7.5), with B replaced by V . This is the exact relation between the two calculations, including its nonzero preserved term.

6.8 Adelic data transported without changing the product formula

Proposition 6.20 (Exact transport of a topological adelic curve). *Let $(K, \phi : \Omega \rightarrow M_K, \nu)$ satisfy Sédillot’s definition: Ω is locally compact Hausdorff, ϕ is continuous, and for each $a \in K^*$, $\omega \mapsto \log \phi(\omega)(a)$ is integrable. Then $\tilde{\phi} = j_K \circ \phi$ is continuous into $\mathcal{P}(\mathcal{G}(K))$, and for every supported unit $a \in K^*$,*

$$\int_{\Omega} \log \tilde{\phi}(\omega)(a) \, d\nu(\omega) = \int_{\Omega} \log \phi(\omega)(a) \, d\nu(\omega). \quad (\text{AV8.1})$$

Thus the original product formula is preserved exactly. Conversely, a continuous map into the component $w(e_K) = 0$ with these integrability conditions uniquely recovers the original adelic data. An additional isolated point mapped to χ_K , with any finite nonnegative mass, contributes zero to every integral in (AV8.1).

Proof. Continuity and the inverse assertion are Theorem 6.2. The two integrands in (AV8.1) agree pointwise on their exact original domain K^* , proving integrability and equality without alteration of any values. At the added point, $\chi_K(a) = 1$ for every such a , so $\log \chi_K(a) = 0$. A topological disjoint union with one isolated point is locally compact Hausdorff, and adjoining its finite atomic measure has the asserted integral. No product formula for e_K or τ is claimed, since those elements are not units and their logarithms do not enter the original definition. $\square$

The final construction describes exactly what the support point can and cannot add at this stage: it retains a distinct character and an independent atomic mass, but the logarithms of supported units vanish there. No assertion of a Riemann-hypothesis criterion, a functional equation, or smoothness follows from this transport alone.

7 Prime faces and the complete Weil functional

7.1 The exact semilocalization associated with a monoid prime

For $\Sigma \subseteq \mathcal{P}$, define

$$T_\Sigma = \{n \in \mathbb{Z} \setminus \{0\} : p \nmid n \text{ for every } p \in \Sigma\}, \quad R_\Sigma = T_\Sigma^{-1}\mathbb{Z}.$$

Theorem 7.1. *The complement of the monoid prime J_Σ is T_Σ . Localization followed by the original supported addition gives exactly $G(R_\Sigma)$. Its arithmetic spectrum consists of (0) and the primes $p \in \Sigma$, each with its original residue field. If $F \subset \mathcal{P}$ is finite, then*

$$R_{\mathcal{P} \setminus F} = \mathbb{Z}[p^{-1} : p \in F], \quad R_{\mathcal{P} \setminus F}^\times = \{\pm \prod_{p \in F} p^{n_p} : n_p \in \mathbb{Z}\} = \Gamma_F. \quad (26)$$

Proof. The complement statement follows directly from the definition of J_Σ . A supported fraction a/t obeys the original fraction operations and $\tau/t = \tau$, giving the localization isomorphism in Theorem 1.2. A ring prime survives localization exactly when it misses T_Σ : (p) misses it precisely when $p \in \Sigma$, and (0) always misses it. Its localized residue is still $\mathbb{F}_p$, since the denominators are nonzero modulo p . For the complement of F , the allowed denominators have no prime factors outside F , yielding the first equality in (26). A rational number and its inverse belong to that localization exactly when numerator and denominator have no prime factors outside F , proving the unit formula. $\square$

The arithmetic trace space associated with these units is

$$\mathbb{A}_F = \mathbb{R} \times \prod_{p \in F} \mathbb{Q}_p, \quad Y_F = \mathbb{A}_F / \Gamma_F.$$

This is the precise set-of-places and scaling-group construction used by Connes [5, labels YQS, GL1QS]. Equation (26) identifies its group with the original multiplicative-prime localization, without replacing that spectrum by the scalar semiring spectrum.

7.2 Every prime-power, Gamma and pole term

Use the multiplicative convention

$$\widehat{h}(s) = \int_0^\infty h(x) x^{-is} \frac{dx}{x}, \quad g^*(x) = \overline{g(x^{-1})}, \quad (g_1 * g_2)(x) = \int_0^\infty g_1(y) g_2(x/y) \frac{dy}{y}.$$

For $h \in C_c^\infty(\mathbb{R}_+^*)$, set

$$W_p(h) = (\log p) \sum_{m \geq 1} p^{-m/2} (h(p^m) + h(p^{-m})), \quad (27)$$

$$\begin{aligned} W_\infty(h) &= (\log(4\pi) + \gamma_E) h(1) \\ &\quad + \int_1^\infty \left(h(x) + h(x^{-1}) - 2x^{-1/2} h(1) \right) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}, \end{aligned} \quad (28)$$

$$D_F(h) = \widehat{h}(i/2) + \widehat{h}(-i/2) - W_\infty(h) - \sum_{p \in F} W_p(h). \quad (29)$$

The cancellation in the integrand at $x = 1$ is retained; beyond the support of h , its subtraction term is also retained. These are the explicit-formula conventions in [5, `wp`, `arch`, `sectweilpos`]. In this notation W_∞ is that source's $W_{\mathbb{R}}$, without changing its sign. The full Weil functional is $D_{\mathcal{P}}$, whose prime sum is finite for each such h .

Theorem 7.2 (The entire prime-face valuation). *For finite sets of rational primes F, E ,*

$$D_{F \cup E}(h) + D_{F \cap E}(h) = D_F(h) + D_E(h), \quad (30)$$

$$D_{F \cup \{p\}}(h) - D_F(h) = -W_p(h) \quad (p \notin F). \quad (31)$$

*If $\text{supp } g \subseteq [\lambda^{-1}, \lambda]$ and $h = g * g^*$, every prime term with $p > \lambda^2$ vanishes. Thus for $F_\lambda = \{p : p \leq \lambda^2\}$,*

$$D_{F_\lambda}(g * g^*) = D_{\mathcal{P}}(g * g^*). \quad (32)$$

The full Weil positivity criterion therefore concerns these compatible complete finite-place returns. Equations (30)–(31) specify exactly what changes when a prime face is omitted.

Proof. In (29), both pole terms and the archimedean term are independent of F . Each prime occurs with the indicator $1_F(p)$, and $1_{F \cup E} + 1_{F \cap E} = 1_F + 1_E$, proving the first formula. Subtraction gives the second with its minus sign. In a nonzero convolution integrand for $h(x)$, both y and x/y belong to $[\lambda^{-1}, \lambda]$, since the involution preserves that interval. Hence $x \in [\lambda^{-2}, \lambda^2]$. For $p > \lambda^2$ and $m \geq 1$, neither p^m nor p^{-m} belongs to that interval, so every summand in (27) is zero. This proves (32). The classical equivalence of positivity of this full functional and RH is Weil's criterion as stated in the cited source; the finite-face identities themselves have been proved here without assuming RH. $\square$

Theorem 7.3 (Prime-face increments have both signs, even with both poles killed). *Fix a rational prime p , put $a = \log p$, and choose a nonzero real $\phi \in C_c^\infty((-\eta, \eta))$ with $0 < 4\eta < a$. Set*

$$\psi = \phi'' - \frac{1}{4}\phi, \quad f_\theta(u) = \psi(u) + e^{i\theta}\psi(u - a), \quad g_\theta(x) = f_\theta(\log x).$$

*Then $\widehat{g}_\theta(i/2) = \widehat{g}_\theta(-i/2) = 0$, and for $h_\theta = g_\theta * g_\theta^*$,*

$$W_p(h_\theta) = 2(\log p)p^{-1/2} \cos \theta \|\psi\|_{L^2(\mathbb{R})}^2. \quad (33)$$

In particular the change (31) is strictly negative at $\theta = 0$ and strictly positive at $\theta = \pi$. Retaining residue norm p does not make prime-face removal a sign-neutral operation.

Proof. If $\psi = 0$, then $\phi'' = \phi/4$. Multiplying by ϕ and integrating gives $-\int |\phi'|^2 = \frac{1}{4} \int |\phi|^2$, forcing $\phi = 0$, a contradiction. Thus $\|\psi\|^2 > 0$. Integration by parts twice gives $\int \psi(u)e^{\pm u/2} du = 0$. Translation multiplies these integrals by $e^{\pm a/2}$, so both Mellin values remain zero for g_θ .

In logarithmic coordinates, $h_\theta(e^v) = \int f_\theta(u)f_\theta(u - v) du$. At $v = a$, only the right bump of the first factor and the left bump of the second overlap; the other three products vanish since $2\eta < a$. Their integral is $e^{i\theta}\|\psi\|^2$. At $v = -a$ it is its conjugate. For $|m| \geq 2$, the shifts ma exceed the difference support $[-a - 2\eta, a + 2\eta]$, so the correlations are zero. Substitution in the full sum (27) proves (33). The pole conditions hold exactly, without deleting either pole from the definition of D_F . $\square$

This calculation is a signed arithmetic consequence of the prime-face map. It does not assign positivity to the complete functional: the other prime terms and the archimedean term in (29) remain part of that calculation. It supplies an explicit reason that no argument may infer the full Weil sign from an unchanged list of residue cardinalities.

8 The support defect of the Fourier map

Let K be $\mathbb{R}$ or $\mathbb{Q}_p$, with its self-dual additive Haar measure and the same basic character in both Fourier transforms. Write

$$(\mathcal{F}f)(\xi) = \int_K f(x)\psi_K(x\xi) dx, \quad \mathcal{F}^2 f(x) = f(-x).$$

Then $\mathcal{F}f(0) = \int_K f$ and $\int_K \mathcal{F}f = f(0)$ for $f \in \mathcal{S}(K)$. Give $G(K)$ the disjoint-union topology $K \sqcup \{\tau\}$ and define its test space as

$$\mathcal{S}_\tau(K) = \mathcal{S}(K) \oplus \mathbb{C}\delta_\tau.$$

The pullback of the ring reflection and the support-preserving Fourier operator are respectively

$$Jf = (f, f(0)), \quad \mathcal{F}_\tau(f, c) = (\mathcal{F}f, c).$$

These are explicitly defined operators on test spaces; the scalar on the absent point is not confused with the value at supported zero.

Theorem 8.1 (Exact Fourier defect and its stable kernel). *For every original test function f ,*

$$(\mathcal{F}_\tau J - J\mathcal{F})f = \left(0, f(0) - \int_K f(x) dx\right). \quad (34)$$

For the original unweighted scaling $U_a f(x) = f(a^{-1}x)$, its defect scalar is

$$(U_a f)(0) - \int_K U_a f = f(0) - |a|_K \int_K f. \quad (35)$$

The largest scaling-invariant vector subspace on which (34) vanishes is exactly

$$\mathcal{S}(K)_{00} = \{f : f(0) = 0, \int_K f = 0\}. \quad (36)$$

It is also Fourier invariant. The quotient by this subspace is the two-dimensional scaling representation $\mathbb{C} \oplus \mathbb{C}(| \cdot |_K)$, with quotient map $f \mapsto (f(0), \int_K f)$.

Proof. The first components of the two sides of the commutator agree, and their absent components are $f(0)$ and $\mathcal{F}f(0) = \int f$, proving (34). Change variables in the integral to prove (35). If a scaling-invariant subspace lies in the defect kernel, every f in it satisfies $f(0) = \int f$ and $f(0) = |a|_K \int f$ for every a . Choosing $|a|_K \neq 1$ forces both to vanish. Conversely both conditions are preserved by scaling, and Fourier exchanges the two, so (36) has all claimed invariances.

The map to $\mathbb{C}^2$ is onto. Over $\mathbb{R}$, a smooth bump supported away from zero has zero evaluation and may be chosen to have nonzero integral; an independent bump with value one at zero supplies the other coordinate. Over $\mathbb{Q}_p$, use indicators of a compact open ball away from zero and of a compact open ball containing zero. Subtract a suitable multiple of the first function from the second to set its integral to zero while leaving value one. Thus the kernel is exactly (36) and the quotient is two-dimensional. The transformation rule is $(b, m) \mapsto (b, |a|_K m)$, proving its asserted characters. $\square$

The same two maps $f(0)$ and $\int f$ and the characters 1 and $|a|$ occur in Connes's original adelic construction [4, Section III, (6)–(11)]. Here they have arisen by an explicit commutator with the split-zero ring reflection. This proves the connecting map; it does not identify a local construction with the complete global spectral quotient.

8.1 All mixed support faces

For a finite set V of local fields, use the algebraic tensor test space $\bigotimes_{v \in V} \mathcal{S}(K_v)$; every claim below is on this explicitly specified domain. Tensoring the disjoint-sum test spaces gives

$$\bigotimes_{v \in V} \mathcal{S}_\tau(K_v) = \bigoplus_{A \subseteq V} \bigotimes_{v \in A} \mathcal{S}(K_v).$$

The A component means supported coordinates on A and absent coordinates off A . The ring-reflection pullback J_V has component

$$(J_V f)_A = \text{ev}_{V \setminus A} f,$$

where evaluation sets precisely the missing coordinates equal to their supported zeros. Let $\mathcal{F}_V = \bigotimes_v \mathcal{F}_v$ and let $\mathcal{F}_\tau$ act on face A by $\mathcal{F}_A$.

Theorem 8.2 (Every mixed Fourier defect). *For every face $A \subseteq V$,*

$$((\mathcal{F}_\tau J_V - J_V \mathcal{F}_V) f)_A = \mathcal{F}_A \left(\text{ev}_{V \setminus A} f - \int_{\prod_{v \notin A} K_v} f \right). \quad (37)$$

The largest subspace invariant under independent local scalings on which all these defects vanish is

$$\bigcap_{v \in V} \ker(\text{ev}_v) \cap \ker(\text{int}_v). \quad (38)$$

All conditions here are function-valued in the coordinates not removed.

Proof. For a pure tensor the left term is Fourier transform in A of evaluations in its complement. In the right term, evaluation of Fourier transform in a missing coordinate is integration in that coordinate. These operations commute with Fourier transforms in A , proving the formula for pure tensors and then finite sums. No mixed face is omitted.

Taking $A = V \setminus \{v\}$ and using invertibility of $\mathcal{F}_A$ gives $\text{ev}_v f = \text{int}_v f$. Scaling only coordinate v leaves the former unchanged and multiplies the latter by $|a_v|_v$. Invariance for a scale of nonunit absolute value forces both functions to vanish. This proves necessity for each v . Conversely these separate vanishing conditions imply both terms in every proper-face difference vanish, and the full-face difference is zero identically. They are preserved by scaling. Fourier in coordinate v exchanges its evaluation and integration; Fourier in every other coordinate commutes with them. Thus the subspace is also Fourier invariant. $\square$

The independent-scaling domain in Theorem 8.2 is explicit. Restriction to a diagonal rational action has fewer independent scales and can retain additional defect vectors; that restriction must use its actual character relations. The full mixed-face formula (37) remains exact under such a restriction and provides those relations without collapsing the support lattice to one Boolean branch.

9 All adelic boundary faces and the two pole terms

We now restrict the preceding Fourier construction to the actual diagonal rational action. The calculation keeps every mixed face before taking a quotient. The quotient and its contracting homotopy will be computed, rather than inferred from a support count. The local zero and integral maps are those in Connes's original construction [4, Section III, (6)–(11)]. The product-formula defect is also the defect in Sédillot's definition of a topological adelic curve [8, `def:topological_adelic_curve`].

9.1 The exact finite boundary representation

Fix a nonempty finite set F of rational primes, write $d = |F|$, and put

$$V = \{\infty\} \sqcup F, \quad K_\infty = \mathbb{R}, \quad K_p = \mathbb{Q}_p, \quad \Gamma_F = \mathbb{Z}[F^{-1}]^\times = \{\pm \prod_{p \in F} p^{n_p} : n_p \in \mathbb{Z}\}. \quad (\text{AB1})$$

Use the unchanged absolute values $|p|_p = p^{-1}$ and $|a|_\infty = |a|$, and self-dual additive Haar measures. All tensor products of Schwartz spaces in this section are algebraic tensor products over $\mathbb{C}$. Set $\mathcal{T}_V = \bigotimes_{v \in V} \mathcal{S}(K_v)$ and

$$b_v : \mathcal{S}(K_v) \longrightarrow \mathbb{C}^2, \quad f \longmapsto (f(0), \int f), \quad B_V = \bigotimes_{v \in V} \mathbb{C}^2 = \bigoplus_{A \subseteq V} \mathbb{C} e_A. \quad (\text{AB2})$$

The symbol e_A is a basis vector, not the scalar $e = 0_{\mathbb{Z}}$. The coordinate of $b = \bigotimes b_v$ indexed by A integrates the coordinates in A and evaluates at their supported zeros the coordinates in $V \setminus A$:

$$b_A(f) = \text{int}_A \text{ev}_{V \setminus A} f. \quad (\text{AB3})$$

In terms of the preceding split test space it is exactly the integral of the A face of $J_V f$. Thus these scalar boundary coordinates are explicit receivers of the original function-valued faces, with no identification of an absent coordinate with a supported zero.

Theorem 9.1 (Complete diagonal character calculation). *The map b is onto and its kernel is*

$$\ker b = \sum_{v \in V} \left(\mathcal{S}(K_v)_{00} \otimes \bigotimes_{w \neq v} \mathcal{S}(K_w) \right), \quad \mathcal{S}(K_v)_{00} = \ker b_v. \quad (\text{AB4})$$

For the original unweighted action $(U_\gamma f)(x) = f(\gamma^{-1}x)$, $bU_\gamma = D_\gamma b$, where

$$D_\gamma e_A = \chi_A(\gamma) e_A, \quad \chi_A(\gamma) = \prod_{v \in A} |\gamma|_v = \prod_{p \in F} p^{n_p(\mathbf{1}_{\infty \in A} - \mathbf{1}_{p \in A})}. \quad (\text{AB5})$$

The characters are distinct except for $\chi_\emptyset = \chi_V = 1$. Every nonempty proper face has a nontrivial character. Fourier transform induces the map $Ce_A = e_{V \setminus A}$ and $CD_\gamma C = D_\gamma^{-1}$.

Proof. Each b_v is onto by the explicit two-bump construction in Theorem 8.1. Choose functions $h_{v,0}, h_{v,1}$ with $b_v h_{v,j}$ equal to the j th coordinate vector. Then $\mathcal{S}(K_v) = \ker b_v \oplus \text{span}(h_{v,0}, h_{v,1})$. Tensoring these direct sums proves surjectivity and (AB4): every summand having at least one kernel factor maps to zero and the remaining summand maps isomorphically to B_V . This also proves that (AB4) is a sum, with its stated domain; it is not a replacement of the earlier intersection of function-valued conditions.

Evaluation at zero is fixed by U_γ and change of variables multiplies integration at v by $|\gamma|_v$. This proves the first part of (AB5). Since γ is an F -unit, its absolute value at every finite place outside F is one. Its exponents at infinity and at p give the last expression. If $\chi_A = \chi_B$, evaluating at each generator p gives

$$\mathbf{1}_{\infty \in A} - \mathbf{1}_{p \in A} = \mathbf{1}_{\infty \in B} - \mathbf{1}_{p \in B}. \quad (\text{AB6})$$

If the two infinite-place indicators agree, all finite indicators agree and $A = B$. If they differ, say by 1, all finite indicators must differ by 1, so $A = V$ and $B = \emptyset$. The other sign gives the reverse pair. This proves the entire character classification, including nontriviality. Fourier exchanges local evaluation and integration, hence complements A . Finally the product formula gives $\chi_A \chi_{V \setminus A} = \prod_{v \in V} |\gamma|_v = 1$, proving the operator identity. $\square$

The logarithm of (AB5) is the exact restricted product-formula defect

$$\log \chi_A(\gamma) = \sum_{v \in A} \log |\gamma|_v. \quad (\text{AB7})$$

The finite places and the Archimedean place are both present. In particular, the complement is the opposite defect, not a separately discarded boundary.

9.2 Original generators, projectors and higher cohomology

For $p \in F$ set $T_p = D_p$. With respect to the original face basis,

$$\det(I - tT_p) = (1 - pt)^{2^{d-1}}(1 - p^{-1}t)^{2^{d-1}}(1 - t)^{2^d}. \quad (\text{AB8})$$

Indeed, the indicators at ∞, p are respectively $(1, 0)$, $(0, 1)$ or equal; the other $d - 1$ indicators are unrestricted. This proves every multiplicity. Define the following polynomial in the actual generators:

$$\Pi = \prod_{p \in F} \frac{(T_p - pI)(T_p - p^{-1}I)}{(1 - p)(1 - p^{-1})}. \quad (\text{AB9})$$

All denominators are nonzero. On a character line the p factor is one exactly when its p eigenvalue is one, and otherwise is zero. Theorem 9.1 therefore proves

$$\Pi B_V = \mathbb{C}e_\emptyset \oplus \mathbb{C}e_V = B_V^{\Gamma_F}, \quad B_V / \text{span}\{(D_\gamma - I)c\} = \Pi B_V. \quad (\text{AB10})$$

The quotient is identified by $[c] \mapsto \Pi c$, with inverse $c \mapsto [c]$ on ΠB_V . For each proper nonempty face, choosing p where its character differs from one expresses e_A as $(\chi_A(p) - 1)^{-1}(T_p - I)e_A$. This proves the kernel assertion in (AB10). In the Hermitian norm making the original face vectors orthonormal, Π is an orthogonal projector of norm one.

There is also an exact calculation in every cohomological degree. Order $F = \{p_1, \dots, p_d\}$ and let $\theta_1, \dots, \theta_d$ be the dual coordinate basis of $\mathbb{C}^d$. The Koszul cochain complex is

$$\mathcal{C}^j = B_V \otimes \Lambda^j \mathbb{C}^d, \quad \delta = \sum_{i=1}^d (T_{p_i} - I) \otimes (\theta_i \wedge -). \quad (\text{AB11})$$

It computes cohomology of $\mathbb{Z}^d$ on B_V : the group algebra is $\mathbb{C}[t_1^{\pm 1}, \dots, t_d^{\pm 1}]$, and its Koszul resolution at $t_i - 1$ is free and exact. To verify exactness, quotienting successively by the first j of these elements leaves a Laurent polynomial ring in $d - j$ variables, where the next $t_{j+1} - 1$ is not a zero divisor. The one-variable resolution is injective multiplication followed by evaluation at 1; induction by its tensor mapping cone proves the asserted Koszul resolution. Applying module Hom gives (AB11).

On a proper face A , choose $i(A)$ with $a_{i(A), A} = \chi_A(p_{i(A)}) - 1 \neq 0$. Let ι_i be contraction by the vector dual to θ_i and set

$$h_A = a_{i(A), A}^{-1} \iota_{i(A)}. \quad (\text{AB12})$$

The exterior identity $\iota_i(\theta_j \wedge \eta) + \theta_j \wedge \iota_i \eta = \delta_{ij} \eta$ proves $h_A \delta + \delta h_A = I$ on that face. Its coefficient has modulus at most 2, since its denominator is either $p - 1 \geq 1$ or $1 - p^{-1} \geq 1/2$. The direct-sum homotopy, set equal to zero on the two trivial faces, has norm at most 2 and satisfies

$$h\delta + \delta h = I - \Pi \otimes I. \quad (\text{AB13})$$

On the remaining two faces $\delta = 0$. The sign subgroup $\{\pm 1\}$ acts trivially on the entire boundary space, and its invariants functor on complex representations is exact by the averaging projector $(I + D_{-1})/2$. Consequently

$$H^j(\Gamma_F, B_V) \simeq (\mathbb{C}e_\emptyset \oplus \mathbb{C}e_V) \otimes \Lambda^j \mathbb{C}^d, \quad 0 \leq j \leq d, \quad (\text{AB14})$$

with zero groups above d . Equations (AB9)–(AB14) compute how every mixed scalar boundary face reaches rational-invariant cohomology, including an explicit uniformly bounded homotopy for the other faces.

9.3 Pairing, Fourier duality and the actual pole receiver

Define the Hermitian form

$$\mathcal{B}(c, d) = \sum_{A \subseteq V} \overline{c_A} d_{V \setminus A}. \quad (\text{AB15})$$

Complementation is an involution without fixed subsets because V is nonempty. Each pair has matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so the signature is $(2^d, 2^d)$. Formula (AB5) and the product formula prove $\mathcal{B}(D_\gamma c, D_\gamma d) = \mathcal{B}(c, d)$. Moreover $\mathcal{B}(c, d) = \langle c, Cd \rangle$ in the original face norm. The invariant two-dimensional subspace is orthogonal for $\mathcal{B}$ to the sum of proper face lines, and its restriction has signature $(1, 1)$. This is an explicitly constructed boundary form; its positivity is not being identified with positivity of the full arithmetic spectral form.

For the unchanged multiplicative test space $\mathcal{D} = C_c^\infty(\mathbb{R}_{>0})$ use $d^\times x = dx/x$, involution $g^*(x) = \overline{g(x^{-1})}$, and multiplicative convolution. Put

$$\widehat{g}(s) = \int_0^\infty g(x) x^{-is} d^\times x, \quad P(g) = \widehat{g}(-i/2) e_\emptyset + \widehat{g}(i/2) e_V. \quad (\text{AB16})$$

Then the two pole terms in the symmetric explicit formula are exactly

$$\widehat{g * g^*}(i/2) + \widehat{g * g^*}(-i/2) = \mathcal{B}(Pg, Pg) = 2 \operatorname{Re} \left(\widehat{g}(-i/2) \overline{\widehat{g}(i/2)} \right). \quad (\text{AB17})$$

Indeed Fubini's theorem applies to compact supports, $\widehat{g * g^*}(s) = \widehat{g}(s) \overline{\widehat{g}(\overline{s})}$, and substitution of the two imaginary arguments gives (AB17). These are precisely the pole conventions used in [5, sectweilpos, wp].

The map P is onto. Choose a nonnegative nonzero $g_0 \in \mathcal{D}$. Its two moments $a_- = \widehat{g_0}(-i/2)$ and $a_+ = \widehat{g_0}(i/2)$ are positive. For $t > 0$, $t \neq 1$, let $g_t(x) = g_0(x/t)$. The two moment columns of g_0, g_t have determinant $a_- a_+ (t^{1/2} - t^{-1/2}) \neq 0$. This proves surjectivity and the exact sequence

$$0 \longrightarrow \{g : \widehat{g}(-i/2) = \widehat{g}(i/2) = 0\} \longrightarrow \mathcal{D} \xrightarrow{P} \Pi B_V \longrightarrow 0. \quad (\text{AB18})$$

The relation to the original unweighted adelic scaling can also be stated without changing any test function: define the explicitly different representation $M_t g(x) = t^{1/2} g(x/t)$ on $\mathcal{D}$. Direct change of variables gives $P(M_t g) = \operatorname{diag}(1, t) P(g)$. On ΠB_V , an idele of modulus t acts by exactly $\operatorname{diag}(1, t)$. Thus P is an intertwiner for these specified representations. The $t^{1/2}$ is part of this stated intertwining map, not an omitted factor in U_γ or in (AB17).

Finally, the chosen $h_{v,j}$ define a concrete section $j_B : B_V \rightarrow \mathcal{T}_V$ by $j_B(e_A) = \bigotimes_{v \in A} h_{v,1} \otimes \bigotimes_{v \notin A} h_{v,0}$, with factors in their fixed order. Thus $b j_B = I$ and

$$b(j_B P(g)) = P(g), \quad \Pi b(j_B P(g)) = P(g). \quad (\text{AB19})$$

This supplies the exact lift from the pole receiver to original local test functions. The section depends on its explicitly chosen functions and is not asserted to be equivariant. The quotient, its characters and its pairing are independent of that choice.

9.4 Passing to every prime with the original reference functions

The finite-place calculation has a specific global continuation. If $F \subset F'$ and $V' = \{\infty\} \sqcup F'$, the original restricted Schwartz tensor product uses $\mathbf{1}_{\mathbb{Z}_p}$ at $p \in F' \setminus F$. Its evaluation and integral are both one for the fixed Haar measure. The induced injective boundary map is therefore

$$j_{FF'} : B_V \longrightarrow B_{V'}, \quad c \longmapsto c \otimes \bigotimes_{p \in F' \setminus F} (e_0 + e_1). \quad (\text{AB20})$$

There is no omitted scalar in this formula. The ordering of the added tensor factors is the ordering of the places. These maps compose exactly and commute with the original Fourier

$$\begin{array}{ccc}
\begin{array}{c} \mathcal{T}_V \\ \text{original local test functions} \end{array} & \xrightarrow{b \text{ (AB3)}} & \begin{array}{c} B_V = \bigoplus_{A \subseteq V} \mathbb{C} e_A \\ \text{all } 2^{d+1} \text{ mixed faces} \end{array} \\
\uparrow j_B P & & \downarrow \Pi \text{ (AB9)} \\
\begin{array}{c} C_c^\infty(\mathbb{R}_{>0}) \\ \text{original multiplicative tests} \end{array} & \xrightarrow{P \text{ (AB16)}} & \begin{array}{c} \Pi B_V = \mathbb{C} e_\emptyset \oplus \mathbb{C} e_V \\ \text{two pole characters} \end{array}
\end{array}$$

Figure 4: The mixed boundary calculation, with its actual maps. The square commutes by (AB19). Every proper face has the character (AB5) and the contracting homotopy (AB12); both surviving faces enter the pole pairing (AB17). This diagram concerns the computed boundary quotient, not an asserted replacement for the full zeta cohomology.

involutions. They are equivariant for Γ_F , since its elements have absolute value one at each added prime. Let

$$B_{\text{all}} = \varinjlim_F B_{\{\infty\} \cup F}. \quad (\text{AB21})$$

The nonempty finite F form a cofinal set, so the preceding results apply. The whole group $\mathbb{Q}^\times$ acts on this direct limit: for any γ , first enlarge F to include its prime factors and then use (AB5). Equivariance at every further enlargement proves independence of choices and the group law. The restricted algebraic Schwartz space

$$\mathcal{T}_{\text{all}} = \mathcal{S}(\mathbb{R}) \otimes \bigotimes_p' (\mathcal{S}(\mathbb{Q}_p), \mathbf{1}_{\mathbb{Z}_p}) \quad (\text{AB22})$$

maps onto B_{all} by the b_v , because every finite-level map is onto and their transition maps agree by (AB20). This is the original reference-function convention of [4, Section III].

Theorem 9.2 (The global boundary quotient and its invariant vectors). *Let $\pi_F : B_V \rightarrow \mathbb{C}^2$ take the empty and full coordinates, in that order. The maps π_F induce a surjective $\mathbb{Q}^\times$ -invariant map $\pi_{\text{all}} : B_{\text{all}} \rightarrow \mathbb{C}^2$, and*

$$(B_{\text{all}})_{\mathbb{Q}^\times} \xrightarrow[\pi_{\text{all}}]{\sim} \mathbb{C}^2, \quad (B_{\text{all}})^{\mathbb{Q}^\times} = 0. \quad (\text{AB23})$$

The composite from $\mathcal{T}_{\text{all}}$ is exactly $f \mapsto (f(0), \int f)$ with its original adelic measure. The two-dimensional quotient consequently has no equivariant linear section into B_{all} .

Proof. Each endpoint coordinate of (AB20) is the same endpoint coordinate of c , so $\pi_{F'} j_{FF'} = \pi_F$. An element $\gamma \in \mathbb{Q}^\times$ can be tested in a stage containing all its prime factors. Both endpoint characters there equal one by the product formula, proving invariance. Surjectivity holds at any finite stage.

If $\pi_{\text{all}} c = 0$, represent c at a finite stage. Its two endpoint coordinates are zero, so it is a sum of proper-face basis vectors. For each such vector the explicit identity following (AB10) expresses it as $(D_p - I)$ of a vector at that same stage. Hence c is in the span of the coinvariant relations. Conversely invariance of π_{all} kills every such relation. This proves the quotient assertion.

For the invariant-vector assertion, let a nonzero c be represented in B_V . Choose a prime $p \notin F$ and view it in the next stage through $j_{F, F \cup \{p\}}$. For each nonzero coefficient c_A , the two extended faces $(A, 0)$ and $(A, 1)$ both have that same nonzero coefficient. Under p , their eigenvalues are respectively $p^{1_{\infty \in A}}$ and $p^{1_{\infty \in A} - 1}$. Exactly one equals one; the other is p or p^{-1} , and therefore differs from one. Different A yield different coordinate positions, so these nonzero changes cannot cancel. Thus $(D_p - I) j_{F, F \cup \{p\}} c \neq 0$. All later maps are injective, so this remains

nonzero in the direct limit. No nonzero vector is invariant. If an equivariant section from the trivial representation $\mathbb{C}^2$ existed, its image would consist of invariant vectors, contradicting this result. Finally both evaluation and integral multiply by one on each reference factor, proving the formula on pure tensors and then on finite sums. $\square$

Both pairings can be tracked through these same maps. From (AB15), for $r = |F' \setminus F|$ one obtains

$$\mathcal{B}_{V'}(j_{FF'}c, j_{FF'}d) = 2^r \mathcal{B}_V(c, d), \quad \mathcal{B}_{V'}(\Pi_{F'}j_{FF'}c, \Pi_{F'}j_{FF'}d) = \mathcal{B}_V(\Pi_F c, \Pi_F d). \quad (\text{AB24})$$

For the first equality each added local factor $(1, 1)$ has value two under its complement pairing, so tensoring gives exactly 2^r . For the second equality both sides equal $\overline{c_\emptyset}d_V + \overline{c_V}d_\emptyset$. Consequently the pole pairing (AB17) is compatible at every stage, whereas the full mixed pairing has the explicitly calculated factor 2^r . No unrecorded change of measure or scalar has been made.

This establishes a global quotient from the actual local test spaces. It also identifies exactly what passage to all primes changes: the finite invariant endpoint vectors do not survive as global invariant vectors, while the two pole coordinates survive as a quotient. The kernel of that quotient, including its original arithmetic cohomology, is retained as a separate mathematical object throughout the construction.

10 The mixed-support extension and its arithmetic determinant

The full mixed boundary constructed above contains more information than its two-dimensional quotient. We now compute a nonzero extension class, its local differentials, and their exact contribution to the prime terms of Weil's formula. The primes and their absolute values are those of the original arithmetic stratum $V(e) \subset \text{Spec } G(\mathbb{Z})$. All tests, Haar measures and the prime terms W_p remain those of Section 7. The classical trace-formula source is Connes [4, Section III and Section V], and the exact explicit-formula conventions are those read in [5, **wp**, **arch**]. The extension and determinant calculations below are proved here.

10.1 A nonzero class carried by all the mixed faces

For a finite set F of primes, including $F = \emptyset$, write $B_F = \bigotimes_{v \in \{\infty\} \sqcup F} \mathbb{C}^2$ as in (AB2). Use exactly the bonding maps

$$j_{FF'}c = c \otimes \bigotimes_{p \in F' \setminus F} (e_0 + e_1). \quad (\text{ME1})$$

They come from the original reference function $\mathbf{1}_{\mathbb{Z}_p}$, whose value at zero and integral both equal one. Put $B = \varinjlim_F B_F$. For $\gamma \in \mathbb{Q}^\times$, choose F containing every prime factor of its numerator and denominator. The action on B_F is $D_\gamma e_A = \chi_A(\gamma)e_A$ with (AB5), and enlarging F makes this an action on B .

There is a canonical complex-parameter deformation of this particular action: because all its eigenvalues are positive real numbers, define

$$D_\gamma(s)e_A = \exp(s \log \chi_A(\gamma))e_A, \quad s \in \mathbb{C}. \quad (\text{ME2})$$

The logarithm here is the real logarithm of a positive number. Thus $D_\gamma(1) = D_\gamma$, no eigenvalue or absolute value is changed, and $D_{\gamma\eta}(s) = D_\gamma(s)D_\eta(s)$ follows from the real logarithm's additivity. The new coordinates in (ME1) have character one on Γ_F , proving compatibility for every s . Denote this module by $B(s)$. Fourier complementation satisfies $CD_\gamma(s)C = D_{\gamma^{-1}}(s)$.

The two endpoint coordinates give a surjection $\epsilon : B(s) \rightarrow \mathbb{C}^2$, independent of F and s . Define $K(s) = \ker \epsilon$. The target has the trivial rational action, by the product formula. Hence there is an exact sequence of actual modules

$$0 \longrightarrow K(s) \longrightarrow B(s) \xrightarrow{\epsilon} \mathbb{C}^2 \longrightarrow 0. \quad (\text{ME3})$$

Let $b_0, b_1 \in B(s)$ be the two basis vectors at $F = \emptyset$; they give the specified vector-space section of ϵ .

Theorem 10.1 (Explicit global mixed-support extension). *For $\Re s > 0$, $B(s)^{\mathbb{Q}^\times} = 0$, and (ME3) has no equivariant section. Its two connecting cocycles are*

$$c_j(\gamma; s) = D_\gamma(s)b_j - b_j \in K(s), \quad j = 0, 1. \quad (\text{ME4})$$

They determine linearly independent elements of $H^1(\mathbb{Q}^\times, K(s))$. For γ with prime factors in F , their complete coordinates are

$$c_0(\gamma; s) = \sum_{A \subseteq F} \left(\prod_{p \in A} |\gamma|_p^s - 1 \right) e_A, \quad (\text{ME5})$$

$$c_1(\gamma; s) = \sum_{A \subseteq F} \left(|\gamma|_\infty^s \prod_{p \in A} |\gamma|_p^s - 1 \right) e_{\{\infty\} \cup A}. \quad (\text{ME6})$$

In particular, on the original four faces at $\{\infty, p\}$,

$$c_0(p; s) = (p^{-s} - 1)e_{\{p\}}, \quad c_1(p; s) = (p^s - 1)e_{\{\infty\}}. \quad (\text{ME7})$$

These are nonzero mixed-face coordinates, not endpoint coordinates.

Proof. At stage F , (ME1) sends b_0 to $\sum_{A \subseteq F} e_A$ and b_1 to $\sum_{A \subseteq F} e_{\{\infty\} \cup A}$. Acting diagonally and subtracting gives (ME5)–(ME6). Their empty and full coordinates vanish, the latter by the full product formula, so they lie in $K(s)$.

If $z \neq 0$ is represented in B_F , choose a prime $p \notin F$. Its image in $B_{F \cup \{p\}}$ has equal coefficients z_A on A and $A \cup \{p\}$. If $\infty \notin A$, the first has eigenvalue one under $D_p(s)$ and the second has eigenvalue p^{-s} . If $\infty \in A$, the eigenvalues are p^s and one. Since $\Re s > 0$, the nonunit eigenvalue never equals one. At least one nonzero coefficient is therefore changed. Every bonding map is injective, since tensoring by the nonzero vector $e_0 + e_1$ is injective. Thus z cannot become invariant at a later stage either. This proves the assertion about invariants and excludes an equivariant section.

Multiplying the operators gives $c_j(\gamma\eta) = D_\gamma(s)c_j(\eta) + c_j(\gamma)$, the cocycle identity. Suppose $a_0c_0 + a_1c_1$ is a coboundary $D_\gamma(s)k - k$ with $k \in K(s)$. Then $a_0b_0 + a_1b_1 - k$ is invariant. The preceding argument forces it to vanish, and applying ϵ gives $(a_0, a_1) = 0$. This proves linear independence, using the definition of group cohomology rather than an assumed vanishing or comparison theorem. Specializing (ME5)–(ME6) to $F = \{p\}$ proves (ME7). $\square$

At the original value $s = 1$, the complement pairing of (AB15) gives

$$\mathcal{B}(c_0(p; 1), c_1(p; 1)) = (p^{-1} - 1)(p - 1) = -(\sqrt{p} - 1/\sqrt{p})^2. \quad (\text{ME8})$$

This follows by pairing the two complementary face vectors, each with coefficient one in the original pairing. It is an evaluated pairing of the extension, not an assignment of a Weil sign to every test.

10.2 The determinant of the actual local differential

Restrict to the infinite cyclic subgroup generated by p and the invariant mixed-face subspace

$$M_p = \mathbb{C}e_{\{p\}} \oplus \mathbb{C}e_{\{\infty\}} \subset B_{\{p\}}. \quad (\text{ME9})$$

The standard two-term cochain complex in this case is obtained directly from the free resolution $0 \rightarrow \mathbb{C}[t, t^{-1}] \xrightarrow{t-1} \mathbb{C}[t, t^{-1}] \rightarrow \mathbb{C} \rightarrow 0$. The first map is injective because a Laurent polynomial

ring is a domain; the cokernel is evaluation at one. Applying module Hom proves, in the displayed order of basis vectors, that the differential is

$$d_p(s) = D_p(s) - I = \begin{pmatrix} p^{-s} - 1 & 0 \\ 0 & p^s - 1 \end{pmatrix} : M_p \longrightarrow M_p. \quad (\text{ME10})$$

It is the same matrix that evaluates the two cocycles (ME7) on the generator. It is invertible for $\Re s > 0$. Its determinant, with both orientations and the original ordered basis retained, is

$$\det d_p(s) = (p^{-s} - 1)(p^s - 1) = -p^s(1 - p^{-s})^2. \quad (\text{ME11})$$

For clarity, the determinant-line calculation can be specified without a torsion convention: in a one-dimensional complex $L^0 \xrightarrow{a} L^1$ with named bases, the map $\det L^0 \otimes (\det L^1)^* \rightarrow \mathbb{C}$ induced by the differential sends the named tensor to a . Its inverse scalar is a^{-1} . For the finite-place line in (ME10) this scalar is $(p^{-s} - 1)^{-1} = -(1 - p^{-s})^{-1}$. Thus the exact arithmetic Euler factor is

$$Z_p(s) = \frac{1}{1 - p^{-s}} = -(\det d_p(s)|_{\mathbb{C}e_{\{p\}}})^{-1}. \quad (\text{ME12})$$

The minus sign is displayed, not absorbed into a basis change. The complementary line has inverse determinant $(p^s - 1)^{-1} = p^{-s}Z_p(s)$. Their product is therefore $-p^{-s}Z_p(s)^2$, in agreement with (ME11).

The extraction of this local line has an exact chain map from all the finite-place data. Fix $p \in F$. Define $r_{F,p} : B_F(s) \rightarrow B_{\{p\}}(s)$ by retaining the coefficients of e_A only when $A \subseteq \{\infty, p\}$ and setting every spectator coordinate to the evaluation component. Then

$$r_{F,p}D_p(s) = D_p(s)r_{F,p}, \quad r_{F,p}j_{\{p\},F} = I. \quad (\text{ME13})$$

Indeed $|p|_q = 1$ at every spectator prime q , and exactly the empty spectator choice survives the projection of the tensor in (ME1). Projection of the full Koszul complex (AB11) to degrees zero and one along the p direction, followed by $r_{F,p}$, is a cochain map to the cyclic complex: all other exterior generators project to zero. Finally projection onto the two mixed faces and then onto the finite-place line commutes with $D_p(s) - I$. These identities prove the whole chain map that extracts (ME12). They do not identify this determinant with the torsion of the entire higher-rank Koszul complex.

Proposition 10.2 (The full prime product in its domain of convergence). *For $\Re s > 1$, the product of the factors (ME12), one for each original finite prime, converges absolutely and equals*

$$\prod_p Z_p(s) = \sum_{n \geq 1} n^{-s} = \zeta(s). \quad (\text{ME14})$$

Proof. Write $\sigma = \Re s > 1$. Then $\sum_p \sum_{m \geq 1} p^{-m\sigma}/m \leq (1 - 2^{-\sigma})^{-1} \sum_{n \geq 2} n^{-\sigma} < \infty$, so the logarithms of the product converge absolutely. For a finite set of primes the geometric-series expansion is absolutely convergent and enumerates exactly the positive integers with prime factors in that set, once each by unique prime factorization. Its omitted terms are dominated by the summable series $\sum n^{-\sigma}$. Passing to all primes proves (ME14). This derivation concerns the convergent Euler product; it makes no assertion about zeros in its meromorphic continuation. $\square$

10.3 An exact receiver in the original Weil prime term

Define the logarithmic derivative of the same finite-place differential by

$$L_p(s) = -\frac{d}{ds} \log Z_p(s) = \frac{(\log p)p^{-s}}{1 - p^{-s}} = (\log p) \sum_{m \geq 1} p^{-ms}, \quad \Re s > 0. \quad (\text{ME15})$$

There is no branch issue: the first expression means $-Z'_p/Z_p$, and the last equality is an absolutely convergent geometric series. Let $g \in C_c^\infty(\mathbb{R}_{>0})$, let $h = g * g^*$ use the original multiplicative Haar measure dx/x , and put $\widehat{g}(t) = \int_0^\infty g(x)x^{-it}dx/x$. Thus $\widehat{h}(t) = |\widehat{g}(t)|^2$ for real t .

Theorem 10.3 (Mixed-differential formula for every prime-power term). *The exact prime term of the original explicit formula is*

$$\begin{aligned} W_p(h) &= (\log p) \sum_{m \geq 1} p^{-m/2} (h(p^m) + h(p^{-m})) \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 (L_p(\tfrac{1}{2} + it) + L_p(\tfrac{1}{2} - it)) dt. \end{aligned} \quad (\text{ME16})$$

Consequently, with $r = p^{-1/2}$ and $P_r(\theta) = (1 - r^2)/(1 - 2r \cos \theta + r^2)$,

$$W_p(g * g^*) = \frac{\log p}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 (P_{p^{-1/2}}(t \log p) - 1) dt. \quad (\text{ME17})$$

Both pole terms, every prime power and the Archimedean term are retained in the resulting finite-place Weil functional:

$$\begin{aligned} D_F(g * g^*) &= \mathcal{B}(Pg, Pg) - W_{\infty}(g * g^*) \\ &\quad + \sum_{p \in F} (\log p) \left(\|g\|_{L^2(dx/x)}^2 - \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 P_{p^{-1/2}}(t \log p) dt \right). \end{aligned} \quad (\text{ME18})$$

Here P is precisely the two-pole map (AB16); no new source or metric has been substituted for the original test.

Proof. In logarithmic coordinates the Fourier inversion formula is $h(e^u) = (2\pi)^{-1} \int \widehat{h}(t) e^{itu} dt$. The two series in (ME15), evaluated at $1/2 \pm it$, have total absolute sum at most $2(\log p)p^{-1/2}/(1 - p^{-1/2})$ uniformly in t . Since $\widehat{h}$ is a Schwartz function, absolute integrability justifies interchanging the integrals and the series. Fourier inversion at $u = \pm m \log p$ proves (ME16), including the factor $1/(2\pi)$. For $z = re^{-i\theta}$ direct multiplication gives

$$\frac{z}{1-z} + \frac{\bar{z}}{1-\bar{z}} = \frac{2r \cos \theta - 2r^2}{1 - 2r \cos \theta + r^2} = P_r(\theta) - 1. \quad (\text{ME19})$$

This proves (ME17). Finally the full pole pairing was computed in (AB17), D_F is the original pole term minus the Archimedean term and minus $\sum_{p \in F} W_p$, and Fourier Plancherel in the same logarithmic coordinate gives $(2\pi)^{-1} \int |\widehat{g}|^2 = \|g\|^2$. Substitution proves (ME18). $\square$

Each Poisson energy in (ME18) is positive for $g \neq 0$: its integrand is nonnegative, P_r is strictly positive, and Plancherel excludes a zero Fourier transform for nonzero g . Its subtraction from $\|g\|^2$ has the exact sign already tested by the two-bump examples of Theorem 7.3. Thus the formula retains the competing energies; it does not replace their difference by a positivity assertion. If $\text{supp } g \subset [\lambda^{-1}, \lambda]$ and $p > \lambda^2$, every $h(p^{\pm m})$ vanishes, so (ME16) proves that the corresponding Poisson energy is *exactly* $\|g\|^2$. Consequently (ME18) becomes the full prime return at $F = \{p : p \leq \lambda^2\}$, without summing separately divergent norm and Poisson-energy series.

The use of the new support geometry in this section is specific: a global nonsplit mixed extension supplies a local differential; its determinant and logarithmic derivative supply the Euler factor and the complete prime-power term. Ordinary endpoint coinvariants alone would not retain that differential. The remaining calculation for the RH programme is the pairing of the full interior cohomology with these retained relative terms and the Archimedean term in (ME18). No coefficient of an earlier large-degree determinant has been used to stand in for that pairing.

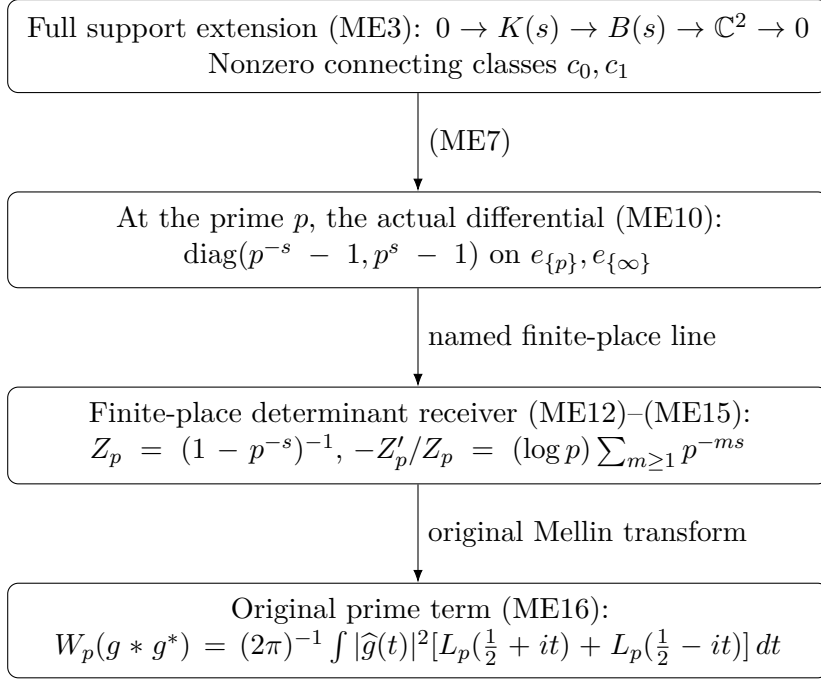


Figure 5: The calculated route from all support faces to each arithmetic prime-power contribution. The complementary line and its minus sign remain in (ME11)–(ME12); the local extraction is the explicit chain map (ME13). Theorem 10.3 proves the last arrow in the conventions of Connes [4, 5]. These are exact maps and equalities, not a diagram asserting the still-unproved full Weil positivity.

11 Relative cohomology at the full supported prime

11.1 The full space and the receiving categories

Let X be a nonempty irreducible topological space. Put $Y = X^+ = X \sqcup \{\eta_\tau\}$. Its nonempty opens are the singleton $\{\eta_\tau\}$ and $U^+ = U \cup \{\eta_\tau\}$ for nonempty open $U \subset X$. Write $i : X \hookrightarrow Y$ and $j : \{\eta_\tau\} \hookrightarrow Y$. All nonempty opens of X intersect. In the first four parts of this chapter, sheaves have values in modules over a fixed commutative coefficient ring Λ ; taking $\Lambda = \mathbb{Z}$ gives abelian sheaves. These are sheaves on the entire underlying prime space. They are not automatically additive-group modules over its semiring structure sheaf. Section 11.4 proves the exact distinction and the resulting morphisms.

For a module B , let cB be the sheaf on X with value B on every nonempty open and identity restriction between such opens. This is a sheaf because any two members of a nonempty open cover intersect, so compatible locally constant values agree. Define

$$g\mathcal{A} = \varinjlim_{\emptyset \neq U \subset X} \mathcal{A}(U), \quad \epsilon_{\mathcal{A}} : \mathcal{A} \longrightarrow c(g\mathcal{A}). \quad (\text{RP1})$$

The indexing order is restriction to smaller nonempty opens. It is filtered by irreducibility. If X has a generic point ξ , then every nonempty open contains ξ , and $g\mathcal{A} = \mathcal{A}_\xi$.

Lemma 11.1 (Exact generic-section adjunction). *The functors g and c are exact, g is left adjoint to c , and $gcB = B$. In particular c preserves injective modules as injective sheaves, and*

$$R\Gamma(X, cB) = B[0]. \quad (\text{RP2})$$

Proof. A morphism $\mathcal{A} \rightarrow cB$ is a family of maps $\mathcal{A}(U) \rightarrow B$ agreeing under all restrictions. The filtered colimit is exactly their universal receiving module, proving the adjunction. The assertions

$gcB = B$ and exactness of c follow from its displayed sections or stalks. To check surjectivity of g on an epimorphism of sheaves, represent a generic section by a section on a nonempty U . Sheaf surjectivity supplies a lift on an open cover of U ; choose a nonempty member and take its generic germ. For exactness in the middle, if a section has zero image as a generic germ, restrict to a nonempty smaller open on which its image is zero, then use the kernel sheaf. This proves exactness of g .

The adjunction and exactness of g imply that c takes injectives to injectives: applying $\text{Hom}(-, cI) = \text{Hom}(g(-), I)$ to an exact sequence is exact if I is injective. Apply c to an injective resolution of B . It remains an injective resolution, and its sections on X are the original resolution. This proves (RP2). $\square$

Theorem 11.2 (All sheaves and all gluing maps). *The category $\text{Sh}(Y, \Lambda)$ is equivalent to the category of triples*

$$\mathcal{F} = (\mathcal{A}, B, \phi), \quad \phi : \mathcal{A} \longrightarrow cB, \quad (\text{RP3})$$

where $\mathcal{A} \in \text{Sh}(X, \Lambda)$ and $B \in \Lambda\text{-Mod}$. Its sections are $\mathcal{F}(U^+) = \mathcal{A}(U)$ and $\mathcal{F}(\{\eta_\tau\}) = B$; restriction to the singleton is ϕ_U . A morphism is a pair (a, b) satisfying $\phi'a = c(b)\phi$. Kernels and cokernels are computed componentwise. Equivalently, the gluing is an arbitrary map $\bar{\phi} : g\mathcal{A} \rightarrow B$.

Proof. For a sheaf $\mathcal{F}$ on Y put $B = \mathcal{F}(\{\eta_\tau\})$ and $\mathcal{A}(U) = \mathcal{F}(U^+)$ on nonempty opens. If U is covered by nonempty opens U_a , their pairwise intersections are nonempty, and $(U_a \cap U_b)^+ = U_a^+ \cap U_b^+$. Thus the sheaf axiom for $\mathcal{F}$ is exactly the sheaf axiom for these values on X . Restriction to the singleton gives ϕ . This $\mathcal{A}$ is $i^*\mathcal{F}$, since U^+ is the smallest open of Y containing U and no further sheafification changes the displayed sheaf.

Conversely the displayed formulas for a triple define a sheaf. On a cover of U^+ , sections on nonempty arithmetic parts glue by the sheaf axiom for $\mathcal{A}$; their images in B agree, and hence also agree with any section on a singleton member of the cover. The singleton case is immediate. The two constructions are inverse. The compatibility equation is precisely restriction compatibility. Since c is exact, both a componentwise kernel and a componentwise cokernel inherit the indicated arrow and satisfy the categorical universal property. The last assertion is Lemma 11.1. $\square$

11.2 The six elementary functors and their derived terms

Theorem 11.3 (Explicit recollement). *In triple notation the functors are*

$$\begin{aligned} i^*(\mathcal{A}, B, \phi) &= \mathcal{A}, & i_*\mathcal{A} &= (\mathcal{A}, 0, 0), & i^!(\mathcal{A}, B, \phi) &= \ker \phi, \\ j^*(\mathcal{A}, B, \phi) &= B, & j_!\mathcal{A} &= (0, B, 0), & j_*\mathcal{A} &= (cB, B, \text{id}). \end{aligned} \quad (\text{RP4})$$

The adjunctions are $i^* \dashv i_* \dashv i^!$ and $j_! \dashv j^* \dashv j_*$. All displayed functors except possibly $i^!$ are exact. There is in addition an exact left adjoint to i^* ,

$$U\mathcal{A} = (\mathcal{A}, g\mathcal{A}, \epsilon_{\mathcal{A}}), \quad U \dashv i^*. \quad (\text{RP5})$$

For any triple, there is a canonical short exact sequence

$$0 \longrightarrow j_!\mathcal{A} \longrightarrow \mathcal{F} \longrightarrow i_*\mathcal{A} \longrightarrow 0. \quad (\text{RP6})$$

Proof. A map $\mathcal{F} \rightarrow i_*\mathcal{C}$ is precisely a map $\mathcal{A} \rightarrow \mathcal{C}$. A map $i_*\mathcal{C} \rightarrow \mathcal{F}$ is a map $\mathcal{C} \rightarrow \mathcal{A}$ whose composite with ϕ is zero, hence a map to $\ker \phi$. A map $j_!\mathcal{D} \rightarrow \mathcal{F}$ is any map $D \rightarrow B$. A map $\mathcal{F} \rightarrow j_*\mathcal{D}$ has its arithmetic component forced to be $c(b)\phi$, so is any map $b : B \rightarrow D$. These calculations prove the four adjunctions and all formulas. Exactness follows from componentwise exactness and exactness of c .

A map $U\mathcal{C} \rightarrow \mathcal{F}$ has arithmetic part $a : \mathcal{C} \rightarrow \mathcal{A}$ and must have generic part the unique map $g\mathcal{C} \rightarrow B$ corresponding to ϕa under the adjunction of Lemma 11.1. This proves $U \dashv i^*$; its exactness follows from exactness of g . Finally the maps in (RP6) have components $(0, \text{id}_B)$ and $(\text{id}_{\mathcal{A}}, 0)$, proving its exactness. $\square$

Complexes use cohomological degrees: $B[-1]$ has B in degree one. All derived statements below can be read in the bounded-below derived categories. No assertion requires an unbounded resolution convention.

Theorem 11.4 (Derived closed restriction and relative cohomology). *For $\mathcal{F} = (\mathcal{A}, B, \phi)$ one has canonical identifications*

$$Ri^!\mathcal{F} \simeq [\mathcal{A} \xrightarrow{\phi} cB] = \text{Cone}(\phi)[-1], \quad (\text{RP7})$$

where the two entries are in degrees zero and one. Thus $i^!\mathcal{F} = \ker \phi$, $R^1i^!\mathcal{F} = \text{coker } \phi$, and $R^n i^!\mathcal{F} = 0$ for $n \geq 2$. Furthermore

$$R\Gamma(Y, \mathcal{F}) = R\Gamma(X, \mathcal{A}), \quad R\Gamma_X(Y, \mathcal{F}) \simeq \text{Cone}(R\Gamma(X, \mathcal{A}) \rightarrow B)[-1]. \quad (\text{RP8})$$

Here $R\Gamma_X(Y, -)$ means $R\Gamma(X, Ri^!(-))$, the derived functor of sections with support in the closed subset X . The corresponding localization triangle is

$$i_* Ri^!\mathcal{F} \longrightarrow \mathcal{F} \xrightarrow{(\phi, \text{id})} j_* B \longrightarrow i_* Ri^!\mathcal{F}[1]. \quad (\text{RP9})$$

Proof. There are enough injectives in the triple category. Indeed embed $\mathcal{A}$ into an injective sheaf I and B into an injective module J . The map

$$(\mathcal{A}, B, \phi) \longrightarrow i_* I \oplus j_* J$$

with arithmetic component $(\mathcal{A} \hookrightarrow I, c(B \hookrightarrow J)\phi)$ and generic component $B \hookrightarrow J$ is injective. The target is injective because i_* and j_* are right adjoints of exact functors.

If $\mathcal{I} = (\mathcal{A}_I, B_I, \phi_I)$ is injective, then ϕ_I is split surjective. To see this, the monomorphism $j_! B_I \hookrightarrow j_* B_I$ and the map $j_! B_I \rightarrow \mathcal{I}$ with generic part the identity extend to a map $j_* B_I \rightarrow \mathcal{I}$. Its arithmetic part is a right inverse of ϕ_I . Also $\ker \phi_I$ is injective on X because $i^!$ is right adjoint to the exact functor i_* . The functors i^* and j^* preserve injectives: their respective left adjoints U and $j_!$ are exact.

For an injective resolution of $\mathcal{F}$, therefore, there is a termwise split exact sequence of complexes $0 \rightarrow i^!\mathcal{I}^\bullet \rightarrow i^*\mathcal{I}^\bullet \rightarrow c(j^*\mathcal{I}^\bullet) \rightarrow 0$. The latter two complexes resolve $\mathcal{A}$ and cB by injectives. Its kernel computes $Ri^!$ and is the indicated shifted cone. This proves (RP7). The same termwise calculation in the triple category proves (RP9).

Ordinary global sections equal $\Gamma(X, \mathcal{A})$, and i^* preserves injectives, proving the first identification in (RP8). Apply $R\Gamma(X, -)$ to (RP7) and use (RP2) for the second. At degree zero the kernel is exactly the sections whose restriction to the open point is zero, which proves the stated support interpretation. $\square$

Corollary 11.5 (A nonzero relative object and its connecting map). For every nonzero B , the sheaf $j_! B$ is nonzero and satisfies

$$R\Gamma(Y, j_! B) = 0, \quad Ri^! j_! B = cB[-1], \quad R\Gamma_X(Y, j_! B) = B[-1]. \quad (\text{RP10})$$

The sequence

$$0 \longrightarrow j_! B \longrightarrow j_* B \longrightarrow i_* cB \longrightarrow 0 \quad (\text{RP11})$$

does not split. Its localization boundary identifies the actual module B at the new point with $H_X^1(Y, j_! B)$, with the sign convention fixed by the order of maps in this exact sequence.

Proof. The three calculations are substitutions of $(0, B, 0)$ into Theorem 11.4; its stalk at the new point proves it is nonzero. A section of the surjection in (RP11) would be a triple morphism $(cB, 0, 0) \rightarrow (cB, B, \text{id})$ with arithmetic part the identity. The compatibility equation would instead force that part to be zero. This contradicts $B \neq 0$. In the localization long exact sequence, both global cohomology groups of $j_! B$ adjoining this boundary vanish, so the boundary $B \rightarrow H_X^1(Y, j_! B)$ is the stated isomorphism. $\square$

11.3 Cross extensions, relative traces, and exact pairing maps

Theorem 11.6 (The new cross-extension groups). *For $n \geq 1$,*

$$\mathrm{Ext}_Y^n(i_*\mathcal{A}, j_!B) = \mathrm{Ext}_X^{n-1}(\mathcal{A}, cB) = \mathrm{Ext}_\Lambda^{n-1}(g\mathcal{A}, B), \quad \mathrm{Hom}_Y(i_*\mathcal{A}, j_!B) = 0. \quad (\mathrm{RP12})$$

All groups $\mathrm{Ext}_Y^n(j_!B, i_\mathcal{A})$ vanish, including $n = 0$. Under the degree-one identification in (RP12), the class of (RP6) is exactly $\bar{\phi}$. In particular the universal coupling $U\mathcal{A}$ represents $\mathrm{id}_{g\mathcal{A}}$ and is nonsplit if $g\mathcal{A} \neq 0$.*

Proof. The exact functor i_* and its right adjoint $i^!$ give, using injective resolutions,

$$\mathrm{RHom}_Y(i_*\mathcal{A}, j_!B) = \mathrm{RHom}_X(\mathcal{A}, Ri^!j_!B).$$

Substitute $Ri^!j_!B = cB[-1]$. The exact adjunction $g \dashv c$, with c preserving injectives, then gives $\mathrm{RHom}_X(\mathcal{A}, cB) = \mathrm{RHom}_\Lambda(g\mathcal{A}, B)$. This proves (RP12). For the reverse direction, use $j_! \dashv j^*$, exactness of $j_!$, and $j^*i_* = 0$.

The extension (RP6) is the pullback of (RP11) along $i_*\phi : i_*\mathcal{A} \rightarrow i_*cB$: the fibre product has exactly the triple $(\mathcal{A}, B, \phi)$. The universal sequence corresponds to id_{cB} under the derived adjunction, fixing its sign; pullback therefore corresponds to ϕ , and then to $\bar{\phi}$. For $U\mathcal{A}$ that map is the identity. $\square$

For a scheme with generic point, (RP12) says exactly that this direct cross-extension channel receives the generic stalk. A sheaf whose generic stalk is zero has no such cross-extension. This fact does not remove its other supported-prime data. The universal coupling in Section 11.5 gives the exact relative object in which the quotients by those prime stalks occur.

Proposition 11.7 (Full mapping complex). *For $\mathcal{F} = (\mathcal{A}, B, \phi)$ and $\mathcal{F}' = (\mathcal{A}', B', \phi')$, their derived mapping complex is the fibre of*

$$\mathrm{RHom}_X(\mathcal{A}, \mathcal{A}') \oplus \mathrm{RHom}_\Lambda(B, B') \longrightarrow \mathrm{RHom}_\Lambda(g\mathcal{A}, B'), \quad (a, b) \longmapsto \bar{\phi}' g(a) - b \bar{\phi}. \quad (\mathrm{RP13})$$

Proof. Resolve the target by injective triples. Their arithmetic and generic components are injective, as proved in Theorem 11.4, and their gluing arrows split. In each degree, the map of ordinary Hom groups in (RP13) is surjective: lift any map $\mathcal{A} \rightarrow cB'_I$ through a splitting of ϕ'_I and take $b = 0$. Its kernel is exactly the compatible pairs defining a triple morphism. Thus the complex of triple morphisms is the kernel of a surjective map of complexes computing the displayed derived Homs. It is therefore their homotopy fibre. Lemma 11.1 identifies the last mapping complex with the stated module-valued one. $\square$

Now take a coefficient field k . If V, W are finite-dimensional k -vector spaces, define

$$E(V, W) = \mathrm{Ext}_Y^1(i_*cV, j_!W) = \mathrm{Hom}_k(V, W). \quad (\mathrm{RP14})$$

There is a canonical perfect pairing

$$E(V, W) \times E(W, V) \longrightarrow k, \quad (\phi, \psi) \longmapsto \mathrm{tr}_V(\psi\phi) = \mathrm{tr}_W(\phi\psi). \quad (\mathrm{RP15})$$

Indeed matrices give both equal traces as $\sum_{a,b} \phi_{ba}\psi_{ab}$; pairing a matrix unit with its transpose shows nondegeneracy. This is a pairing of two explicitly identified extension spaces with exchanged coefficient spaces. It is not a Yoneda product through a nonexistent reverse cross-extension: those reverse groups vanish by Theorem 11.6. For $k = \mathbb{C}$ and specified positive Hermitian metrics, $\mathrm{tr}(\phi^*\psi)$ gives the associated Hermitian form and $\mathrm{tr}(\phi^*\phi) = \sum_{a,b} |\phi_{ba}|^2$ in orthonormal bases. The metrics are additional data; topology alone has not supplied them.

If $T : B \rightarrow B$ is an endomorphism of a finite-dimensional coefficient space, its relative alternating trace is

$$\sum_n (-1)^n \operatorname{tr}(T | H_X^n(Y, j_! B)) = -\operatorname{tr}(T). \quad (\text{RP16})$$

This follows from (RP10), with its entire cohomology in degree one. For two such sheaves the relative cup product on their degree-one groups is zero: its target is $H_X^2(Y, j_!(B \otimes_k D)) = 0$. Their tensor sheaf is $j_!(B \otimes D)$ because tensor products are computed on stalks and both arithmetic stalks vanish. Thus the nonzero relative degree-one object and the trace pairing (RP15) have precise different maps; identifying that trace pairing with a degree-two cup pairing would give a false result.

11.4 What the actual split structure sheaf retains

Let X now be an irreducible locally ringed space with nonzero local rings. For a ring R , $\mathbf{G}(R) = \{\tau\} \sqcup R$, with τ the additive identity and multiplicative absorber; write e_R for its supported coefficient zero. Supported addition and multiplication are those of R , so $e_R + e_R = e_R$ and $e_R a = e_R$ for every supported a . The full semiring sheaf is

$$\mathcal{S}(U^+) = \mathbf{G}(\mathcal{O}_X(U)), \quad \mathcal{S}(\{\eta_\tau\}) = \mathbb{B} = \{0, 1\}, \quad 1 + 1 = 1, \quad (\text{RP17})$$

with support restriction to the singleton. Gluing is exact because compatible sections agree in support at the common singleton; absent sections glue to τ , and supported coefficients glue in $\mathcal{O}_X$. Its stalks are $\mathbf{G}(\mathcal{O}_{X,x})$ and $\mathbb{B}$. The global section e has arithmetic germs $e_{\mathcal{O}_{X,x}}$ and generic germ 1 at the additional point. For an affine domain A , the prime ideals of $\mathbf{G}(A)$ are $\{\tau\}$ and $\{\tau\} \sqcup \mathfrak{p}$, $\mathfrak{p} \in \operatorname{Spec} A$: an ideal containing a supported element contains e_A and its supported elements form a ring ideal by multiplication by -1 . Products of supported elements remain supported, proving primality of $\{\tau\}$. This identifies $V(e)$ with the entire closed arithmetic stratum.

Theorem 11.8 (Exact additive loss and joint recovery). *The additive group completion of $\mathcal{S}$ is $i_*\mathcal{O}_X$. Every sheaf of $\mathcal{S}$ -semimodules with abelian-group addition is uniquely an $i_*\mathcal{O}_X$ -module; its generic stalk at η_τ is zero. In particular neither nonzero $j_!B$ nor the universal coupling $U\mathcal{O}_X$ with nonzero generic fibre is an additive-group module over $\mathcal{S}$ with its stated unit action.*

Nevertheless the joint map of semiring sheaves

$$(\chi, q) : \mathcal{S} \longrightarrow j_*\mathbb{B} \times i_*\mathcal{O}_X \quad (\text{RP18})$$

is injective. On U^+ it identifies the sections with $(0, 0)$ and all $(1, a)$, $a \in \mathcal{O}_X(U)$. Here χ retains support and q retains the exact coefficient, including zero.

Proof. An additive map from $\mathbf{G}(R)$ to an abelian group sends e_R to zero, since $f(e_R) + f(e_R) = f(e_R)$. Its supported restriction is then a group homomorphism from R , and the image of τ is zero. This proves the universal group-completion property with quotient R . At the Boolean stalk the relation $1 + 1 = 1$ kills its sole nonzero generator, so its group completion is zero. Sheafification commutes with this local universal property, yielding $i_*\mathcal{O}_X$.

For an additive-group semimodule, the relation $em + em = em$ gives $em = 0$. Thus its scalar action factors through the indicated ring quotient. At the extra stalk $e = 1$, so this says $m = 0$ for every element. This proves the module and nonexistence assertions. In (RP18), the absent section gives $(0, 0)$ and the supported zero gives $(1, 0)$, which are distinct. All other coefficients are distinguished by their second coordinate. The same checks on operations prove it is a morphism and injective. $\square$

Proposition 11.9 (A multiplicative linearization that retains support). *Let $\mathcal{M}$ be the sheaf of supported multiplicative elements of $\mathcal{S}$: its values on U^+ are the multiplicative monoid of*

$\mathcal{O}_X(U)$, including its supported zero, and its value on the singleton is $\{1\}$. Let $\mathcal{L} = \mathbb{Z}[\mathcal{M}]$ be the sheafification of its integral monoid algebra. Write $E = [e]$. Then E is a central idempotent, and

$$\mathcal{L} = E\mathcal{L} \oplus (1 - E)\mathcal{L}, \quad E\mathcal{L} \cong j_*\mathbb{Z}, \quad (1 - E)\mathcal{L} \cong i_*\mathcal{C} \quad (\text{RP19})$$

as rings with the two factor identities E and $1 - E$. Here $\mathcal{C}_x = (1 - [0])\mathbb{Z}[(\mathcal{O}_{X,x}, \cdot)]$. There is a surjection $\mathcal{L} \rightarrow i_*\mathcal{O}_X$, $[a] \mapsto a$, whose kernel is the sheaf of ideals generated by $[a + b] - [a] - [b]$. It kills the whole factor $E\mathcal{L}$. The factor $E\mathcal{L}$ contains the canonical nonzero submodule $j_*\mathbb{Z}$, and its quotient is $i_*\mathbb{Z}_X$ as in (RP11).

Proof. Multiplicatively $e^2 = e$ and $ea = e$ for every supported element. Consequently $E^2 = E$ and $E[a] = E$. The augmentation $\varepsilon : \mathcal{L} \rightarrow j_*\mathbb{Z}$, $[a] \mapsto 1$, is a ring map. Multiplication by E sends a section to its augmentation times E , so its image is the free rank-one ring sheaf with identity E . This proves its identification with $j_*\mathbb{Z}$. The complementary factor has zero stalk at the added point, since $E = 1$ there; the triple description therefore places it on the closed arithmetic stratum and gives its stated stalks. The central idempotent identity proves the direct-product ring decomposition.

The quotient by the displayed additive relations has exactly the universal property of a ring receiving the original supported addition and multiplication. At an arithmetic stalk this is the original ring; at the singleton the relation $[1] - [1] - [1] = -1$ makes the quotient zero. Thus the quotient is precisely $i_*\mathcal{O}_X$. Taking $a = b = 0$ gives $-E$ among these relations, so its entire factor is killed. The last assertion is (RP11) on that factor. All computations commute with restrictions and so prove the sheaf assertions. $\square$

This construction specifies both what survives and what is lost. After tensoring with a coefficient field k , the E factor is the constant ring sheaf j_*k on the full space Y . Its module category is the category of all k -sheaves used above. The complementary factor is supported on X . Every $\mathcal{L} \otimes k$ -module splits as $EM \oplus (1 - E)M$, since these orthogonal idempotents sum to one; the same decomposition of every map makes the module category a product. Its cross-factor Hom and Ext groups are zero, by resolving within each product component. Each supported scalar $[a]$ acts as the identity on the E factor. Thus its rank-one relative trace for any such scalar is -1 , by (RP16); the support factor alone has not encoded a prime norm.

More specifically, if arithmetic coefficients are required to act through q and the additional fibre through ε , equivariance of a gluing map would give

$$\phi(ea) = e\phi(a), \quad \text{hence } 0 = \phi(a). \quad (\text{RP20})$$

The exact space of couplings in the underlying abelian-sheaf category is still $\text{Hom}(g\mathcal{A}, B)$ by (RP12); the specified module compatibility selects its zero subspace. The universal coupling (RP5) retains the entire arithmetic sheaf and gives its identity generic coupling in that explicit receiving category. The next calculation uses this object and does not assert the incompatible semiring-module action.

11.5 Principal parts at the supported primes

Theorem 11.10 (Principal parts are the relative object of the new point). *Let X be an integral scheme with function field F , viewed with its actual structure sheaf. Let $F_X = cF$ be its sheaf of rational functions and define $\mathcal{P}_X = F_X/\mathcal{O}_X$. For the canonical coupled abelian sheaf*

$$\mathcal{U}_X = U\mathcal{O}_X = (\mathcal{O}_X, F, \mathcal{O}_X \hookrightarrow F_X) \quad (\text{RP21})$$

one has

$$Ri^!\mathcal{U}_X = \mathcal{P}_X[-1], \quad (R^1i^!\mathcal{U}_X)_x = F/\mathcal{O}_{X,x}. \quad (\text{RP22})$$

In particular a supported valuation-ring stalk V contributes exactly F/V . The ordinary generic stalk gives zero, while nonfield valuation stalks give nonzero quotients. The maps are the original inclusions $\mathcal{O}_{X,x} \hookrightarrow F$ and their exact cokernels.

Proof. An integral scheme embeds its structure sheaf in its function field: on an affine domain chart, each local ring and each regular section maps injectively into its fraction field, and these maps agree on overlaps. Its generic stalk is F . Hence (RP5) gives exactly (RP21). Substitute this injective arrow into (RP7). The two-term complex has zero degree-zero cohomology and cokernel $\mathcal{P}_X$ in degree one, proving its indicated quasi-isomorphism and every stalk formula. If $V \neq F$, an element of $F \setminus V$ gives a nonzero quotient class; for $V = F$ the quotient is zero. $\square$

Theorem 11.11 (The complete integral prime receiver). *For $X = \operatorname{Spec} \mathbb{Z}$, one has*

$$\mathcal{P}_X(U) = \bigoplus_{p \in U \text{ closed}} \mathbb{Q}/\mathbb{Z}_{(p)} \quad (\text{RP23})$$

for every nonempty open U , with restriction the projection discarding the primes removed from the open. At the generic point the stalk is zero. Each closed-prime stalk admits the canonical isomorphism

$$\mathbb{Q}/\mathbb{Z}_{(p)} \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p. \quad (\text{RP24})$$

The resulting relative cohomology is exactly

$$H_X^1(Y, \mathcal{U}_X) = \mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Q}_p/\mathbb{Z}_p, \quad H_X^n(Y, \mathcal{U}_X) = 0 \quad (n \neq 1). \quad (\text{RP25})$$

Proof. A nonempty open of $\operatorname{Spec} \mathbb{Z}$ omits a finite set S of closed primes and has ring of regular functions $\mathbb{Z}[S^{-1}]$. Indeed it contains a basic open $D(n)$, so its complement is contained in the finitely many prime divisors of n ; a rational number is regular there exactly when its denominator has no prime factors outside S . The map

$$\mathbb{Q} \longrightarrow \bigoplus_{p \notin S} \mathbb{Q}/\mathbb{Z}_{(p)} \quad (\text{RP26})$$

sends a rational number to its indicated quotient classes. It has finite support, since its denominator has finitely many prime factors, and kernel $\mathbb{Z}[S^{-1}]$. It is surjective. For a prescribed finite family of classes, represent its p component by a_p/p^{n_p} with $a_p \in \mathbb{Z}$: any denominator prime to p is invertible modulo p^{n_p} , which supplies this representative. Then the sum $\sum_p a_p/p^{n_p}$ has the desired component at each p , because all the other summands are p -integral. This proves surjectivity.

The direct-sum formula with projection restrictions defines a sheaf. Every open here is quasi-compact: a cover by basic opens has a finite subcover because the ideal generated by their defining integers has a finite set of generators. Compatible families therefore have a finite-subcover union of finite supports, and glue to a finite-support family. They glue uniquely componentwise. Its stalk at p is the displayed p group, because the finitely many other support primes can be excluded by a smaller neighbourhood. Its generic stalk is zero by excluding its finite support. Thus the stalk maps from the quotient sheaf $\mathbb{Q}_X/\mathcal{O}_X$ identify it with the displayed sheaf, proving (RP23).

The inclusion $\mathbb{Q} \subset \mathbb{Q}_p$ induces (RP24): its kernel is the rational numbers with nonnegative p valuation, namely $\mathbb{Z}_{(p)}$. For surjectivity, write a p -adic element as $p^{-n}a$ with $a \in \mathbb{Z}_p$ and choose an integer b with $a - b \in p^n\mathbb{Z}_p$. Then b/p^n has the same class modulo $\mathbb{Z}_p$.

The sheaf in (RP23) is flasque: every restriction projection is surjective, by extension by zero at the omitted primes. Therefore its higher sheaf cohomology vanishes. This elementary acyclicity criterion follows, for example, by the standard injective resolution argument: injective sheaves are flasque, a quotient of a flasque sheaf by a flasque subsheaf is flasque, and global sections of such a quotient lift by extending compatible local lifts; the extension is obtained by adjoining one open set at a time and taking unions of compatible extensions. Induction on a resolution then makes the global-sections complex exact in positive degrees. Applying (RP22) now leaves only its degree-one global sections. Formula (RP26) for $S = \emptyset$ identifies those with $\mathbb{Q}/\mathbb{Z}$ and with the displayed direct sum, proving (RP25). $\square$

11.6 The exact filtration, finite pairings, and length weights

Fix a prime p . Put $C_p = \mathbb{Q}_p/\mathbb{Z}_p$ and

$$C_{p,n} = p^{-n}\mathbb{Z}_p/\mathbb{Z}_p \quad (n \geq 0), \quad C_{p,0} = 0. \quad (\text{RP27})$$

These are the original subquotients at the supported prime, not their numerical replacements.

Theorem 11.12 (All filtration maps and dual maps). *The map $a \bmod p^n \mapsto a/p^n \bmod \mathbb{Z}_p$ identifies $\mathbb{Z}/p^n\mathbb{Z}$ with $C_{p,n}$. In these coordinates the inclusion $C_{p,n} \hookrightarrow C_{p,n+1}$ is $a \mapsto pa$. For $n \geq 1$ there is the exact sequence*

$$0 \longrightarrow C_{p,n-1} \longrightarrow C_{p,n} \xrightarrow{\rho_{p,n}} \mathbb{F}_p \longrightarrow 0, \quad \rho_{p,n}(a/p^n) = a \bmod p. \quad (\text{RP28})$$

Thus $C_p = \bigcup_n C_{p,n}$, its n th layer has length one and cardinality p , and $\text{length}_{\mathbb{Z}_p} C_{p,n} = n$, $|C_{p,n}| = p^n$.

The finite pairing

$$C_{p,n} \times \mathbb{Z}/p^n\mathbb{Z} \longrightarrow \mathbb{C}^*, \quad (a/p^n, b) \longmapsto \exp(2\pi iab/p^n) \quad (\text{RP29})$$

is perfect. Dual to the filtration inclusion is reduction $\mathbb{Z}/p^{n+1}\mathbb{Z} \rightarrow \mathbb{Z}/p^n\mathbb{Z}$. Hence the character group of C_p , with its discrete topology, is $\mathbb{Z}_p$. Its annihilator of $C_{p,n}$ is $p^n\mathbb{Z}_p$, and the induced perfect layer pairing is

$$(C_{p,n}/C_{p,n-1}) \times (p^{n-1}\mathbb{Z}_p/p^n\mathbb{Z}_p) \longrightarrow \mathbb{C}^*, \quad (a/p^n, p^{n-1}b) \longmapsto \exp(2\pi iab/p). \quad (\text{RP30})$$

Proof. Changing a by a multiple of p^n changes a/p^n by an integer; these are exactly the changes giving zero in the quotient. Every $a \in \mathbb{Z}_p$ has an integer representative modulo p^n , proving the first isomorphism. Expressing the same fraction with denominator p^{n+1} multiplies its numerator by p , proving the inclusion map. Reduction of the numerator modulo p is well defined for $n \geq 1$. Its kernel consists of $a = pb$, which gives exactly the preceding filtration group, and it is surjective. This proves the exact sequence, the layer, and, by induction, both length and cardinality. Every element of $\mathbb{Q}_p$ has a finite negative valuation, proving the union.

The exponential is unchanged by changing either residue representative. If $a \not\equiv 0 \pmod{p^n}$, pairing it with $b = 1$ is not one; the same argument holds in the other variable. A character of the cyclic group is determined by its value on $1/p^n$, an arbitrary p^n th root of unity, and these are exactly the displayed values. This proves perfectness, not merely nondegeneracy. Substituting pa at level $n+1$ gives the level- n formula with b reduced modulo p^n ; thus compatible characters are exactly the inverse limit $\mathbb{Z}_p$. A p -adic integer kills every element of $C_{p,n}$ exactly when it is zero modulo p^n . Restricting that assertion to successive annihilators gives precisely (RP30), whose perfection also follows from the same cyclic argument at order p . $\square$

Corollary 11.13 (The same residue layer is the exact degree of multiplication). Multiplication by p gives the exact sequences

$$0 \longrightarrow \mathbb{F}_p \xrightarrow{a \mapsto a/p} C_{p,n+1} \xrightarrow{p} C_{p,n} \longrightarrow 0 \quad (n \geq 0), \quad 0 \longrightarrow \mathbb{F}_p \longrightarrow C_p \xrightarrow{p} C_p \longrightarrow 0. \quad (\text{RP30a})$$

In finite numerator coordinates the middle map is reduction $\mathbb{Z}/p^{n+1}\mathbb{Z} \rightarrow \mathbb{Z}/p^n\mathbb{Z}$. Its dual is multiplication by p , and the dual map on $\mathbb{Z}_p$ is multiplication by p . Every fibre on C_p has exactly p elements. Consequently on $\ell^2(C_p)$ with counting measure the actual pullback $A_p f(d) = f(pd)$ obeys

$$\|A_p f\|^2 = p \|f\|^2. \quad (\text{RP30b})$$

Thus the same one-layer cardinality produces both the logarithmic weight $\log p$ below and the norm factor $\sqrt{p}$ in the companion degree-operator calculation (PD7)–(PD10).

Proof. One has $p(a/p^{n+1}) = a/p^n$, giving the coordinate reduction and its surjectivity. Its kernel consists of the fractions a/p modulo $\mathbb{Z}_p$, exactly the stated copy of $\mathbb{F}_p$. Every element of C_p lies at a finite level, so passing to the union gives the second exact sequence and all fibre sizes. Substitution in (RP29) proves that the dual finite map is $b \mapsto pb$; equivalently on the inverse limit, $\langle pd, z \rangle = \langle d, pz \rangle$. For finite-support f , group the sum $\sum_d |f(pd)|^2$ by its image pd . Each image has exactly p preimages, so the sum is $p \sum_u |f(u)|^2$. Density of finite-support functions extends the pullback as a bounded operator and proves the same equality on its whole Hilbert domain. This uses the original group map and its exact kernel, not a substituted scalar map on a residue field. $\square$

Taking the direct sum of the prime groups in (RP25), a character is independently determined on each summand. Thus its character group is $\prod_p \mathbb{Z}_p = \widehat{\mathbb{Z}}$. Equivalently, for $a/n \bmod \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ and $z \in \widehat{\mathbb{Z}}$ the global pairing is

$$\langle a/n, z \rangle = \exp(2\pi i a(z \bmod n)/n). \quad (\text{RP31})$$

It is independent of the representative because the reductions $z \bmod n$ are compatible under divisibility. Perfection follows from the cyclic finite-level calculation and its inverse limit. At any finite level $N = p^n$, the exact orthogonality identity is

$$\frac{1}{N} \sum_{b \bmod N} \left| \sum_{a \bmod N} f(a) \exp(2\pi i ab/N) \right|^2 = \sum_{a \bmod N} |f(a)|^2. \quad (\text{RP32})$$

Expanding the square reduces this to $\sum_b \exp(2\pi i(a - a')b/N) = N$ for $a = a'$ and zero otherwise; the latter is a finite geometric series with ratio different from one. Thus the original residue labels and their finite pairing are retained, in addition to the length receiver constructed next.

Theorem 11.14 (The weighted filtration receiver is the prime logarithmic derivative). *Let $K_0^{\text{fl}}(\mathbb{Z}_p)$ be the Grothendieck group of finite length $\mathbb{Z}_p$ -modules, with the relations $[M] = [M'] + [M'']$ from short exact sequences. The length map is an isomorphism $K_0^{\text{fl}}(\mathbb{Z}_p) \cong \mathbb{Z}$, taking $[\mathbb{F}_p]$ to one. The canonical additive weight is*

$$w_p([M]) = \log |M| = (\log p) \text{length}_{\mathbb{Z}_p} M. \quad (\text{RP33})$$

For the exact original filtration (RP27), its successive-layer series therefore satisfies

$$w_p \left(\sum_{m \geq 1} [C_{p,m}/C_{p,m-1}] T^m \right) = (\log p) \frac{T}{1-T}. \quad (\text{RP34})$$

For $\Re s > 0$, put $T = p^{-s}$. Then this convergent series is exactly

$$(\log p) \sum_{m \geq 1} p^{-ms} = -\frac{d}{ds} \log \frac{1}{1-p^{-s}} = L_p(s). \quad (\text{RP35})$$

It equals the expression (ME15), label `eq:ME-log-derivative`, in the companion complete proof `11_mixed_extension_prime_trace.tex`, lines 210–215 of the 22 September 2026 working edition.

Proof. Every simple module over the local ring $\mathbb{Z}_p$ is its residue field: the annihilator of a nonzero generator is a maximal ideal, and the unique maximal ideal is (p) . A composition series of a finite length module therefore gives $[M] = \text{length}(M)[\mathbb{F}_p]$. Length is additive on a short exact sequence, as one sees by adjoining a composition series of the submodule to the inverse images of a composition series of the quotient. Consequently it gives a homomorphism back to $\mathbb{Z}$, inverse

to $1 \mapsto [\mathbb{F}_p]$. Each simple factor has p elements. Cardinalities multiply in a short exact sequence of finite groups, since all fibres of the quotient have the size of its kernel. Induction on the composition series proves $|M| = p^{\text{length}(M)}$, establishing (RP33).

By (RP28), every layer has class $[\mathbb{F}_p]$. Summing their formal series gives (RP34). For $\Re s > 0$, $|p^{-s}| < 1$, so it is absolutely convergent and is the ordinary geometric series. With $q = p^{-s}$ one has $dq/ds = -(\log p)q$ and $d \log(1 - q)^{-1}/ds = -(\log p)q/(1 - q)$, proving the exact sign in (RP35). Thus the receiving weight per layer is $\log p$, while the cumulative weight through level n is $n \log p$; no factor m has been lost or added. $\square$

The receiver in (RP34) is a Grothendieck-group receiver. It does not tensor the actual finite residue group with $\mathbb{C}$: $\mathbb{F}_p \otimes_{\mathbb{Z}} \mathbb{C} = 0$, since multiplication by p is invertible on $\mathbb{C}$. The full finite groups and their pairings remain in (RP27)–(RP32). The map through K_0 records their exact length and cardinality and is separately identified as such.

11.7 A trace operator receiving the original prime-power Weil term

Let $\mathcal{H}_p = \ell^2(\mathbb{N}_{\geq 1})$ with distinguished orthonormal vectors $\lambda_{p,m}$ representing the length generator $[C_{p,m}/C_{p,m-1}] = [\mathbb{F}_p]$ in grade m . Define

$$D_p(s)\lambda_{p,m} = p^{-ms}\lambda_{p,m} \quad (\Re s > 0). \quad (\text{RP36})$$

The exact domain is this complex length receiver; it is not the $\mathbb{F}_p$ -module on which a nonintegral complex scalar would be undefined. Its trace norm is $\sum_{m \geq 1} p^{-m\Re s} < \infty$, because finite diagonal truncations have that trace norm and form a Cauchy sequence in trace norm. Therefore

$$(\log p) \text{tr } D_p(s) = L_p(s). \quad (\text{RP37})$$

This proves an exact map from the arithmetic principal-parts filtration through its additive logarithmic-cardinality weight to the logarithmic derivative of the mixed prime differential.

For completeness the resulting receiver in the original prime-power test can be displayed without changing its measure or coefficients. Let $g \in C_c^\infty(\mathbb{R}_{>0})$, let $g^*(x) = \overline{g(x^{-1})}$, use convolution for dx/x , and put $h = g * g^*$ and $\widehat{g}(t) = \int_0^\infty g(x)x^{-it} dx/x$. Then

$$\begin{aligned} W_p(h) &= (\log p) \sum_{m \geq 1} p^{-m/2} (h(p^m) + h(p^{-m})) \\ &= \frac{\log p}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 \text{tr} \left(D_p\left(\frac{1}{2} + it\right) + D_p\left(\frac{1}{2} - it\right) \right) dt. \end{aligned} \quad (\text{RP38})$$

The first line is the unchanged prime-power expression (ME16) in the same companion proof, and the second line substitutes (RP37) into its exact Mellin receiver. One can also verify the substitution directly: in the coordinate $u = \log x$, Fourier inversion gives $h(e^u) = (2\pi)^{-1} \int |\widehat{g}(t)|^2 e^{itu} dt$. The sum of absolute diagonal coefficients in the two traces is $2p^{-1/2}/(1 - p^{-1/2})$, uniformly in t . Since $\widehat{g}$ is rapidly decreasing, this bound justifies exchanging the integral and the trace series. Substituting $u = \pm m \log p$ gives exactly the first line, including $1/(2\pi)$ and every $p^{-m/2}$ coefficient.

Thus the extra-point universal coupling gives the relative principal parts; the supported prime gives its entire $p^{-n}\mathbb{Z}_p$ filtration; its successive residue layers give the exact $\log p$ weight in the original Weil prime term. The independent finite residue pairing is retained alongside this trace receiver. This computation does not assert positivity of the complete Weil functional: the prime term in the explicit formula retains its original sign, and its pole and Archimedean contributions remain the ones in the companion calculation. Nor does it assert an additive-group module action that contradicts (RP20).

$$\begin{array}{ccc}
\mathcal{U}_X & \xrightarrow{Ri^!} & \mathcal{P}_X[-1] \xrightarrow{\text{stalk at } (p)} (\mathbb{Q}_p/\mathbb{Z}_p)[-1] \\
& & \uparrow \\
& & C_{p,n}[-1] \\
C_{p,n}/C_{p,n-1} \cong \mathbb{F}_p & \xrightarrow{\text{class}} & [\mathbb{F}_p] \in K_0^{\text{fl}}(\mathbb{Z}_p) \xrightarrow{w_p} \log p
\end{array}$$

Figure 6: Exact routes from the full prime space to its arithmetic receiver. The first row uses the universal coupling in the explicitly specified abelian-sheaf category. Its supported-prime stalk retains every filtration group and the finite pairing (RP29)–(RP30). The lower row records the separate length receiver; summing its successive-layer weights with p^{-ns} gives (RP35), exactly (ME15), and hence the unchanged prime-power Weil expression (RP38). Theorems 11.10, 11.11, 11.12, and 11.14 prove every object and map shown.

11.8 Provenance and exact dependence

The earlier complete derivation `split_zero_adelic_bridge.tex`, Theorem `thm:fullspectral`, Proposition `prop:fullspectralfunctor`, and Corollary `cor:fullrefinement`, constructs the full prime space, its local semiring sheaf, and the valuation-refinement morphism. The construction needed here is stated and checked again in (RP17). The original-author source ledger in `WORKLOG.md` was read before this derivation. Antoine Sédillot’s arXiv:2503.20156v2, original author-TeX lines 623–629 and 831–835, supplies the prior valuation and residue-field context. The recollement, universal coupling, principal-parts receiver, finite pairings, and filtration trace maps in this note are proved here; they are not attributed to that paper. No external literature claim is used as a substitute for these proofs. The companion (ME15)–(ME16) source is included in the same working collection; its use is bounded to the displayed prime logarithmic derivative and exact prime-power test, whose connecting calculation has also been proved in (RP35)–(RP38).

12 The prime-degree operator and its exact Weil pairing

We now use the relative principal-parts object produced by the full spectral extension, rather than only comparing it with older determinant bounds. At $\text{Spec } \mathbb{Z}$ its arithmetic part is $\mathbb{Q}/\mathbb{Z}$, and its p -primary part is

$$D_p = \mathbb{Z}[1/p]/\mathbb{Z} \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p. \quad (\text{PD1})$$

The relative cohomological construction and its receiving category are proved in the preceding recollement chapter. The operators and pairings on (PD1) will all be calculated here. The absolute value remains $|p|_p = p^{-1}$, and the additive Haar measure remains $\mu(\mathbb{Z}_p) = 1$. The use of local Fourier transforms follows the same convention as Connes [4, Section III]. No spectral interpretation of the zeros is assumed.

12.1 The actual filtration, degree and finite character pairing

Put $D_{p,n} = p^{-n}\mathbb{Z}/\mathbb{Z}$, for $n \geq 0$. This is a subgroup of D_p of order p^n , and $D_p = \bigcup_n D_{p,n}$. Mapping rational classes into $\mathbb{Q}_p/\mathbb{Z}_p$ is injective: an element a/p^n lying in $\mathbb{Z}_p$ has $p^n \mid a$ after cancellation. It is onto: the finite negative part of any p -adic expansion differs from the number by an element of $\mathbb{Z}_p$. This proves (PD1), including its filtration. Multiplication by p gives

$$0 \longrightarrow D_{p,1} \longrightarrow D_p \xrightarrow{p} D_p \longrightarrow 0, \quad |D_{p,1}| = p. \quad (\text{PD2})$$

Surjectivity follows by replacing a/p^n by a/p^{n+1} ; the kernel consists of the p classes a/p modulo integers. Every fibre therefore has exactly p elements. The exact layer maps are

$$D_{p,n}/D_{p,n-1} \xrightarrow{\sim} \mathbb{F}_p, \quad [a/p^n] \mapsto a \pmod{p}, \quad n \geq 1. \quad (\text{PD3})$$

They are well defined because changing the representative modulo $D_{p,n-1}$ changes a by a multiple of p . They are injective and onto by the same formula. Thus each layer has module length one and logarithmic cardinality $\log p$.

For $d = a/p^n \pmod{\mathbb{Z}}$ and $x \in \mathbb{Z}_p$, define

$$\chi_d(x) = \exp\left(2\pi i \frac{a(x \bmod p^n)}{p^n}\right). \quad (\text{PD4})$$

Changing representatives of either residue changes the exponent by an integer; passing to denominator p^{n+1} gives the same character. At finite level this is a perfect pairing of $D_{p,n}$ and $\mathbb{Z}/p^n\mathbb{Z}$: a nonzero a is detected by $x = 1$, and a nonzero x is detected by $a = 1$. The geometric-series identity gives

$$p^{-n} \sum_{x \bmod p^n} \chi_d(x) \overline{\chi_{d'}(x)} = \mathbf{1}_{d=d'} \quad (d, d' \in D_{p,n}). \quad (\text{PD5})$$

The Haar measure of each residue class is p^{-n} , so these are also the integrals over $\mathbb{Z}_p$.

Let $\ell^2(D_p)$ use counting measure. Formula (PD5) implies that

$$\mathcal{F}_p : \ell^2(D_p) \longrightarrow L^2(\mathbb{Z}_p, dx), \quad \mathcal{F}_p \delta_d = \chi_d \quad (\text{PD6})$$

extends isometrically from finite sums. It is onto: the p^n orthogonal characters at level n form a basis of the p^n -dimensional space of functions constant on residue classes modulo p^n . The union of these spaces is dense in $L^2(\mathbb{Z}_p)$. One way to see this last assertion is to approximate a continuous function uniformly using uniform continuity on the compact space $\mathbb{Z}_p$ and then use the density of continuous functions for the finite regular Haar measure. In particular $\mathcal{F}_p \delta_0 = \mathbf{1}_{\mathbb{Z}_p}$, with norm exactly one.

12.2 The square-root factor is fixed by the prime degree

The pullback of the actual map (PD2) on finite-support functions is

$$(A_p f)(d) = f(pd), \quad d \in D_p. \quad (\text{PD7})$$

Each fibre has p elements, so direct summation proves $\|A_p f\|^2 = p\|f\|^2$. Consequently its bounded extension has norm exactly $\sqrt{p}$. Define $V_p = p^{-1/2}A_p$ and keep both operators distinguished. The factor in V_p is forced by this degree calculation. Their adjoints and the Fourier transforms are

$$(V_p^* f)(u) = p^{-1/2} \sum_{pd=u} f(d), \quad (\text{PD8})$$

$$(\mathcal{F}_p V_p \mathcal{F}_p^{-1} F)(x) = \sqrt{p} \mathbf{1}_{p\mathbb{Z}_p}(x) F(x/p), \quad (\text{PD9})$$

$$(\mathcal{F}_p V_p^* \mathcal{F}_p^{-1} F)(x) = p^{-1/2} F(px). \quad (\text{PD10})$$

Proof. Taking the counting-measure inner product in (PD7) and collecting terms over each fibre proves (PD8). For finite f , the Fourier transform of $A_p f$ is $\sum_u f(u) \sum_{pd=u} \chi_d(x)$. The inner character sum is zero unless $x \in p\mathbb{Z}_p$: translation of the fibre by $D_{p,1}$ multiplies it by a nontrivial p th root of unity otherwise. For $x = py$ it is $p\chi_u(y)$. This proves (PD9) on the dense finite-support subspace and hence on the whole Hilbert space. Finally changing variables $x = py$, with $dx = p^{-1}dy$, in the inner product for (PD9) proves (PD10). No counting measure or Haar factor has been omitted. $\square$

In particular $V_p^* V_p = I$. In the Fourier representation $V_p^n (V_p^*)^n$ is multiplication by $\mathbf{1}_{p^n \mathbb{Z}_p}$. Their decreasing intersection has measure zero, since $\mu(p^n \mathbb{Z}_p) = p^{-n}$. Thus $\bigcap_n V_p^n \ell^2(D_p) = 0$. The distinction between this isometry and an invertible evolution is resolved by the following explicit extension, not by an assumed inverse.

Theorem 12.1 (Exact unitary dilation and its original vector). *On the actual local space $L^2(\mathbb{Q}_p, dx)$ define*

$$(\mathcal{U}_p F)(x) = \sqrt{p} F(x/p), \quad (\mathcal{U}_p^{-1} F)(x) = p^{-1/2} F(px). \quad (\text{PD11})$$

This is unitary. Extension by zero embeds $L^2(\mathbb{Z}_p)$ isometrically, and $\mathcal{U}_p$ restricts there to (PD9). The union of $\mathcal{U}_p^{-n} L^2(\mathbb{Z}_p)$ is dense in $L^2(\mathbb{Q}_p)$. For the original vector $\xi_p = \mathbf{1}_{\mathbb{Z}_p}$,

$$\langle \mathcal{U}_p^n \xi_p, \xi_p \rangle = p^{-|n|/2}, \quad n \in \mathbb{Z}. \quad (\text{PD12})$$

Its scalar spectral measure on the unit circle is

$$d\nu_p(\theta) = \frac{1}{2\pi} \frac{1 - p^{-1}}{1 - 2p^{-1/2} \cos \theta + p^{-1}} d\theta. \quad (\text{PD13})$$

Proof. Change of variables gives $\int p |F(x/p)|^2 dx = \int |F(y)|^2 dy$, and the two formulas (PD11) compose to the identity. A function supported in $\mathbb{Z}_p$ is carried to one supported in $p\mathbb{Z}_p$, proving the restriction. The inverse images are exactly the spaces supported in $p^{-n}\mathbb{Z}_p$. These sets exhaust $\mathbb{Q}_p$, so truncation of any L^2 function proves density. For $n \geq 0$, $\mathcal{U}_p^n \xi_p = p^{n/2} \mathbf{1}_{p^n \mathbb{Z}_p}$; pairing with $\mathbf{1}_{\mathbb{Z}_p}$ gives $p^{n/2} p^{-n} = p^{-n/2}$. Unitarity gives the complex conjugate value for $-n$, here the same real number. This proves (PD12).

For $0 < r < 1$, absolutely convergent geometric series give $\sum_{n \in \mathbb{Z}} r^{|n|} e^{in\theta} = (1 - r^2)/(1 - 2r \cos \theta + r^2)$. Its constant coefficient is one, and its density is positive. Setting $r = p^{-1/2}$ shows that (PD13) has exactly the moments (PD12). On any Laurent polynomial a , direct expansion then gives $\|a(\mathcal{U}_p) \xi_p\|^2 = \int |a(e^{i\theta})|^2 d\nu_p(\theta)$. Density of Laurent polynomials identifies the cyclic Hilbert space with $L^2(\nu_p)$ and $\mathcal{U}_p$ with multiplication by $e^{i\theta}$ there. This constructs the asserted scalar spectral measure directly. $\square$

12.3 The original prime term is the energy of this operator

Write $a_p = \log p$ and $f(u) = g(e^u)$ for the original multiplicative test g . Define a vector-valued function on the interval $[0, a_p]$ by

$$(\mathcal{E}_p g)(x) = \sum_{n \in \mathbb{Z}} f(x + na_p) \mathcal{U}_p^n \xi_p \in L^2(\mathbb{Q}_p), \quad 0 \leq x < a_p. \quad (\text{PD14})$$

For compact support of f , only finitely many n occur, uniformly in this bounded interval. Thus this is an explicitly defined map into $L^2([0, a_p], dx; L^2(\mathbb{Q}_p))$.

Theorem 12.2 (The relative-prime Hilbert pairing in Weil's formula). *For every $g \in C_c^\infty(\mathbb{R}_{>0})$, with $h = g * g^*$,*

$$\|\mathcal{E}_p g\|^2 = \sum_{j \in \mathbb{Z}} p^{-|j|/2} h(p^j) \quad (\text{PD15})$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{g}(t)|^2 P_{p^{-1/2}}(t \log p) dt. \quad (\text{PD16})$$

In particular the complete arithmetic prime term, including every prime power, is exactly

$$W_p(g * g^*) = (\log p) \left(\|\mathcal{E}_p g\|^2 - \|g\|_{L^2(dx/x)}^2 \right). \quad (\text{PD17})$$

The full finite-place return therefore becomes

$$D_F(g * g^*) = \mathcal{B}(Pg, Pg) - W_\infty(g * g^*) + \sum_{p \in F} (\log p) \left(\|g\|_{L^2(dx/x)}^2 - \|\mathcal{E}_p g\|^2 \right). \quad (\text{PD18})$$

Proof. Use the inner product linear in its first variable. Expanding (PD14) and using (PD12) gives

$$\sum_{n,m} \int_0^{a_p} f(x + na_p) \overline{f(x + ma_p)} p^{-|n-m|/2} dx.$$

Set $j = n - m$ and $u = x + na_p$. For fixed j , the half-open intervals $[na_p, (n+1)a_p)$ partition $\mathbb{R}$ up to measure zero, so the sum is $\sum_j p^{-|j|/2} \int_{\mathbb{R}} f(u) \overline{f(u - ja_p)} du$. The integral is precisely $h(p^j)$ with the original multiplicative convolution, proving (PD15). There are only finitely many nonzero correlations for this test. Fourier inversion of the correlations and the absolutely convergent Poisson series prove (PD16); the dominating sum is $(1+r)/(1-r)|\widehat{g}(t)|^2$ with $r = p^{-1/2} < 1$, which is integrable. The $j = 0$ term of (PD15) is $h(1) = \|g\|^2$. Multiplying the remaining terms by $\log p$ gives the original $W_p(h)$, proving (PD17). Substitution into the unmodified explicit formula proves (PD18). $\square$

12.4 The same determinant and the exact finite support guard

The scalar resolvent on this original vector is also explicit:

$$\langle (I - z\mathcal{U}_p)^{-1} \xi_p, \xi_p \rangle = \sum_{n \geq 0} z^n p^{-n/2} = \frac{1}{1 - zp^{-1/2}}, \quad |z| < 1. \quad (\text{PD19})$$

The operator series is norm convergent in this domain. The scalar series and its last expression continue to $|z| < \sqrt{p}$ without asserting operator resolvent invertibility there. Substituting $z = p^{1/2-s}$ for $\Re s > 1/2$ identifies the scalar with the local determinant receiver $Z_p(s)$ of (ME12); the resulting scalar identity continues to $\Re s > 0$ by the displayed convergent scalar series. This is the exact comparison between the noninvertible prime-degree map, its unitary extension, and the mixed-face determinant.

If $\text{supp } g \subset [\lambda^{-1}, \lambda]$ and $p > \lambda^2$, then $h(p^j) = 0$ for all $j \neq 0$. Formula (PD15) gives $\|\mathcal{E}_p g\|^2 = \|g\|^2$ exactly, so (PD18) agrees with the full prime return once F contains the primes at most λ^2 . It is essential to take the differences before enlarging F ; neither of the two infinite sums of positive energies is asserted to converge separately.

12.5 The precise local weight identity

The unweighted prime pullback A_p has the invertible extension

$$T_p^{\text{rel}} = \sqrt{p} \mathcal{U}_p, \quad (T_p^{\text{rel}} F)(x) = pF(x/p) \quad \text{on } L^2(\mathbb{Q}_p). \quad (\text{PD20})$$

The superscript distinguishes this relative-prime operator from the finite mixed-boundary matrix $T_p = D_p$ of (AB8); the connecting scalar resolvent was proved in (PD19).

Theorem 12.3 (Weight-one norm identity on the computed relative space). *For the actual operator (PD20),*

$$(T_p^{\text{rel}})^* T_p^{\text{rel}} = T_p^{\text{rel}} (T_p^{\text{rel}})^* = pI, \quad \sigma(T_p^{\text{rel}}) = \{z \in \mathbb{C} : |z| = \sqrt{p}\}. \quad (\text{PD21})$$

It has no nonzero Hilbert-space eigenvectors. The spectrum assertion concerns the entire Hilbert spectrum, not a list of zeta eigenvalues.

Proof. The two adjoint identities follow from the proved unitarity of $\mathcal{U}_p$. For the exact spectrum, put $\mathcal{A}_p = \mathbb{Z}_p^\times$ and $H_0 = L^2(\mathcal{A}_p) \subset L^2(\mathbb{Q}_p)$. The measurable disjoint sets $p^n \mathcal{A}_p$, $n \in \mathbb{Z}$, partition $\mathbb{Q}_p \setminus \{0\}$. The missing singleton has Haar measure zero. The summands $\mathcal{U}_p^n H_0$ are therefore mutually orthogonal and their Hilbert direct sum is the whole space. Choose a nonzero unit vector $h \in H_0$; then $h_n = \mathcal{U}_p^n h$ is orthonormal. For $|\lambda| = 1$ put $v_N = (2N+1)^{-1/2} \sum_{n=-N}^N \lambda^{-n} h_n$. Cancellation at the interior indices gives $\|(\mathcal{U}_p - \lambda I)v_N\|^2 = 2/(2N+1)$, while $\|v_N\| = 1$. Hence

every such λ is in the spectrum. For $|\lambda| > 1$, the Neumann series for $-\lambda^{-1}(I - \lambda^{-1}\mathcal{U}_p)^{-1}$ inverts $\mathcal{U}_p - \lambda I$; for $|\lambda| < 1$ use $(I - \lambda\mathcal{U}_p^{-1})^{-1}\mathcal{U}_p^{-1}$. Thus there is no other spectrum. Scaling proves (PD21).

If $\mathcal{U}_p F = \lambda F$, then $|\lambda| = 1$ follows by taking norms unless $F = 0$. The orthogonal components in $\mathcal{U}_p^n H_0$ consequently have equal norms for all n . Square summability forces them all to be zero. Scaling again proves the assertion for T_p^{rel} . $\square$

For comparison, Deligne's actual purity result in Weil II, (3.3.6), concerns the image of compactly supported cohomology in ordinary cohomology, using the upper-weight theorem and its dual lower-weight theorem [9]. Formula (PD21) supplies a complex weight-one norm identity on the relative object explicitly constructed here. The remaining programme step is to calculate the corresponding image for the full split cohomology and its trace, with the maps (ME13), (PD14) and the complete expression (PD18). No identification of that image with $L^2(\mathbb{Q}_p)$, and no full cohomological purity, is asserted by the local calculation.

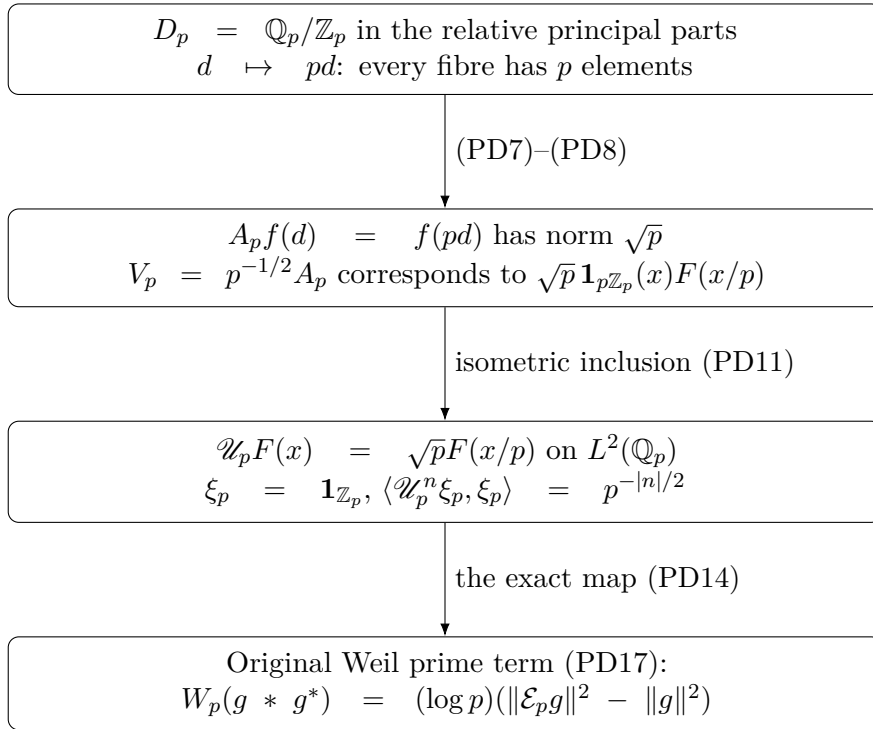


Figure 7: Use of the relative prime data in the original arithmetic pairing. The prime degree fixes every factor of $p^{1/2}$; the original Haar measure fixes every norm. Theorems 12.1 and 12.2 prove the arrows. The formula retains the subtracted diagonal energy, both poles and the Archimedean contribution in (PD18), in the conventions of [4, 5].

These calculations establish a positive local Hilbert representation constructed from the relative prime object and the exact signed map from it into the Weil functional. The full inequality for (PD18) has not been proved: positivity of each individual squared norm does not assign the sign of their displayed difference. The new task supplied by this construction is the full pairing of these relative-prime operators with the interior and Archimedean objects, with the subtraction and the two endpoint terms retained.

13 All integers, including supported zero, on the relative object

The local prime construction extends to the whole relative group $D = \mathbb{Q}/\mathbb{Z}$. We keep its supported zero 0_D and the newly absent element τ_D distinct. In particular the integer $e = 0_{\mathbb{Z}}$ is

included in the scalar action, with its exact domain and codomain.

13.1 Simultaneous arithmetic action and the original integral vector

For $n \in \mathbb{Z} \setminus \{0\}$ the map $d \mapsto nd$ on D is onto with kernel $n^{-1}\mathbb{Z}/\mathbb{Z}$, of cardinality $|n|$. Therefore

$$(A_n f)(d) = f(nd), \quad \|A_n f\|^2 = |n| \|f\|^2, \quad V_n = |n|^{-1/2} A_n \quad (\text{IA1})$$

on $\ell^2(D)$ with counting measure. These formulas follow by counting each fibre, as in (PD7), and $A_m A_n = A_{mn}$ follows by composition. The finite character pairings identify $\ell^2(D)$ with $L^2(\widehat{\mathbb{Z}})$, where $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ has Haar measure one. Indeed every $d = a/N$ gives the character $x \mapsto \exp(2\pi i a(x \bmod N)/N)$; the finite cyclic orthogonality proof (PD5) applies with N in place of p^n . The functions factoring through the finite quotients form a dense subspace, by compactness and the same approximation argument.

Let $\mathbb{A}_f$ denote the restricted product of the $\mathbb{Q}_p$ with respect to $\mathbb{Z}_p$, with its original additive Haar measure $\mu(\widehat{\mathbb{Z}}) = 1$. For a positive rational a define

$$(\mathcal{U}_a F)(x) = \sqrt{a} F(x/a), \quad F \in L^2(\mathbb{A}_f). \quad (\text{IA2})$$

The finite-adelic modulus is $|a|_f = a^{-1}$: this follows first for a prime from $|p|_p = p^{-1}$ and $|p|_q = 1$ for $q \neq p$, and then for all positive rationals by prime factorization. Changing variables in (IA2) proves unitarity, and composition proves $\mathcal{U}_a \mathcal{U}_b = \mathcal{U}_{ab}$. For a positive integer n its restriction to $L^2(\widehat{\mathbb{Z}})$ is the Fourier transform of V_n . The proof is the finite character sum over the kernel of multiplication by n : it is zero off $n\widehat{\mathbb{Z}}$, and equals n times the original character on that subgroup. This gives the formula $\sqrt{n} \mathbf{1}_{n\widehat{\mathbb{Z}}}(x) F(x/n)$, including its factor.

Put $\xi = \mathbf{1}_{\widehat{\mathbb{Z}}}$, of norm one. For $a = m/n > 0$ with $\gcd(m, n) = 1$,

$$\langle \mathcal{U}_a \xi, \xi \rangle = \frac{1}{\sqrt{mn}}. \quad (\text{IA3})$$

In fact $\widehat{\mathbb{Z}} \cap a\widehat{\mathbb{Z}} = m\widehat{\mathbb{Z}}$, as checked at each valuation, and its measure is $1/m$. Multiplication by $\sqrt{m/n}$ proves the displayed equality. In particular, for all positive integers m, n ,

$$\langle \mathcal{U}_m \xi, \mathcal{U}_n \xi \rangle = \frac{\gcd(m, n)}{\sqrt{mn}}. \quad (\text{IA4})$$

This retains every integral scalar, not only primes or prime powers. For a finite set of primes the joint scalar spectral measure is exactly the product of the local measures (PD13): its joint Fourier coefficient at (k_p) is $\prod_p p^{-|k_p|/2}$ by (IA3). Expanding Laurent polynomials proves the equality of their pairings with the actual operators, just as in the proof of (PD13).

The determinant of the matrix (IA4) for $1 \leq m, n \leq N$ is

$$\det \left(\frac{\gcd(m, n)}{\sqrt{mn}} \right)_{1 \leq m, n \leq N} = \prod_{d=1}^N \frac{\varphi(d)}{d} > 0. \quad (\text{IA5})$$

Here $\varphi(d)$ counts the units of $\mathbb{Z}/d\mathbb{Z}$, with $\varphi(1) = 1$. For completeness, sorting the integers $1, \dots, k$ by the value of $\gcd(a, k)$ proves $\sum_{d|k} \varphi(d) = k$: those with $\gcd(a, k) = k/d$ correspond bijectively to residues coprime to d . Thus, for the lower-triangular matrix $E_{md} = \mathbf{1}_{d|m}$,

$$(\gcd(m, n))_{m, n} = E \operatorname{diag}(\varphi(1), \dots, \varphi(N)) E^t.$$

The diagonal of E consists of ones, so its determinant is one. Taking determinants and retaining the row and column factors $m^{-1/2}, n^{-1/2}$ proves (IA5). This determinant identity is classical, associated with H. J. S. Smith [13]; the complete proof is given here. The calculation specific to this construction is its realization as the Gram matrix of the relative-prime operators on the original integral vector.

13.2 Supported zero acts in an explicitly computed distribution space

Let $\tilde{D} = G(D) = D \sqcup \{\tau_D\}$ with its standard split-semimodule scalar action. For supported $n \in \mathbb{Z}$, $n \cdot d = nd$ is supported and $n \cdot \tau_D = \tau_D$. For the absent scalar τ , every product is τ_D . Let

$$\mathcal{T} = \mathbb{C}^{(D)} \oplus \mathbb{C}\delta_{\tau_D}, \quad \mathcal{T}' = \mathbb{C}^D \oplus \mathbb{C}, \quad \langle f, h \rangle = \sum_{d \in D} f(d)h(d) + c_f c_h. \quad (\text{IA6})$$

The first factor of $\mathcal{T}$ consists of finite-support functions, so the displayed pairing is defined for every $h \in \mathcal{T}'$. It is the full algebraic dual pairing. No square-summability condition is imposed on a distribution in $\mathcal{T}'$.

Pullback by the scalar maps defines operators R_a on all functions $\mathcal{T}'$ satisfying $R_a R_b = R_{ab}$ and $R_1 = I$. For nonzero supported n , R_n restricts to (IA1) on D and fixes the absent coordinate. For the two distinct zeros it gives, exactly,

$$R_e(f, c) = (f(0_D)\mathbf{1}_D, c), \quad R_\tau(f, c) = (c\mathbf{1}_D, c). \quad (\text{IA7})$$

In particular

$$R_e\delta_{0_D} = (\mathbf{1}_D, 0), \quad R_\tau\delta_{0_D} = 0, \quad R_e\delta_{\tau_D} = \delta_{\tau_D}, \quad R_\tau\delta_{\tau_D} = (\mathbf{1}_D, 1). \quad (\text{IA8})$$

Every equality follows by evaluating at a supported d and at τ_D . They prove, rather than identify away, the difference between the action of the integer zero and the absent scalar. Composition of the original scalar maps proves the representation law also when a or b is either zero. This is a representation of the *multiplicative scalar monoid*; no false additive identity $R_{a+b} = R_a + R_b$ is asserted for pullbacks of functions.

There is an exact reason to keep the distribution space (IA6). The nonzero function $\mathbf{1}_D$ has infinite counting-measure norm. Thus R_e has no bounded extension on $\ell^2(D)$ agreeing with (IA8). Its full action nonetheless exists on $\mathcal{T}'$ and its restriction $\mathcal{T} \rightarrow \mathcal{T}'$ is given by (IA7). Its Fourier transform on the arithmetic component is

$$f(0_D)\mathbf{1}_D \longleftrightarrow \left(\int_{\widehat{\mathbb{Z}}} \mathcal{F}f(x) dx \right) \delta_0. \quad (\text{IA9})$$

To prove this without a convergence claim, test against a character polynomial $F = \sum_d a_d \chi_d$. The distribution with all Fourier coefficients one takes the value $\sum_d a_d = F(0)$, which is exactly δ_0 . Orthogonality (PD5) gives $f(0_D) = \int \mathcal{F}f$. This proves both sides of (IA9) on every test. The algebraic transpose of R_e on distributions is also represented on finite-support tests: it sends (f, c) to $((\sum_d f(d))\delta_{0_D}, c)$. Direct substitution in (IA6) verifies the transpose identity. Thus the mass and supported-zero evaluation terms are explicit adjoint boundary maps.

The integer 0 has therefore remained in every scalar map. Its action lands at a boundary distribution that the Hilbert completion alone does not contain. The maps (IA7)–(IA9) specify this enlargement and its relation to the positive Hilbert operators (IA1)–(IA4). Any purity or duality argument using that Hilbert realization must retain this computed zero-scalar boundary and its transpose, rather than quietly delete the integer zero from the programme.

14 Complete support cohomology of the original maps

The construction starts with the distinct zeros of $G(R)$ and retains both of them. It takes finite matrices with their original coefficients and declared supports to a diagram of cohomology modules over every compatible support trajectory. This entire diagram carries a canonical action of $G_{\mathcal{L}}(R)$. Its full-support fibre is exactly the original kernel with its original minimum metric. Nonzero composites are computed before a sequence is called a complex.

The foundational input is the original semimodule classification [2], lines 458–945, especially the source labels `thm:fiber-modules`, `thm:reconstruct-semimodule`, `thm:equivalence-semimodules-diagrams`, and `thm:free-semimodule-boolean-cube`. All reconstruction assertions needed here are proved below. The original source and conductor matrices are those of OPG5, CEP2–5, PCL17–21, and GIP1–6 in the pinned source collection [6]. No coordinate support is identified across a basis change without its explicit connecting maps.

14.1 The two zeros and cancellation paths

Let R be a nonzero commutative ring with identity, and let

$$S = G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset S. \quad (\text{SCD1})$$

On the supported copy of R , the operations are the ring operations. The new element τ is the additive identity and multiplicative absorber. Thus $r + (-r) = e$, not τ . The ring projection $p : S \rightarrow R$ sends τ to 0_R and fixes R . The support character $\chi : S \rightarrow \{0, 1\}$ sends τ to zero and every supported scalar to one. Checking whether the factors or summands are supported proves that both are semiring morphisms. These are the same maps as in the baseline note [1].

For finite sets I, J , a declared support for an R -matrix $d : R^I \rightarrow R^J$ is a relation $D \subseteq J \times I$ containing every pair with $d_{ji} \neq 0_R$. It may also contain supported zeros. Define

$$\tilde{d}_{ji} = \begin{cases} d_{ji} \in R \subset S & (j, i) \in D, \\ \tau & (j, i) \notin D, \end{cases} \quad D(A) = \{j : \exists i \in A, (j, i) \in D\}. \quad (\text{SCD2})$$

The sparse declaration contains exactly the nonzero entries; the fully entrywise declaration has $D = J \times I$. All results below retain whichever support is declared.

A vector of S^I is a pair (A, x) with $A \subseteq I$, $x \in R^A$. Zero entries of x are still supported; outside A its entries are τ . Direct matrix multiplication gives

$$\tilde{d}(A, x) = (D(A), d_{D(A), Ax}). \quad (\text{SCD3})$$

Every row with a declared incident edge receives a supported sum, including a sum that cancels to e .

Proposition 14.1. *For $d : R^I \rightarrow R^J$ and $f : R^J \rightarrow R^K$ with supports D, F ,*

$$p(\tilde{f}\tilde{d}) = fd, \quad \chi(\tilde{f}\tilde{d})_{ki} = \bigvee_{j \in J} (\mathbf{1}_F(k, j) \wedge \mathbf{1}_D(j, i)). \quad (\text{SCD4})$$

If $fd = 0_R$, the lifted composite has entry e at every declared length-two path and τ elsewhere. It is the global zero matrix over S exactly when there are no such paths.

Proof. A product in the sum is supported exactly when both entries are supported. Its ring value is $f_{kj}d_{ji}$, even when that product is zero. A nonempty sum of supported terms is supported and has the ring sum as its value. Only an empty supported sum is τ . This proves every assertion, including rings with zero divisors. The cancellation certificate $\sum_j f_{kj}d_{ji} = 0_R$ therefore retains all of its individual paths. $\square$

In particular,

$$d_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad d_1 = (1 \quad -1), \quad d_1 d_0 = 0_R, \quad \tilde{d}_1 \tilde{d}_0 = (e). \quad (\text{SCD5})$$

The exact morphism p carries this e to 0_R . It does not identify e and τ in S .

14.2 Every compatible support trajectory

Fix a finite cochain complex with its original based matrices:

$$C^n = R^{I_n}, \quad d_n : C^n \rightarrow C^{n+1}, \quad d_{n+1}d_n = 0_R. \quad (\text{SCD6})$$

Only finitely many I_n are nonempty. Fix every declared support D_n . A support trajectory is a sequence $A = (A_n)_n$, $A_n \subseteq I_n$, with $D_n(A_n) \subseteq A_{n+1}$. Let $\mathcal{L}_D$ be their componentwise ordered set.

Theorem 14.2. *The lattice $\mathcal{L}_D$ is complete and distributive. Its meet and join are componentwise intersection and union, its bottom is the empty trajectory, and its top is $\mathbf{I} = (I_n)_n$. For arbitrary seed sets $U_n \subseteq I_n$, their smallest containing trajectory is*

$$\text{cl}_D(U)_n = U_n \cup \bigcup_{m < n} D_{n-1} \cdots D_m(U_m). \quad (\text{SCD7})$$

This closure is monotone, idempotent, and preserves unions.

Proof. An edge starting in a union starts in one member, whose next component contains its endpoint. An edge starting in an intersection has its endpoint in every next component. Thus arbitrary unions and intersections are trajectories, with the stated empty conventions. Set-theoretic distributivity proves the lattice assertion. Every path endpoint in SCD7 is forced in any containing trajectory. Appending one edge proves that SCD7 itself is a trajectory. This proves minimality and all closure laws. $\square$

Equivalently the graded vertices (n, i) carry directed edges $(n, i) \rightarrow (n+1, j)$ from D_n . Degrees strictly increase, so reachability defines a finite partial order. The trajectories are exactly its upward-closed subsets. Every path remains visible.

Theorem 14.3. *For $A \in \mathcal{L}_D$ put*

$$C_A^n = R^{A_n}, \quad d_{n,A} = (d_n)_{A_{n+1}, A_n}. \quad (\text{SCD8})$$

These are cochain complexes. Extension by zero $j_{AB}^n : C_A^n \rightarrow C_B^n$ for $A \subseteq B$ is an injective cochain map. Consequently

$$Z_A^n = \ker d_{n,A}, \quad B_A^n = \text{im } d_{n-1,A}, \quad H_A^n = Z_A^n / B_A^n, \quad h_{AB}^n[x] = [j_{AB}^n x] \quad (\text{SCD9})$$

defines a diagram of R -modules, with exact composition of its maps.

Proof. If $i \in A_n$ and $d_{n,ji} \neq 0$, then $j \in A_{n+1}$. Thus the restricted intermediate sum removes no nonzero term:

$$(d_{n+1,A}d_{n,A})_{ki} = \sum_{j \in I_{n+1}} d_{n+1,kj}d_{n,ji} = 0_R.$$

Every declared intermediate path is also included in A_{n+1} , so its cancellation is retained. The same support condition says that applying d_n to a vector supported in A_n produces a vector in $R^{A_{n+1}}$. Therefore extension by zero commutes with the differential. It maps cycles and boundaries to cycles and boundaries; this proves the quotient map is well defined. Linearity, identity, and composition follow from zero extension. $\square$

For $A \subseteq B$, identify the smaller coordinate modules with their zero-extended copies. The exact kernels and cokernels are

$$\ker h_{AB}^n = (Z_A^n \cap B_B^n) / B_A^n, \quad \text{coker } h_{AB}^n = Z_B^n / (Z_A^n + B_B^n). \quad (\text{SCD10})$$

Indeed a smaller cycle becomes zero precisely when it becomes a larger boundary, and the image is $(Z_A^n + B_B^n) / B_B^n$. For SCD5 there are exactly six trajectories: the empty one; the last vertex; either middle vertex with the last; both middle vertices with the last; and the full set. The penultimate one has $H^1 = R(1, 1)$, which becomes a boundary in the full exact complex. The last-vertex fibre has $H^2 = R$. Thus exactness at full support does not discard the nontrivial cohomology at smaller supports.

14.3 Reconstruction and the full $G_{\mathcal{L}_D}(R)$ base

Define the disjoint union, retaining its index even when a module is zero,

$$\mathcal{H}_D^n = \bigsqcup_{A \in \mathcal{L}_D} H_A^n. \quad (\text{SCD11})$$

For elements in two fibres use

$$(A, [x]) + (B, [y]) = (A \cup B, h_{A, A \cup B}[x] + h_{B, A \cup B}[y]). \quad (\text{SCD12})$$

A supported $r \in R$ acts within its fibre, and τ acts by the bottom-fibre zero. Define $\mathcal{C}_D^n, \mathcal{Z}_D^n, \mathcal{B}_D^n$ from C_A^n, Z_A^n, B_A^n in the same way.

Theorem 14.4. *These operations make $\mathcal{H}_D^n$ a $G(R)$ -semimodule. Its support lattice is exactly $\mathcal{L}_D$, its fibres are exactly H_A^n , and its transports are h_{AB}^n . It is the quotient of $\mathcal{Z}_D^n$ by the congruence*

$$(A, x) \sim (B, y) \iff A = B \text{ and } x - y \in B_A^n. \quad (\text{SCD13})$$

Thus different support indices are never identified.

Proof. Three summands may all be transported to $A \cup B \cup C$. Functoriality of h makes either parenthesization equal to the same sum in that module. Commutativity and the bottom identity follow in the same way. Linearity of transport proves both scalar distributive laws. In particular $(r + s)x = rx + sx$ occurs in the same fibre, also when $r + s = 0_R$. The rules involving τ follow from its action. The supported zero acts by $e(A, [x]) = (A, 0)$, and $(A, 0) + (B, 0) = (A \cup B, 0)$, identifying the support lattice. Boundaries transport to boundaries, so SCD13 is preserved under addition and scalars. Its fibre quotient is precisely H_A^n . $\square$

There is an exact distinction between this construction and a falsely asserted global square-zero lift. Put

$$\mathcal{D}_n(A, x) = (A, d_{n,A}x), \quad \mathbf{z}_{n,n+2}(A, x) = (A, 0). \quad (\text{SCD14})$$

They are S -linear and $\mathcal{D}_{n+1}\mathcal{D}_n = \mathbf{z}_{n,n+2}$. For nontrivial $\mathcal{L}_D$, this is not the global zero map, whose output has index $\emptyset$. The usual preimage of that global zero is not the union of fibre kernels. Cohomology is taken in the additive category of R -module diagrams on the fixed lattice, as explicitly specified in SCD9, then reconstructed by SCD11–13.

The exact return map to the original free semimodule is

$$E_n : \mathcal{C}_D^n \rightarrow S^{I_n}, \quad E_n(A, x) = (A_n, x). \quad (\text{SCD15})$$

It is S -linear because supports of sums are unions and zero extensions commute with addition. It can fail to be injective: two distinct trajectories may have the same n -th support. The original lifted matrix is related to it by

$$E_{n+1}\mathcal{D}_n(A, x) = T_{D_n(A_n), A_{n+1}}(\tilde{d}_n E_n(A, x)). \quad (\text{SCD16})$$

Here T inserts supported zeros at the additional coordinates, exactly as in the Boolean-cube diagram of $S^{I_{n+1}}$. SCD3 and SCD8 prove this identity at every coordinate.

We now include all support scalars. For the bounded distributive lattice $\mathcal{L} = \mathcal{L}_D$, define

$$G_{\mathcal{L}}(R) = \{(r, \mathbf{I}) : r \in R\} \cup \{z_A = (0_R, A) : A \in \mathcal{L}\} \subset R \times \mathcal{L}, \quad (\text{SCD17})$$

with coordinatewise operations

$$(r, A) + (s, B) = (r + s, A \cup B), \quad (r, A)(s, B) = (rs, A \cap B).$$

It is a subsemiring of the product of the ring R and the lattice semiring $(\mathcal{L}, \cup, \cap)$: closure follows because a nonzero ring coordinate requires full support. The zero is $z_{\emptyset}$, the unit is $(1, \mathbf{I})$, and $(0_R, \mathbf{I}) = z_{\mathbf{I}}$ is the supported e .

Theorem 14.5 (Canonical action of every support scalar). *Every module diagram on $\mathcal{L}$, and in particular the cochain, cycle, boundary and cohomology diagrams above, has the action*

$$(r, \mathbf{I})(A, x) = (A, rx), \quad z_\lambda(A, x) = (\lambda \cap A, 0). \quad (\text{SCD18})$$

This defines a $G_{\mathcal{L}}(R)$ -semimodule. Restricting along

$$G(R) \longrightarrow G_{\mathcal{L}}(R), \quad \tau \longmapsto z_\emptyset, \quad r \longmapsto (r, \mathbf{I}) \quad (\text{SCD19})$$

recovers exactly the preceding $G(R)$ -semimodule. The scalar map is injective whenever $\emptyset \neq \mathbf{I}$.

Proof. The two rules agree at the shared element $(0_R, \mathbf{I}) = z_{\mathbf{I}}$: both give $(A, 0)$. A supported scalar preserves the label and acts linearly. For a zero-support scalar, additivity on vectors is exactly

$$\lambda \cap (A \cup B) = (\lambda \cap A) \cup (\lambda \cap B),$$

because linear transport sends a zero to a zero. Scalar addition of two such elements uses the other distributive identity, $(\lambda \cup \mu) \cap A = (\lambda \cap A) \cup (\mu \cap A)$. If one scalar is $(r, \mathbf{I})$ and the other is z_λ , their sum is $(r, \mathbf{I})$; the vector sum is $(A, rx) + (\lambda \cap A, 0) = (A, rx)$, as required. For two supported scalars, scalar additivity is the ordinary module law, even if their sum is the supported zero.

For scalar multiplication, two zero-support scalars give $(\lambda \cap \mu) \cap A$. A supported scalar followed or preceded by z_λ gives the same zero in $\lambda \cap A$. Two supported scalars satisfy the ordinary module multiplication law, also when their product is zero in R . The unit acts as identity. The global zero maps to the zero of the bottom fibre, hence is the required absorber. All semimodule axioms have now been checked. The homomorphism SCD19 follows directly from SCD17, and its restriction gives the declared action of both τ and every supported scalar. Injectivity follows from the distinct support coordinates. $\square$

The maps in SCD14 and the cohomology quotient are linear for this larger semiring: each takes the zero in a fibre to the zero in the same fibre. Coordinate extraction SCD15 is equivariant along the scalar morphism

$$G_{\mathcal{L}}(R) \rightarrow G_{\mathcal{P}(I_n)}(R), \quad (r, \mathbf{I}) \mapsto (r, I_n), \quad z_A \mapsto z_{A_n},$$

because the coordinate map $A \mapsto A_n$ preserves union, intersection, bottom and top.

Theorem 14.6 (Exact additive group completion and arithmetic receiver). *The map*

$$\epsilon_{\text{full}} : \mathcal{H}_D^n \rightarrow H_{\mathbf{I}}^n, \quad (A, [x]) \longmapsto h_{A, \mathbf{I}}^n[x] \quad (\text{SCD20})$$

is the universal additive group completion. It is also the universal linear map to an R -module, for the scalar projection $\pi : G_{\mathcal{L}}(R) \rightarrow R$, $(r, A) \mapsto r$. Thus the original cohomology is recovered exactly on the arithmetic quotient $G_{\mathcal{L}}(R)/(e) \cong R$, while every other fibre remains in $\mathcal{H}_D^n$.

Proof. The displayed map preserves addition and is onto, using the top fibre. Any additive map from $\mathcal{H}_D^n$ to an abelian group kills every $(A, 0)$, since it is additively idempotent and the target permits cancellation. But

$$(A, [x]) + (\mathbf{I}, 0) = (\mathbf{I}, h_{A, \mathbf{I}}[x]).$$

Applying the map forces its value on $(A, [x])$ to be its value on that transported top-fibre element. The restriction to the top fibre is a group homomorphism. This proves unique factorization through SCD20 for every abelian target, hence the universal group-completion assertion. For R -linear targets the same proof retains scalar linearity.

The scalar projection is a semiring morphism. Its kernel ideal is the full zero-amplitude fibre $\{z_A : A \in \mathcal{L}\}$, which equals the principal ideal (e) : $ez_A = z_A$, while multiplying e by any scalar

and adding such products always has zero amplitude. Moreover addition by $e = z_{\mathbf{I}}$ sends every zero-amplitude scalar to e . The congruence identifying e with the global zero therefore identifies every z_A with zero, and leaves exactly the ring operations on R . Conversely π separates all different ring amplitudes. Thus this quotient is exactly R . Finally SCD20 sends $z_\lambda(A, [x])$, a fibre zero, to zero and sends supported scalars to their usual action, proving its π -linearity. $\square$

14.4 Gluing, matrix changes, and the space defined by curvature

For $A, B \in \mathcal{L}_D$, the original coordinate inclusions give a degreewise exact sequence of complexes

$$0 \rightarrow C_{A \cap B} \xrightarrow{x \mapsto (x, -x)} C_A \oplus C_B \xrightarrow{(x, y) \mapsto x + y} C_{A \cup B} \rightarrow 0. \quad (\text{SCD21})$$

Surjectivity follows by assigning coordinates in A to the first summand and those in $B \setminus A$ to the second. A pair summing to zero has equal and opposite entries supported in the intersection, which proves exactness. SCD8 proves compatibility with the differentials.

If z is a cycle in the union, choose $z = x + y$ in the two summands. Then $d_A x = -d_B y = w$ is an intersection cycle. Set

$$\partial[z] = [w], \quad \cdots \rightarrow H_{A \cap B}^n \rightarrow H_A^n \oplus H_B^n \rightarrow H_{A \cup B}^n \xrightarrow{\partial} H_{A \cap B}^{n+1} \rightarrow \cdots \quad (\text{SCD22})$$

A new splitting changes (x, y) by $(t, -t)$ with t in the intersection, so w changes by dt . Splitting a primitive of a boundary shows independence of the representative of $[z]$. For exactness at the middle term, if a pair of cycles sums to a boundary, subtract a split primitive to make its sum zero. It then comes from an intersection cycle. If $\partial[z] = 0$, subtract an intersection primitive from one splitting term and add it to the other; both become cycles, proving exactness at the union. Finally if an intersection cycle maps to boundaries in both larger spaces, choose primitives with opposite signs. Their sum is a union cycle mapping to that intersection class. These arguments verify all three repeating positions and also that each composite is zero.

For an actual cochain map $F_n : C^n \rightarrow C'^n$, let P_n be a declared support containing its nonzero coefficients, and let D' be the target differential supports. Then

$$F_{\#} A = \text{cl}_{D'}((P_n(A_n))_n) \quad (\text{SCD23})$$

preserves joins and bottom, and the same matrices restrict to cochain maps $C_A \rightarrow C'_{F_{\#} A}$. This follows from $d'F = Fd$ and the defining support closure; hence it induces cohomology maps. If G is another map with declared supports Q_n , and the declared support of GF is contained in $Q_n P_n$ while retaining every nonzero coefficient, then

$$(GF)_{\#} A \subseteq G_{\#} F_{\#} A. \quad (\text{SCD24})$$

Indeed the trajectory on the right contains every composite coordinate before cancellation. Its minimal closed subtrajectory contains the direct product support. The composite cohomology map equals the direct one followed by this inclusion, since their underlying matrices are equal. This is the exact comparison when a cancellation changes the minimal support. An invertible basis change gives an isomorphism at full support; it is not asserted to give a bijection of coordinate support lattices.

The reconstructed map is always $G(R)$ -linear. Its exact compatibility with the additional lattice scalars is the transport of a zero along

$$F_{\#}(\lambda \cap A) \subseteq F_{\#} \lambda \cap F_{\#} A.$$

The left side is the support of the image of $z_\lambda(A, x)$; the right side is the support obtained by applying $z_{F_{\#} \lambda}$ to the image of (A, x) . Both amplitudes are zero, and the stated inclusion gives their exact diagram map. It follows from monotonicity of $F_{\#}$. If $F_{\#}$ also preserves intersections

and top, it induces a semiring morphism on the two $G_{\mathcal{L}}(R)$ bases, and this comparison is equality. Thus an arbitrary matrix change has the proved support transport, without an unproved scalar-base isomorphism.

Now allow arbitrary consecutive matrices with

$$\Omega_n = d_{n+1}d_n \quad (\text{SCD25})$$

possibly nonzero, and retain the same trajectory lattice. For a trajectory A , the restricted composite is exactly $(\Omega_n)_{A_{n+2}, A_n}$, by the intermediate-index proof of SCD8. Define the forbidden vertices to be those (n, i) whose i -th column in Ω_n is not the zero vector. Include all ancestors of these vertices in the forbidden set, and let T be its complement.

Theorem 14.7 (The curvature defines a maximal exact support space). *The set T is a trajectory. The trajectories whose restricted matrices are complexes are exactly*

$$\{A \in \mathcal{L}_D : A \subseteq T\}. \quad (\text{SCD26})$$

They form a complete distributive lattice with top T .

Proof. An edge from a non-forbidden vertex cannot reach a forbidden vertex, since that would make its starting vertex an ancestor of a forbidden vertex. Hence T is forward closed. If column i of Ω_n is nonzero, choose k with $(\Omega_n)_{ki} \neq 0$. At least one summand $d_{n+1,kj}d_{n,ji}$ is nonzero, so there is a declared path from (n, i) to $(n+2, k)$. Every trajectory containing the first contains the last, and its restricted curvature therefore has that nonzero entry. It cannot be a complex. It also cannot contain an ancestor of the first vertex. Conversely, a trajectory contained in T contains no initial column on which Ω_n is nonzero. All restricted composites are therefore zero. Closure under unions and intersections and the top assertion follow directly. $\square$

This retains the full original curvature rather than declaring it zero. There is also an exact correction using submodules beyond coordinate supports. For consecutive maps d, f ,

$$d(\ker(fd)) = \text{im } d \cap \ker f. \quad (\text{SCD27})$$

Both inclusions follow from $f(dx) = 0$. Restricting the first domain to $\ker(fd)$ gives a genuine complex, while retaining the explicit defect fd and the exact connecting inclusion. Thus the obstruction specifies both a coordinate-support lattice and its potentially larger linear submodule receiver.

14.5 The original source, conductor and invariant observation

At every literal original cutoff retain

$$N \in \{q-1, q, 2q-1, 2q\}, \quad \mathcal{P}_N = \mathbb{C}[S]_{\leq N}, \quad E = \mathbb{C}[S]/(\chi_k), \quad U_N : \mathcal{P}_N \rightarrow E, \quad p \mapsto [p]. \quad (\text{SCD28})$$

The polynomial χ_k is the complete original monic root polynomial of degree q . The source inner product is the original one, with its complete mass. Polynomial and primary coordinates are connected by the actual Vandermonde matrix.

The observation is exactly the original OPG5 map:

$$P_k = \frac{1}{n+1} \sum_{a=0}^n (C^{-a})^{\otimes k}, \quad L = P_k \Pi^{\otimes k} U^{\otimes k} \iota, \quad B = \text{im } L, \quad j\Lambda = L, \quad K = \ker \Lambda. \quad (\text{SCD29})$$

The U in this tensor formula is the original period-unit map, distinct from the remainder U_N . Every original period and invariant column in P_k, Π, U, ι is kept. See the exact OPG4–5 definition and original cycle action.

Let $Z_{\chi,N}$ be multiplication by the complete $\chi = \chi_k$, including its conversion to the original source coordinates. Use $\mathcal{P}_{N-q} = 0$ when $N = q - 1$. Then

$$\mathcal{P}_{N-q} \xrightarrow{Z_{\chi,N}} \mathcal{P}_N \xrightarrow{\Lambda U_N} B \quad (\text{SCD30})$$

is a complex, with degrees zero, one and two. Its cohomology is

$$H^0 = 0, \quad H^1 \xrightarrow{\sim} K, \quad [p] \mapsto U_N p, \quad H^2 = 0. \quad (\text{SCD31})$$

Multiplication by a monic nonzero polynomial is injective. Division by χ proves $\ker U_N = \chi \mathcal{P}_{N-q}$; U_N is onto since $N \geq q - 1$. A cycle therefore maps to K , every member of K has a lift, and two lifts differ by exactly a relation. The map to B is onto, proving the last assertion. Applying SCD7–20 to these complete matrices supplies the full $G_{\mathcal{L}_D}(\mathbb{C})$ -cohomology object, with precisely K as its terminal arithmetic fibre.

For the conductor graph, retain its original simple-quartet domain $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$. No conclusion about a more general multiplicity is imported from this graph presentation. Retain all coefficients of $A = \sum_{a,b=0}^8 a_{ab} e_{ab}^{(8)}$, all its shifts b_{ab} , and the original lower polynomial $\chi' = \chi_{k-8}$. The prime in χ' denotes that polynomial, not a derivative. The maps are

$$(\mathcal{T}_A \mathcal{P})(S') = \sum_{a,b=0}^8 a_{ab} p(S' + b_{ab}), \quad C_{(i,j),(u,w)} = \begin{cases} a_{u-i,w-j} & 0 \leq u-i, w-j \leq 8, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{SCD32})$$

The root identity $\beta'_{ij} + b_{ab} = \beta_{i+a,j+b}$ gives $\chi' \mid \mathcal{T}_A(\chi h)$ for every h . Thus $\mathcal{T}_A$ induces the map on quotient spaces with matrix $V_{\beta'}^{-1} C V_{\beta}$. The transpose C^T is multiplication by the nonzero biform A , hence injective in the polynomial domain. Therefore C is onto. Its kernel $V_A \subset E$ contains the actual K , by OCF17 and CEP5.

The complete pivot and remaining invariant-image maps are retained:

$$C N_k^T = 0, \quad S_k^T N_k^T = I, \quad \text{im } N_k^T = V_A, \quad J_R = S_k Z_k. \quad (\text{SCD33})$$

These identities hold in the original primary coordinates. Here Z_k contains all additional invariant columns, so after imposing the conductor equation the complete residual equations still include $J_R^T V_{\beta} p = 0$, as in CG29. The actual residual observation is

$$\Xi = \Lambda|_{V_A} : V_A \rightarrow B_A = \Lambda(V_A), \quad \ker \Xi = K.$$

In the original fixed residual frame it is Ξ_k^{fix} ; changing to that frame retains its full invertible target map. No formula below substitutes a historical rank numeral for $\dim K$ or $\dim B_A$.

Let v be the first nonzero translation moment order. The actual invertible source map S_N takes a polynomial to its first v original source coordinates and its $\mathcal{T}_A$ image. To see invertibility, Taylor's finite expansion says $\mathcal{T}_A$ lowers each degree at least v by exactly v and kills exactly $\mathcal{P}_{v-1}$. The first v source coordinates are an isomorphism on that kernel. The source and target dimensions are equal, so the displayed injective map is invertible. Define the original graph maps

$$\begin{aligned} X_N &= \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq N-v-q'}, & D_N &= \text{diag}(I_v, M_{\chi'}), \\ E_N &= U_N S_N^{-1} D_N : X_N \rightarrow V_A, & B_N &= \Xi E_N : X_N \rightarrow B_A. \end{aligned} \quad (\text{SCD34})$$

Here D_N includes the exact conversion to the lower original source coordinates. The equality $S_N^{-1} D_N X_N = U_N^{-1}(V_A)$ follows from divisibility by χ' . Consequently E_N is onto V_A .

The full original relation columns are

$$\begin{aligned} \mathcal{Z}_N h &= \begin{pmatrix} P_{<v} Z_{\chi,N} h \\ \mathcal{U}_{A,N-q} h \end{pmatrix}, & D_N \mathcal{Z}_N &= S_N Z_{\chi,N}, \\ (\mathcal{U}_{A,N-q} h)(S') &= \sum_{a,b=0}^8 a_{ab} d_{ab}(S') h(S' + b_{ab}), & d_{ab}(S') &= \chi(S' + b_{ab}) / \chi'(S'). \end{aligned} \quad (\text{SCD35})$$

Each d_{ab} is the original complete product of root differences. Multiplication by χ' proves the lower equality using SCD32; the upper block is exactly the retained lower source coordinates. Thus $\mathcal{Z}_N$ is injective and $\ker E_N = \text{im } \mathcal{Z}_N$.

The correct graph complex and its full-support cohomology are

$$\mathcal{P}_{N-q} \xrightarrow{\mathcal{Z}_N} X_N \xrightarrow{B_N} B_A, \quad H^1 \xrightarrow{\sim} K, \quad [x] \mapsto E_N x, \quad H^0 = H^2 = 0. \quad (\text{SCD36})$$

Injectivity, surjectivity and $\ker \Xi = K$ prove these statements exactly as in SCD31. The chain map to SCD30 is

$$(I, S_N^{-1} D_N, B_A \hookrightarrow B). \quad (\text{SCD37})$$

Its squares are SCD35 and $\Lambda U_N S_N^{-1} D_N = \Xi E_N$. Its induced middle cohomology map is the identity of K . On support trajectories the exact map is SCD23 with the actual coefficient supports of these three maps. Hence the support comparison is proved as well.

Choose the full original kernel frame I_K with actual rank $r = \dim K$. In original orthonormal source coordinates let $G_N = (U_N U_N^*)^{-1}$, and set

$$W_{K,N} = D_N^{-1} S_N U_N^* G_N I_K, \quad E_N W_{K,N} = I_K.$$

The inverse of D_N is used only on its indicated image; the minimum lift is in $U_N^{-1}(V_A)$. Then the exact kernel is

$$\ker B_N = \text{im}[\mathcal{Z}_N, W_{K,N}], \quad \text{rank}[\mathcal{Z}_N, W_{K,N}] = (N - q + 1) + r. \quad (\text{SCD38})$$

Applying E_N to a relation among those columns first kills the coefficients of the independent I_K . Injectivity of $\mathcal{Z}_N$ then kills the remainder. Conversely subtract the corresponding $W_{K,N}$ lift from any vector in $\ker B_N$; what remains lies in $\ker E_N$. This proves every column and rank assertion, including the empty relation list at $N = q - 1$.

The longer sequence $X_N \xrightarrow{E_N} V_A \xrightarrow{\Xi} B_A$ has the exact curvature

$$\Xi E_N = B_N : X_N \twoheadrightarrow B_A. \quad (\text{SCD39})$$

It is a complex only when the actual B_A is zero. In all cases its zero-curvature submodule is precisely SCD38. Thus the nonzero defect of that proposed sequence is retained, while SCD36 gives the genuine original graph complex.

14.6 Exact return to the original metric and arithmetic action

The full-support identification preserves the actual minimum metric. For a fixed $x \in K$, all representatives in SCD30 differ by χh . Minimizing their complete source norm is precisely the definition of the original quotient norm on E . Thus in the unchanged full kernel frame it is

$$H_N^K = I_K^* G_N I_K. \quad (\text{SCD40})$$

For the graph complex, its actual source metric is

$$H_N^{\text{actual}} = D_N^* S_N^{-1} D_N. \quad (\text{SCD41})$$

The map $S_N^{-1} D_N$ in SCD37 is then an isometry onto its image. Every representative of $x \in K$ lies in $U_N^{-1}(V_A)$, so the same minimum gives exactly SCD40. This proof does not substitute the alternative graph product metric $D_N^* D_N$ for SCD41. For all four cutoffs, the full cohomology determinant is therefore the original determinant, including the original signed return

$$\begin{aligned} & \log \det H_{q-1, \mathbf{I}}^{\text{coh}} + \log \det H_{q, \mathbf{I}}^{\text{coh}} - \log \det H_{2q-1, \mathbf{I}}^{\text{coh}} - \log \det H_{2q, \mathbf{I}}^{\text{coh}} \\ &= \log \det H_{q-1}^K + \log \det H_q^K - \log \det H_{2q-1}^K - \log \det H_{2q}^K. \end{aligned} \quad (\text{SCD42})$$

Use the same I_K at every endpoint and the original empty determinant convention. This exact identity assigns no missing asymptotic coefficient.

The original multiplication $M : E \rightarrow E$ need not preserve K . Its failure to descend through observation is explicitly computable. Keep the full minimum section and observed action

$$Q_N = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_N, \quad P_N = L_N \Lambda, \quad M_{B,N} = \Lambda M L_N.$$

Then

$$\Lambda M - M_{B,N} \Lambda = \Lambda M (I - P_N) = \Lambda M I_K (H_N^K)^{-1} I_K^* G_N. \quad (\text{SCD43})$$

For proof, direct multiplication gives the first equality. The minimum section is orthogonal to $\ker \Lambda = K$; hence $I - P_N$ is the G_N -orthogonal projector onto K . The displayed expression $I_K (H_N^K)^{-1} I_K^* G_N$ is the same projector: multiply to verify idempotence, image, identity on K , and weighted self-adjointness. This proves the final equality without deleting its kernel block.

The exact action-compatible subspace and defect space are

$$0 \rightarrow K \cap M^{-1} K \rightarrow K \xrightarrow{\Lambda M} \text{im}(\Lambda M|_K) \rightarrow 0. \quad (\text{SCD44})$$

The kernel assertion is its definition; the last target is the actual image. Thus this is the exact obstruction to obtaining an action on the original middle cohomology by M .

Retain every higher observation as well:

$$K^{(j)} = \bigcap_{a=0}^j \ker(\Lambda M^a), \quad j \geq 0. \quad (\text{SCD45})$$

The connecting maps are inclusions, and $K^{(j+1)} = K \cap M^{-1} K^{(j)}$. If two successive spaces are equal, the common space is M -invariant: membership in $K^{(j+1)}$ implies that the M -image belongs to $K^{(j)}$. It is then killed by every further observation, so the sequence stabilizes. It has at most $\dim K$ strict drops. Its stable value is the largest M -invariant subspace of K , since every such subspace is killed by every ΛM^a , while the stable value itself is invariant. This is a specified space with exact connecting maps, rather than an assumed descent.

The original metrics, periods and roots are unchanged throughout. The construction provides a complete support object over the absolute base, its exact arithmetic receiver, and the spaces defined by the two original curvature defects SCD39 and SCD43. No support trajectory or additional invariant column is discarded.

14.7 One support base for the four original cutoffs

The four support bases need not be identified by choosing bases again. There is a single exact common construction. For a fixed original source order $s \in \{1, k\}$, retain the original source inclusions $i_{NM} : \mathcal{P}_N \hookrightarrow \mathcal{P}_M$, $N < M$, at the four cutoffs. On relation polynomials use $j_{NM} : \mathcal{P}_{N-q} \hookrightarrow \mathcal{P}_{M-q}$ and on B use the identity. Their original matrices satisfy

$$i_{NM} Z_{\chi,N} = Z_{\chi,M} j_{NM}, \quad U_M i_{NM} = U_N, \quad \Lambda U_M i_{NM} = \Lambda U_N. \quad (\text{SCD46})$$

These are identities on polynomials before passing to any coordinates. Their matrices include whatever original basis conversions are required. They form cochain maps of SCD30 and induce precisely the identity of K in SCD31.

For the graph presentations, the corresponding map is

$$F_{NM}^X = D_M^{-1} S_M i_{NM} S_N^{-1} D_N, \quad E_M F_{NM}^X = E_N, \quad F_{NM}^X Z_N = Z_M j_{NM}. \quad (\text{SCD47})$$

The inverse of D_M is taken only on its actual image: a polynomial whose class lies in V_A still has that property after source inclusion, so S_M of that polynomial has its second component

divisible by χ' . This proves that the map is defined. Cancellation of the displayed inverse maps, followed by SCD35 and SCD46, proves the two identities. Composition for three source cutoffs follows by the same cancellation. The induced cohomology map is again the identity of K .

Now form a finite coefficient graph whose vertices record the cutoff, cochain degree and coefficient index. Include every declared differential edge, every edge of these original horizontal source maps, and, if both presentations are retained together, every edge of the graph-to-polynomial map SCD37. Let $\mathcal{L}_{\text{all}}$ be all subsets closed under these edges. One can include both source orders as separately labelled copies, without identifying their metrics. The proof of SCD7 applies to this finite diagram: unions and intersections remain closed, and forward closure is the least containing closed set. Thus $\mathcal{L}_{\text{all}}$ is a finite bounded distributive lattice with all the original coefficient supports retained.

The coordinate projection to any one of the complexes preserves union, intersection, bottom and top, and takes a master trajectory to a valid local trajectory. Restricting every original matrix at a master trajectory therefore gives a diagram of complexes in which all original squares still commute. To verify a square entry, note that every intermediate index with a nonzero contribution is forced into the master trajectory by its first edge; its second edge is retained as well. Thus the restricted entry is the original full entry, with all its cancellation terms.

For each cutoff reconstruct

$$\mathcal{H}_{\text{all},N}^1 = \bigsqcup_{A \in \mathcal{L}_{\text{all}}} H_{N,A|N}^1. \quad (\text{SCD48})$$

These are modules over the *same* $G_{\mathcal{L}_{\text{all}}}(\mathbb{C})$ by SCD18. The horizontal cohomology maps preserve the master label A , commute with union transports, and take a zero in a fibre to the zero in the same fibre. They are therefore $G_{\mathcal{L}_{\text{all}}}(\mathbb{C})$ -linear: for z_λ both composites give the zero at $\lambda \cap A$; for a supported scalar the assertion is ordinary linearity. Their additive group completions all equal the same K , with identity connecting maps and the four separate original metrics SCD40. This proves a common absolute-base receiver for the complete four-cutoff diagram, including the graph maps, without merging its source metrics.

14.8 Sources and verification scope

The accompanying exact checker verifies the complete six-trajectory cancellation example and its cohomology changes; the $G_{\mathcal{L}}(R)$ scalar laws; and the full source/graph/minimum identities on a specified positive four-root auxiliary example at all four cutoffs. This example tests the proved algebra. It is not a replacement for the original period matrices or an evaluation of a hypothetical zeta zero. No floating-point test proves a native sign or coefficient.

The classification source is read as original LaTeX, not a PDF or a summary. The accompanying source-use ledger records exact hashes, reading coverage and receiving formulas. The elementary group-completion, gluing and cohomology constructions above have complete proofs here; the human semiring reference below is contextual provenance, not a claim that an unread external theorem supplies a missing calculation.

15 The attained metric on every support cohomology fibre

This section works on the fibre complexes constructed from the original coefficient matrices. Their support labels retain which relation columns and ambient coordinates are present. For two nested trajectories $A \subseteq B$, write $K_A \subseteq K_B$ for the cycle spaces and $I_A \subseteq I_B$ for the relation images, all embedded by zero extension in the same original middle coefficient space V . In particular $I_A \subseteq K_A$ and $I_B \subseteq K_B$. No replacement inner product is introduced: let G be the original positive-definite Hermitian form on V .

For a subspace $I \subseteq V$, denote its G -orthogonal projection by P_I . With any full-column-rank frame Z for I ,

$$P_I = Z(Z^*GZ)^{-1}Z^*G. \quad (39)$$

The empty frame defines the zero projection. To verify the formula, its square equals itself, its image is I , and $Z^*G(x - P_I x) = 0$ for every x ; these three properties characterize the orthogonal projection independently of the chosen frame.

Theorem 15.1 (Exact transport and its new boundary kernel). *The quotient $\mathbf{H}_A = K_A/I_A$ has attained norm*

$$\|[x]\|_A^2 = \inf_{z \in I_A} \|x + z\|_G^2 = \|(1 - P_{I_A})x\|_G^2.$$

The inclusion of trajectories induces the well-defined linear map

$$t_{AB} : \mathbf{H}_A \rightarrow \mathbf{H}_B, \quad [x] \mapsto [x], \quad \ker t_{AB} = (K_A \cap I_B)/I_A.$$

For every $x \in K_A$ its entire norm loss is

$$\|[x]\|_A^2 - \|t_{AB}[x]\|_B^2 = x^*G(P_{I_B} - P_{I_A})x = \|(P_{I_B} - P_{I_A})x\|_G^2. \quad (40)$$

The map is a contraction, and its defect has rank at most $\dim I_B - \dim I_A$.

Proof. Write $x = P_{I_A}x + (1 - P_{I_A})x$. For $z \in I_A$, orthogonality gives $\|x + z\|_G^2 = \|P_{I_A}x + z\|_G^2 + \|(1 - P_{I_A})x\|_G^2$. The first term attains zero at $z = -P_{I_A}x$, proving the quotient formula. Because $I_A \subseteq I_B$, two representatives defining the same class at A also define the same class at B . The image vanishes precisely for representatives in $K_A \cap I_B$, giving the exact kernel.

Nested projections obey $P_{I_A}P_{I_B} = P_{I_A} = P_{I_B}P_{I_A}$. The second equality follows because $P_{I_A}x \in I_B$; the first follows by taking adjoints in the G inner product. Thus $P_{I_B} - P_{I_A}$ is the orthogonal projection onto $I_B \cap I_A^\perp$. Subtract the two attained norm formulas to obtain (40); its last expression and its rank follow from this projection description. This proof retains every actual relation vector and its position in the original metric. $\square$

Theorem 15.2 (The complete determinant correction). *Let the columns of U_A be a basis of $K_A \cap I_A^\perp$, so they represent a basis of $\mathbf{H}_A$. Put*

$$\begin{aligned} H_A &= U_A^*GU_A, \\ E_{AB} &= U_A^*G(P_{I_B} - P_{I_A})U_A, \\ H_{B \leftarrow A} &= U_A^*G(1 - P_{I_B})U_A = H_A - E_{AB}. \end{aligned} \quad (41)$$

Here $0 \preceq E_{AB} \preceq H_A$, and $H_{B \leftarrow A}$ is exactly the pullback of the attained quotient metric, including its zero directions. Its kernel is the coordinate representation of $(K_A \cap I_B)/I_A$. When that kernel is zero,

$$\log \det H_{B \leftarrow A} - \log \det H_A = \log \det \left(1 - H_A^{-1/2} E_{AB} H_A^{-1/2} \right). \quad (42)$$

Every eigenvalue of the matrix subtracted from 1 lies in $[0, 1)$; at most $\dim I_B - \dim I_A$ are nonzero. If it has an eigenvalue 1, the corresponding cohomology class becomes a boundary at B and both sides must instead be restricted to a specified surviving subspace before taking an ordinary determinant.

Proof. Theorem 15.1 evaluated on $U_A c$ gives every displayed form identity and inequality. Its zero norm at B is equivalent to membership in I_B , proving the kernel assertion. Congruence by $H_A^{-1/2}$ transforms the metric difference into the matrix in (42); multiplicativity of determinant yields that formula, and positivity proves the eigenvalue claims. Rank cannot increase under congruence or compression. Thus the term records the original relative position of the relations rather than their number alone. $\square$

15.1 Return through all four original cutoffs

Use precisely the four indices and signs

$$(N_1, N_2, N_3, N_4) = (q - 1, q, 2q - 1, 2q), \quad (\sigma_1, \sigma_2, \sigma_3, \sigma_4) = (+1, +1, -1, -1).$$

For each index use its original form G_N , actual relation spaces and attained quotient basis. Whenever the transported spaces have no killed class, define $\Lambda_N = H_{A,N}^{-1/2} E_{AB,N} H_{A,N}^{-1/2}$. Applying the proved identity separately at each index gives the exact signed return

$$\sum_{j=1}^4 \sigma_j \log \det H_{B \leftarrow A, N_j} = \sum_{j=1}^4 \sigma_j \log \det H_{A, N_j} + \sum_{j=1}^4 \sigma_j \log \det(1 - \Lambda_{N_j}). \quad (43)$$

No sign is assigned to the last sum by the separate contraction inequalities: the four signs in the return are retained. If a class is killed, Theorem 15.2 identifies exactly its kernel before a surviving determinant is formed.

Proposition 15.3 (An evaluated example of the full support correction). *Take the complex $\mathbb{C} \xrightarrow{(1,1)^\top} \mathbb{C}^2 \rightarrow 0$. The trajectory with no relation column and both middle coordinates has $I_A = 0$. Activating the relation gives $I_B = \mathbb{C}(1, 1)^\top$. On the original class represented by $x = (1, 0)^\top$ and metric $G_N = \text{diag}(e^N, 1)$,*

$$\|x\|_A^2 = e^N, \quad \|x\|_B^2 = \frac{e^N}{1 + e^N}, \quad \frac{\|x\|_B^2}{\|x\|_A^2} = \frac{1}{1 + e^N}.$$

For m independent copies, the four-cutoff logarithmic correction is exactly

$$2mq + m(\log(1 + e^{-(2q-1)}) + \log(1 + e^{-2q}) - \log(1 + e^{-(q-1)}) - \log(1 + e^{-q})). \quad (44)$$

Its difference from $2mq$ has absolute value at most $m(e^{-(2q-1)} + e^{-2q} + e^{-(q-1)} + e^{-q})$.

Proof. The attained norm is the minimum over $t \in \mathbb{C}$ of $e^N|1 + t|^2 + |t|^2$. Completing the square gives the minimizer $t = -e^N/(1 + e^N)$ and the stated value. The class is not killed because $(1, 0)$ is not a multiple of $(1, 1)$. Orthogonal direct sums multiply the ratios. Take logarithms at all four cutoffs and use $\log(1 + e^N) = N + \log(1 + e^{-N})$; their signed linear combination is $2q$. Finally $0 \leq \log(1 + t) \leq t$ for $t \geq 0$ follows by integrating $(1 + t)^{-1} \leq 1$, proving the error bound. $\square$

This example is completely specified and is not substituted for a native arithmetic coefficient. It demonstrates with an attained minimum that a support transition can contribute at scale kq when m has scale k . Equation (43), evaluated with the original programme spaces, is the receiver required for that programme calculation.

References

- [1] Programme source, *A Note on the Globalization Semiring $G(\mathbb{Z})$* , source bearing the date March 2026, exact public original TeX, especially `cor:prime-ideals`, `thm:zero-prime-not-irr`, `thm:monoid-spec-z`, `cor:monoid-spec-ms`, and `thm:base-extension-product`. This TeX prints no author name. The programme's originator attributes the original supported-zero prime derivation to Gemini 2.5 Pro and states that it predates this reconstruction substantially. The printed date identifies this source version; it does not date the first derivation. The prime theorem is retained prior work, not a new result of this manuscript.
- [2] The Clankers, *A Secondary Note on Split-Zero Globalization: Semimodule Classification, Multiplicative Reconstruction, and All-Orders Zeta Deformation*, complete original TeX, pinned edition, `thm:fiber-modules`, `thm:equivalence-semimodules-diagrams`, and `thm:monoid-prime-explosion`.

- [3] The Clankers, *Split Support Geometry*, complete original version 11 TeX, pinned edition, `def:lattice-split`, `thm:lattice-prime-classification`, `thm:spectral-ordinal-sum`, and `thm:branch-selected-spectrum`.
- [4] Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, arXiv:math/9811068v1, 1998. Original author TeX read: Section III, equations (6)–(11), and Section V, Theorem 3 and its cutoff conventions.
- [5] Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, 2026. Original author source `rhready.tex`, labels `sectweilpos`, `wp`, `arch`, `YQS`, `GL1QS`, and `weilqb`. The criterion stated there is attributed to André Weil; this manuscript does not claim its authorship or reprove that classical equivalence.
- [6] *Retained complete proof sources*, pinned programme edition 6cf5b3aa8ff395eba6b5512b28d117334da5767b. Exact source locators: OCF17, conductor inclusion; CEP2–5, complete translation and primary matrices; GIP1–6, complete source graph and metrics; PCL17–21, pivot and metric transport; CG29–30, additional invariant constraints.
- [7] Jonathan S. Golan, *Semirings and Their Applications*, Kluwer Academic Publishers, 1999. Human-source context for semiring and semimodule language, as cited in the original globalization note. No unproved result from this book is required for the finite constructions here.
- [8] Antoine Sédillot, *Topological adelic curves: Zariski–Riemann spaces, algebraic coverings, Harder–Narasimhan filtrations and heights*, arXiv:2503.20156v2, original author source supplied by the user, `topological_adelic_curves_V2.tex`. Actual reading and uses are recorded in the accompanying source ledger. The exact scope of each imported construction is stated where used.
- [9] Pierre Deligne, *La conjecture de Weil. II*, Publications Mathématiques de l’IHÉS **52** (1980), 137–252, doi:10.1007/BF02684780. The locally supplied D032 English mathematical edition is a transcription/translation, not original author TeX. It has been reread in full; its wording is not substituted for a proof of a programme analogue.
- [10] Jürgen Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften **322**, Springer, 1999. Human provenance for the classical ideal-theoretic arithmetic retained in the foundational source. The fractional-ideal and norm statements used here are proved in Section 5; this citation does not claim a fresh reading of the book.
- [11] Anton Deitmar, *Schemes over $\mathbb{F}_1$* , in *Number Fields and Function Fields—Two Parallel Worlds*, Progress in Mathematics **239**, Birkhäuser, 2005, 87–100. Human provenance retained from the foundational source for monoid-scheme language; all maps used in Section 3 are proved there.
- [12] Oliver Lorscheid, *The geometry of blueprints. Part I: Algebraic background and scheme theory*, Advances in Mathematics **229** (2012), 1804–1846. Human provenance retained from the foundational source for the preaddition construction; the specific prime comparison used here is proved in Section 3.
- [13] Henry J. Stephen Smith, *On the Value of a Certain Arithmetical Determinant*, Proceedings of the London Mathematical Society **s1-7** (1875), 208–213, doi:10.1112/plms/s1-7.1.208. Human provenance for the classical gcd determinant. The publisher metadata has been checked; no original-author TeX is available in that record and no claim of reading the article is made. The exact determinant identity used here is proved in Section 13.